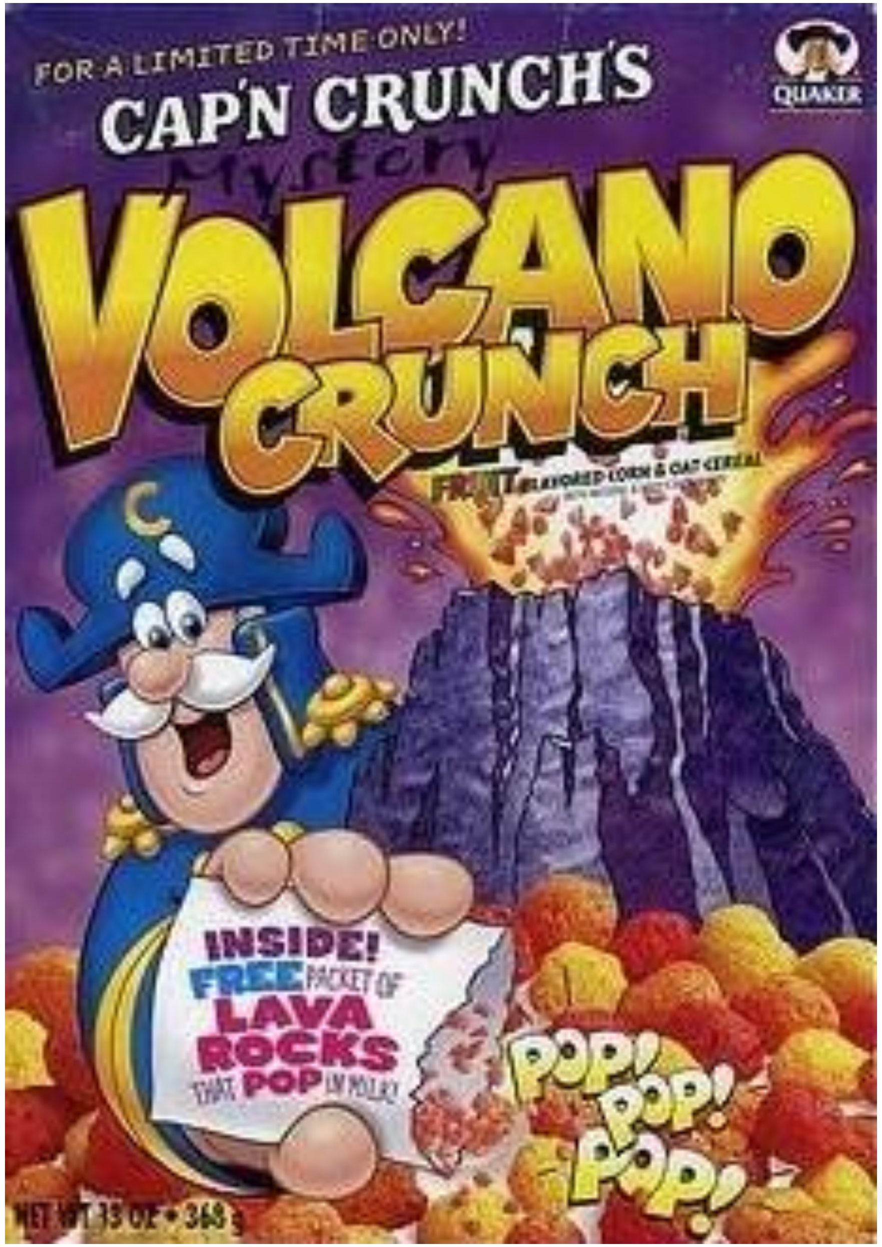
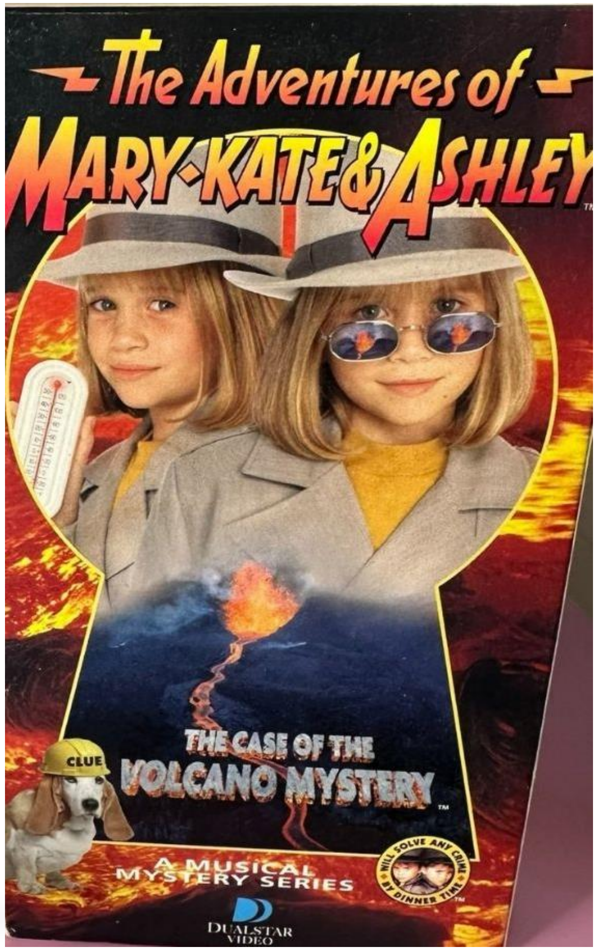


Your *Mystery Volcano* Project
Generic Instructions



Gather information

- From your homework and exercises you should know/have:
  - What volcano you're working on
  - The eruptive history of the volcano
  - The history of lahars (how big? how often? what triggers them?)
  - Any existing geological maps
  - Any existing hazard maps
  - If there is a reason to update the hazard map (different scenario? different audience?)
  - What scenario or scenarios you will focus on (what type of lahars? what volume of lahars? what size eruption, if any?)
  - A DEM of your volcano
  - Some Google Earth images of your volcano
  - A population map if available
  - A topographic or other map that shows roads, rivers, lakes, villages, cities, etc.
  - Who is your target audience (General public? Mining company or transportation agency? Tourists? Emergency managers? Land-use planners?)

Google Earth exploration

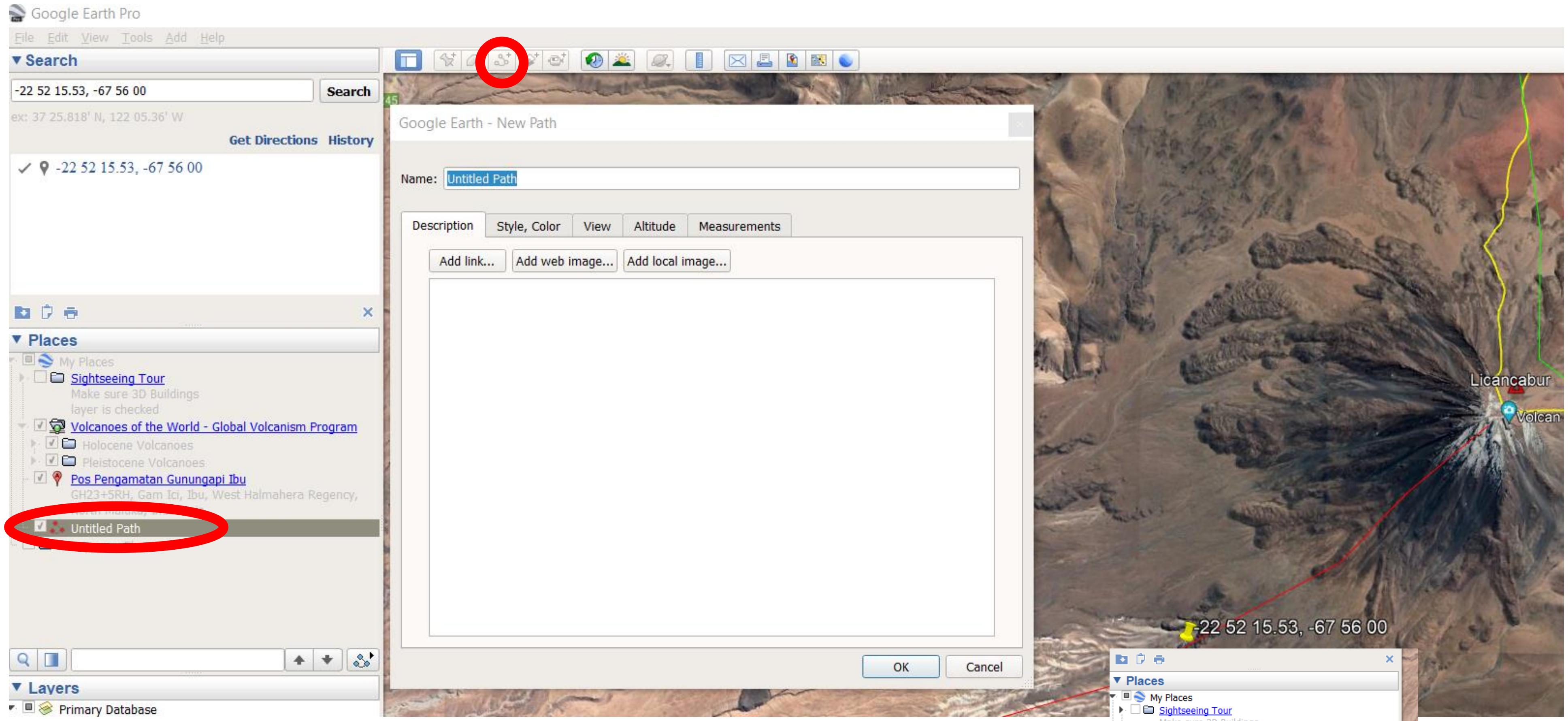
- You need to do some investigating in Google Earth
- Search for your volcano and collect some information
- You’ll need the summit coordinates
- Is your summit nice and pointy? Or complex with multiple peaks or a deep crater?
- You need to make some slope profiles to help decide on an H/L ratio (see next slides as a reminder)
- Look for evidence of pyroclastic flow or other deposit fans and calculate their horizontal distance, change in height (subtract the elevation at the end of the deposit from the elevation of the summit), and calculate the H/L ratio (Look at the streams and cones presentation for a refresh)

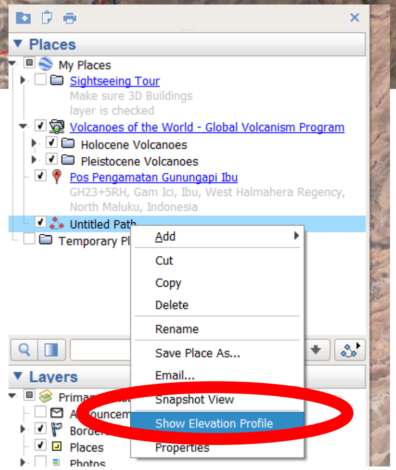
Okay, so how do we choose the H/L value?

PDC deposit fans are useful for choosing an ΔH/L cone/proximal hazard zone

- **Visualize volcano in 3-D:**
  - Find photos (Google image search, GVP)
  - Use Google Earth to get oblique views, fly around, and make a topographic profile.
- **Examine volcano records of past events:**
  - geologic maps; observatory logs; GVP info
- **Look at the geomorphology:**
  - landforms, farming, land use
- **Look for PDC units in the field**
- **Examine satellite imagery**
- **Use global record of analog volcanoes (PDC H/L values)**
- **Start with H/L of 0.3 and refine**

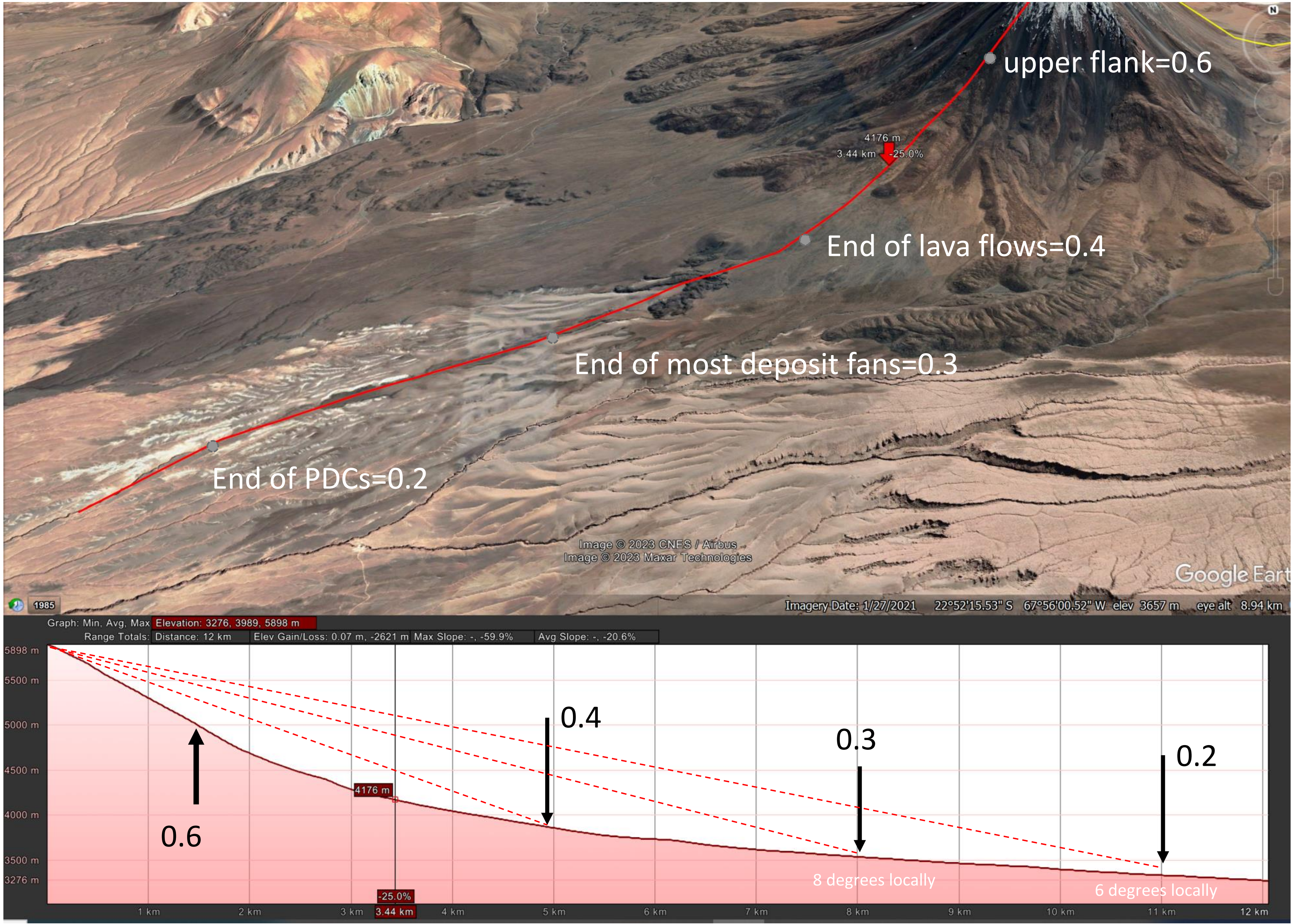
Visualize volcano in 3-D





- Click on the "Add path" button
- Click on the map to make a path
- Name and save the path
- Right click on the path name under "Places"
- Choose "show elevation profile"
- Be careful: the slope values given in Google Earth are local, not H/L from the summit

Visualize volcano in 3-D

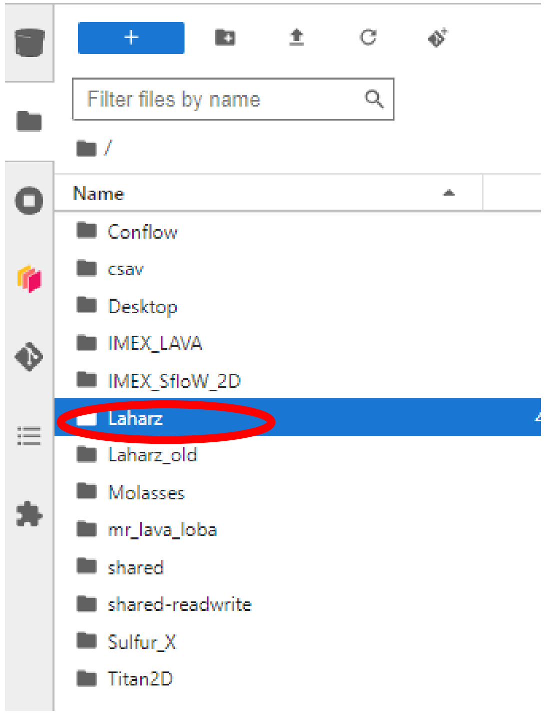


Special Cases: not a nice pointy volcano?
Add elevation in LAHARZ

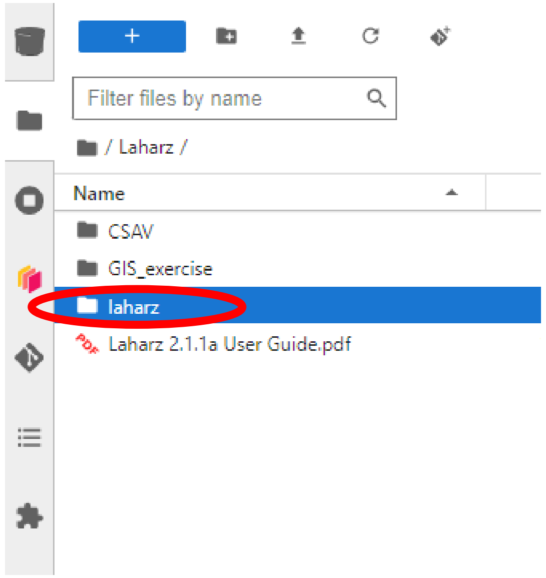
Gather more information and write it down

- What are the coordinates of your summit (search in Google Earth):
  - Latitude:
  - Longitude:
- What volumes will you model:
- What stream network thresholds will you try:
- What H/L ratio(s) will you try:
- Is your volcano pointy or complex:
  - If your volcano is complex, how much elevation do you think you need to add? (check a side profile in Google Earth):

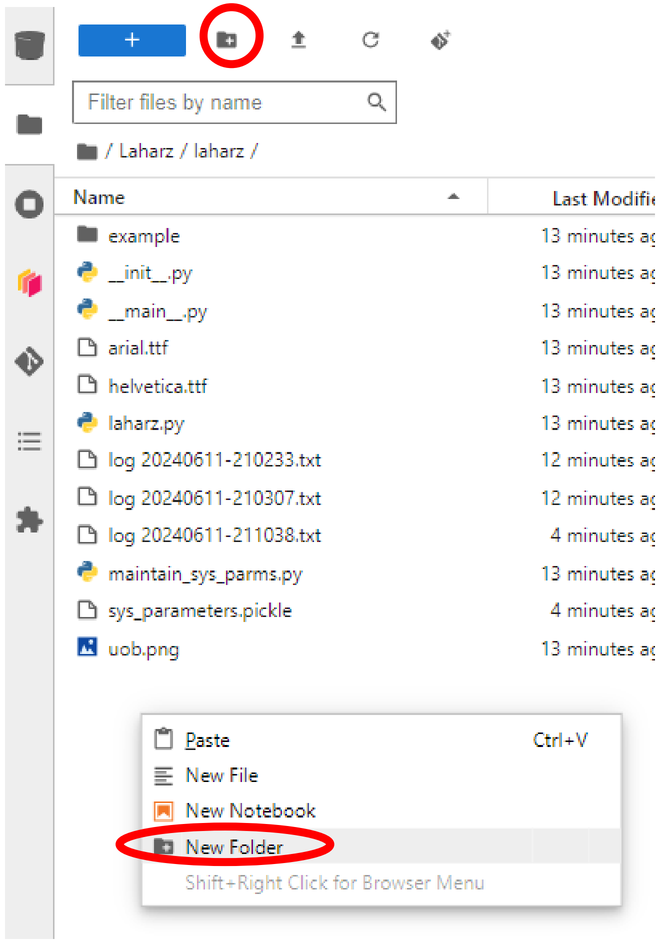
Make a folder to work in and add your DEM



Double click on “Laharz”

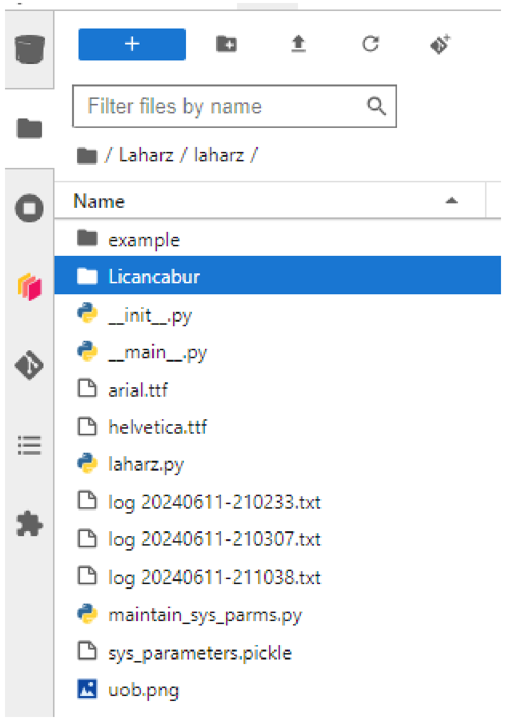


Double click on “laharz”



Right click in a blank space and choose “New Folder”

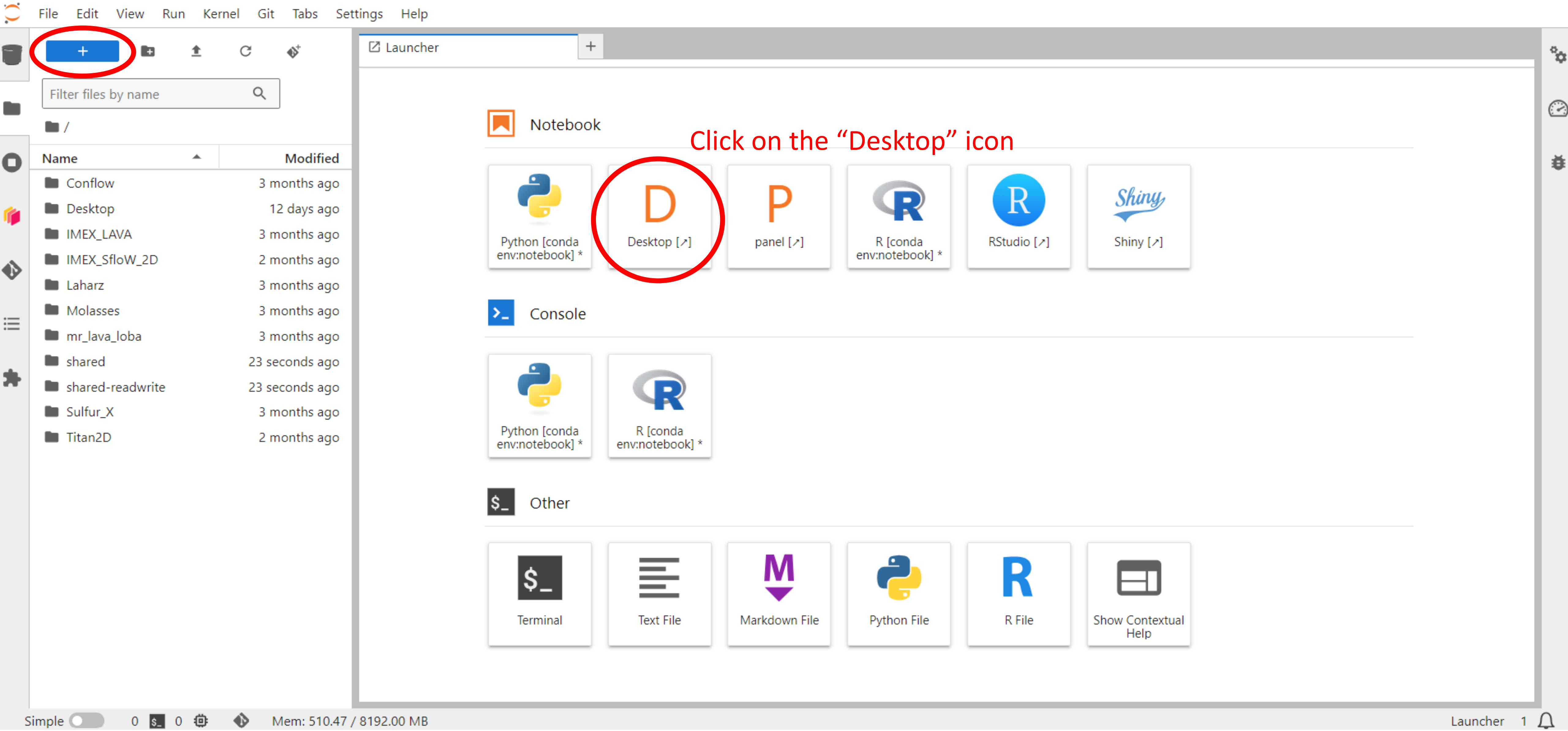
Or click the “New folder“ icon at the top



Name the new folder with your volcano name “mystery_volcano”

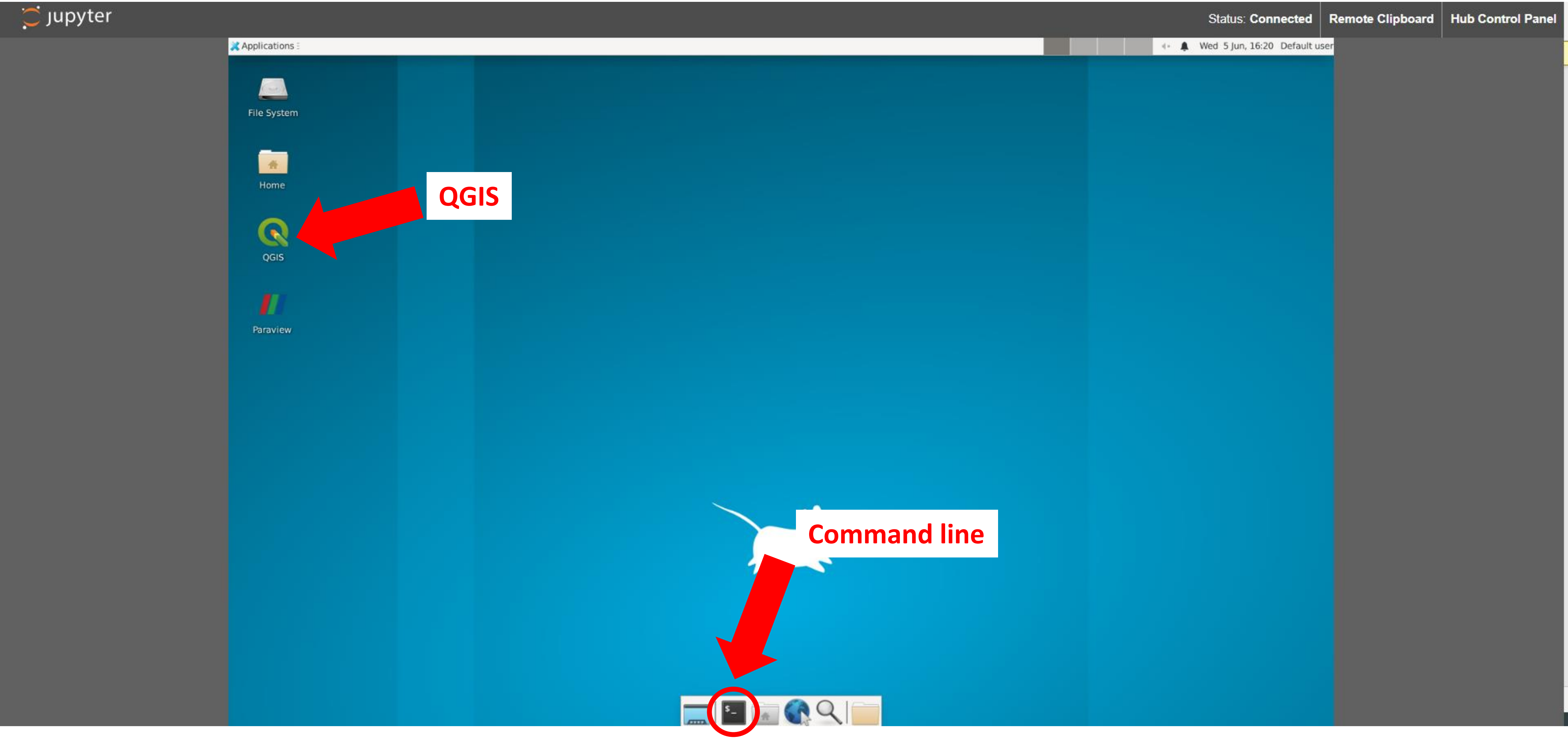
Add your DEM to the new folder!

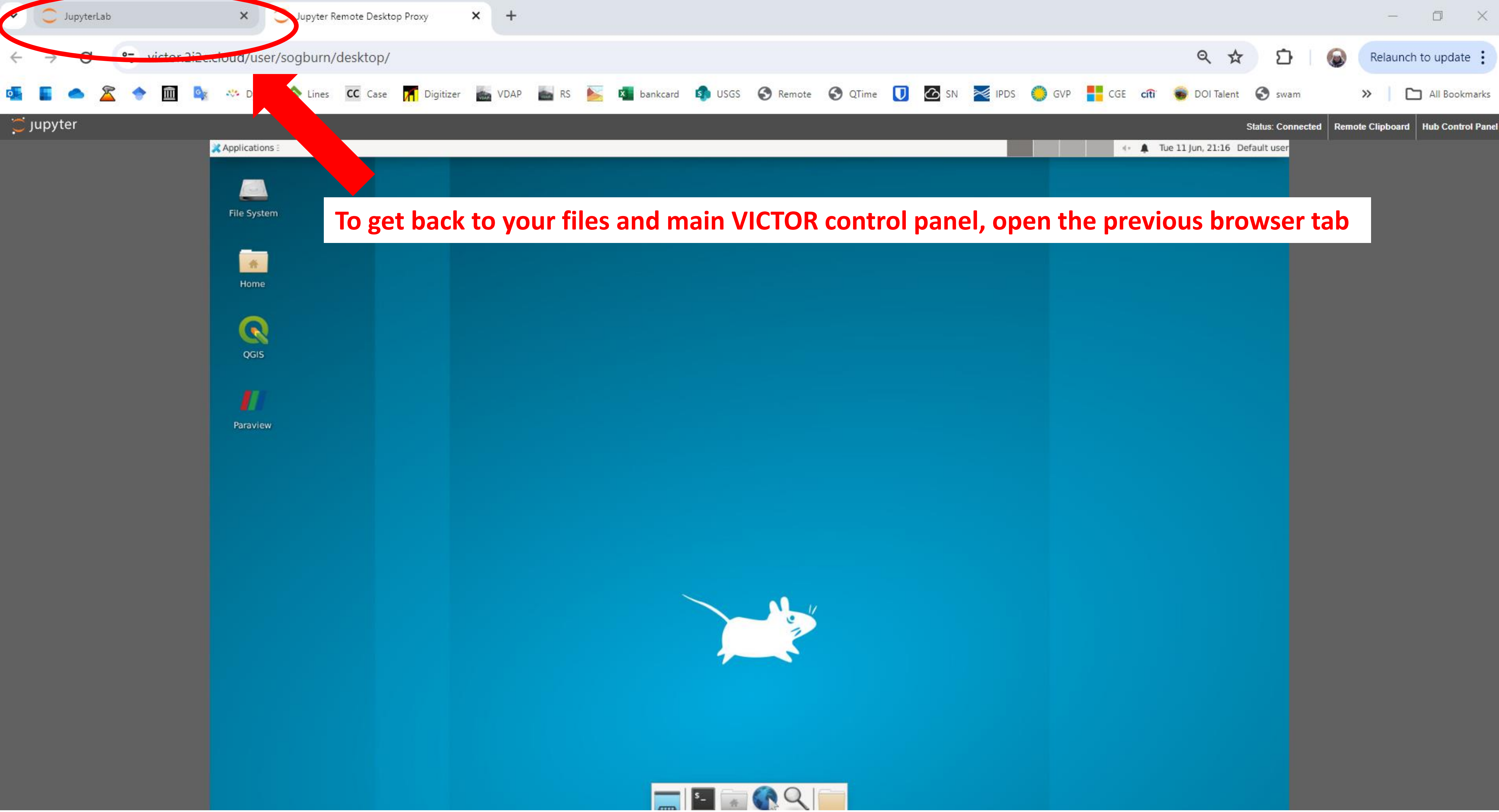
Use QGIS to make a hillshade and a filled DEM



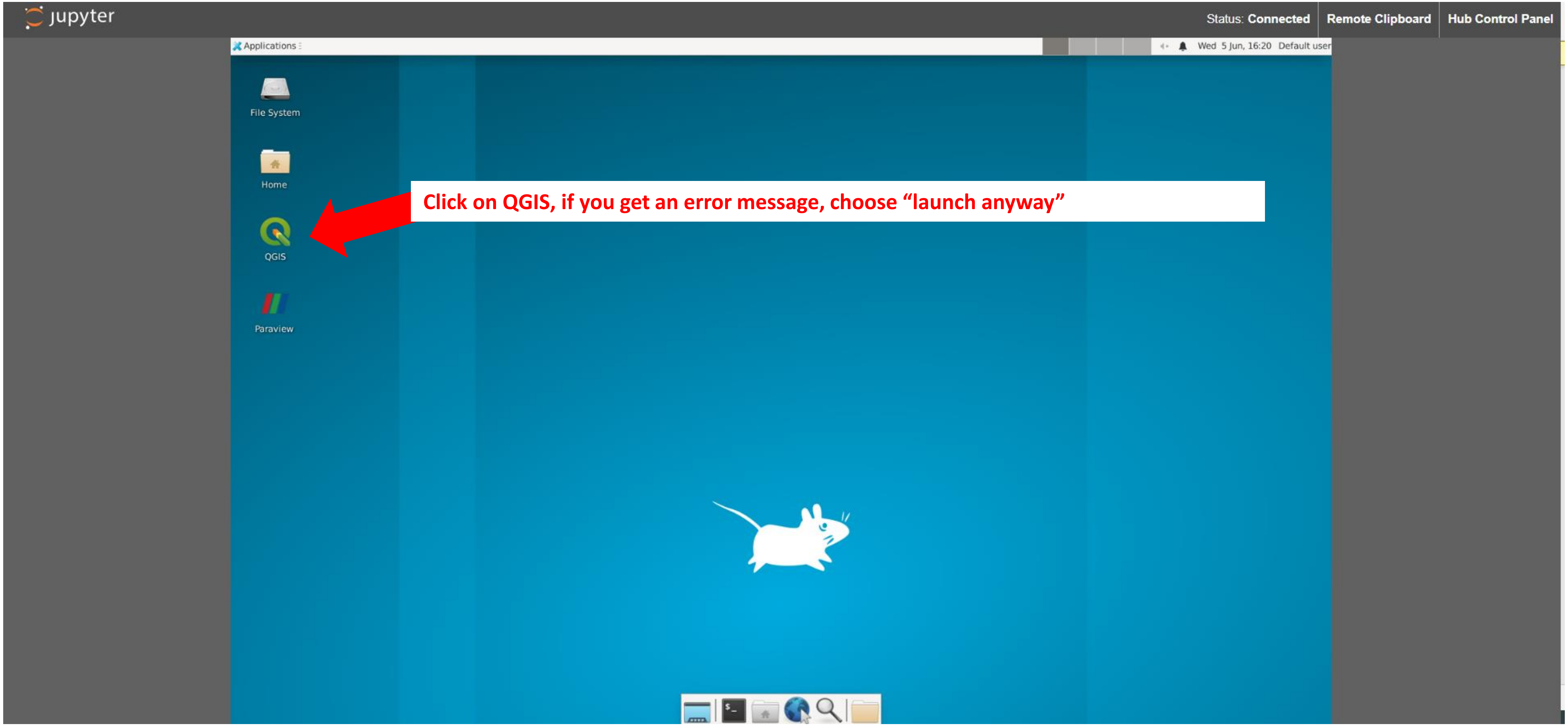
Click on the "Desktop" icon

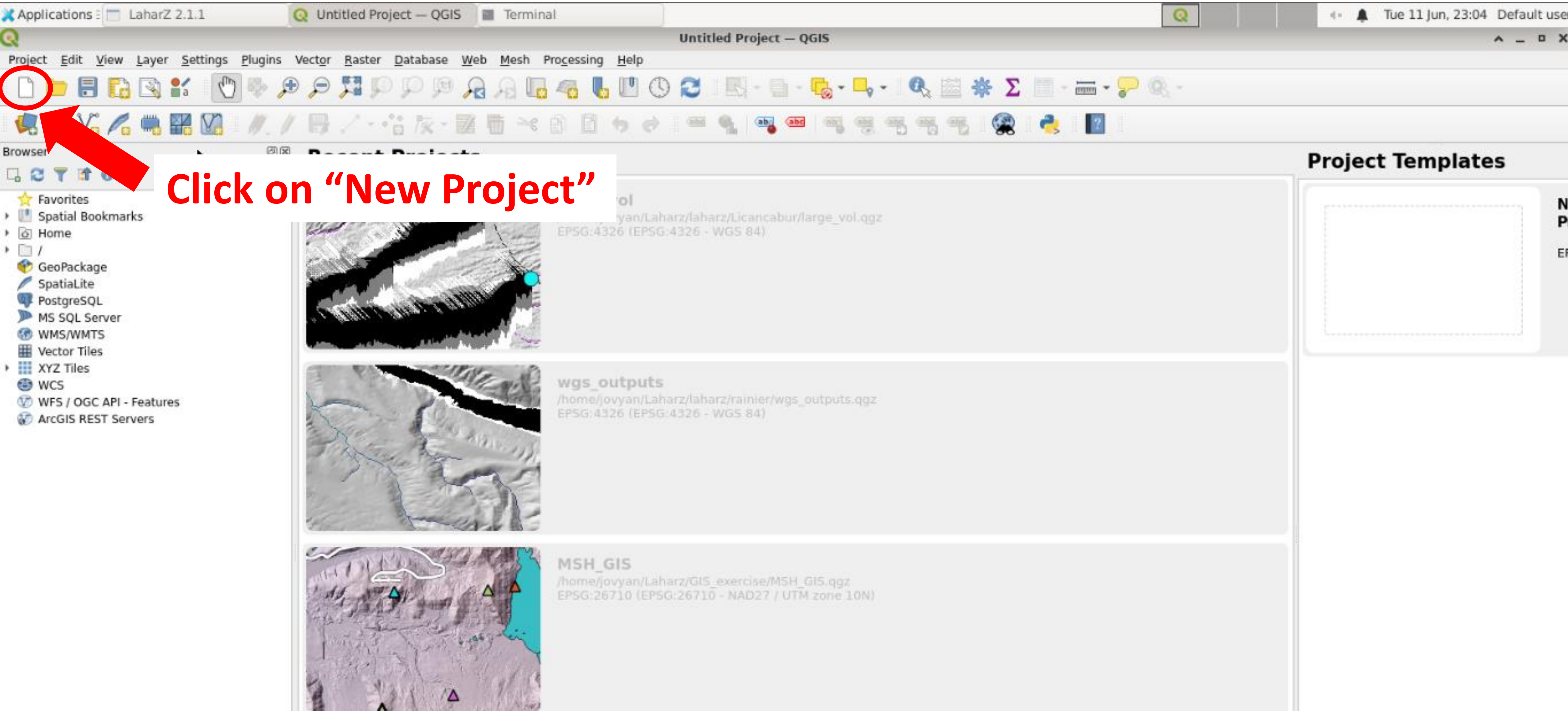
Tip: If you lose your "Launcher" screen, click the "plus" icon at the top left for "New Launcher"

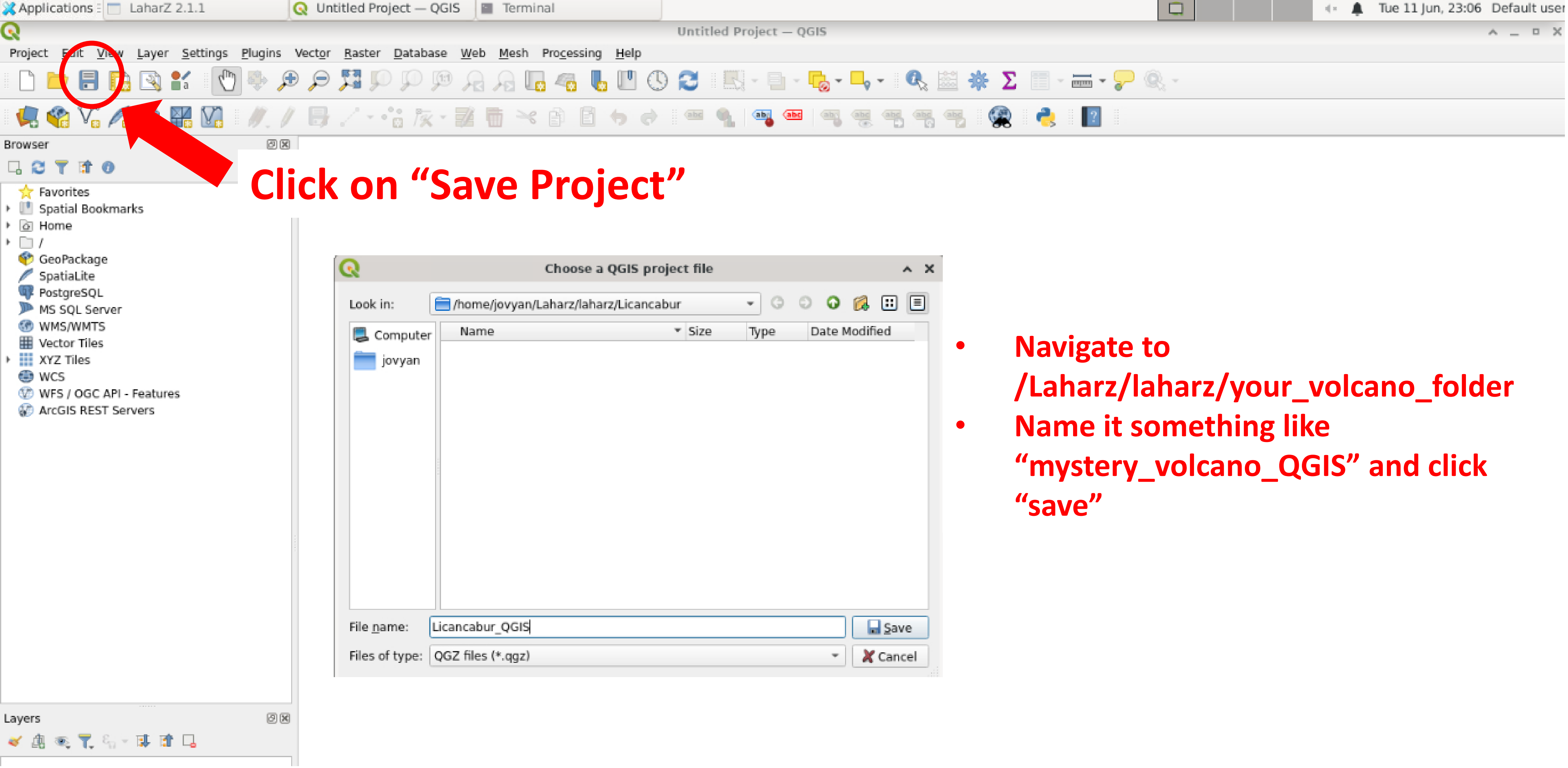

jupyter
Status: Connected
Remote Clipboard
Hub Control Panel
Applications
Wed 5 Jun, 16:20 Default user
File System
Home
QGIS
QGIS
Paraview
Command line



To get back to your files and main VICTOR control panel, open the previous browser tab

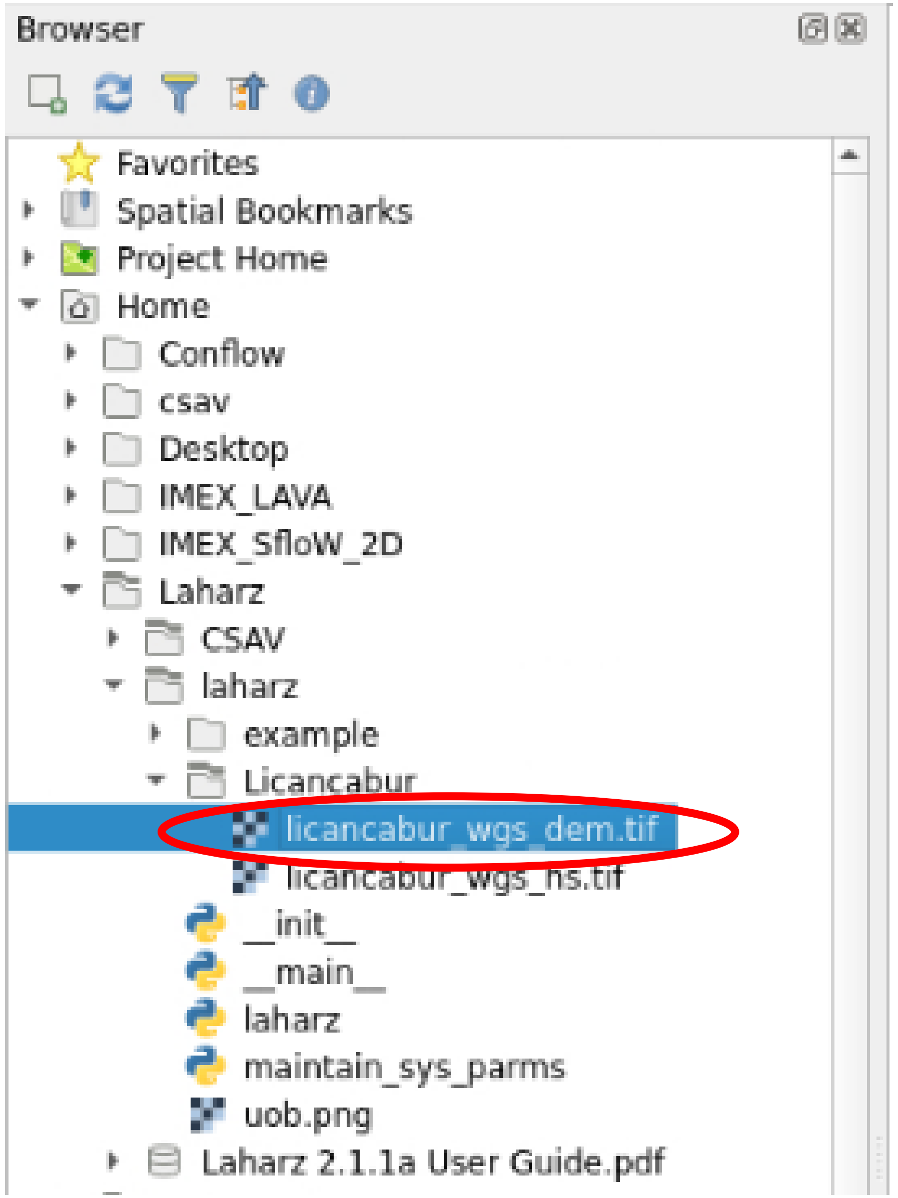

jupyter
Status: Connected
Remote Clipboard
Hub Control Panel
Applications
Wed 5 Jun, 16:20 Default user
File System
Home
QGIS
Paraview
Click on QGIS, if you get an error message, choose “launch anyway”


Click on “New Project”
Applications
LaharZ 2.1.1
Untitled Project — QGIS
Terminal
Tue 11 Jun, 23:04
Untitled Project — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Browser
Project Templates
Favorites
Spatial Bookmarks
Home
/
GeoPackage
SpatiaLite
PostgreSQL
MS SQL Server
WMS/WMTS
Vector Tiles
XYZ Tiles
WCS
WFS / OGC API - Features
ArcGIS REST Servers
EPSG:4326 (EPSG:4326 - WGS 84)
wgs_outputs
/home/jovyan/Laharz/laharz/rainier/wgs_outputs.qgz
EPSG:4326 (EPSG:4326 - WGS 84)
MSH_GIS
/home/jovyan/Laharz/GIS_exercise/MSH_GIS.qgz
EPSG:26710 (EPSG:26710 - NAD27 / UTM zone 10N)



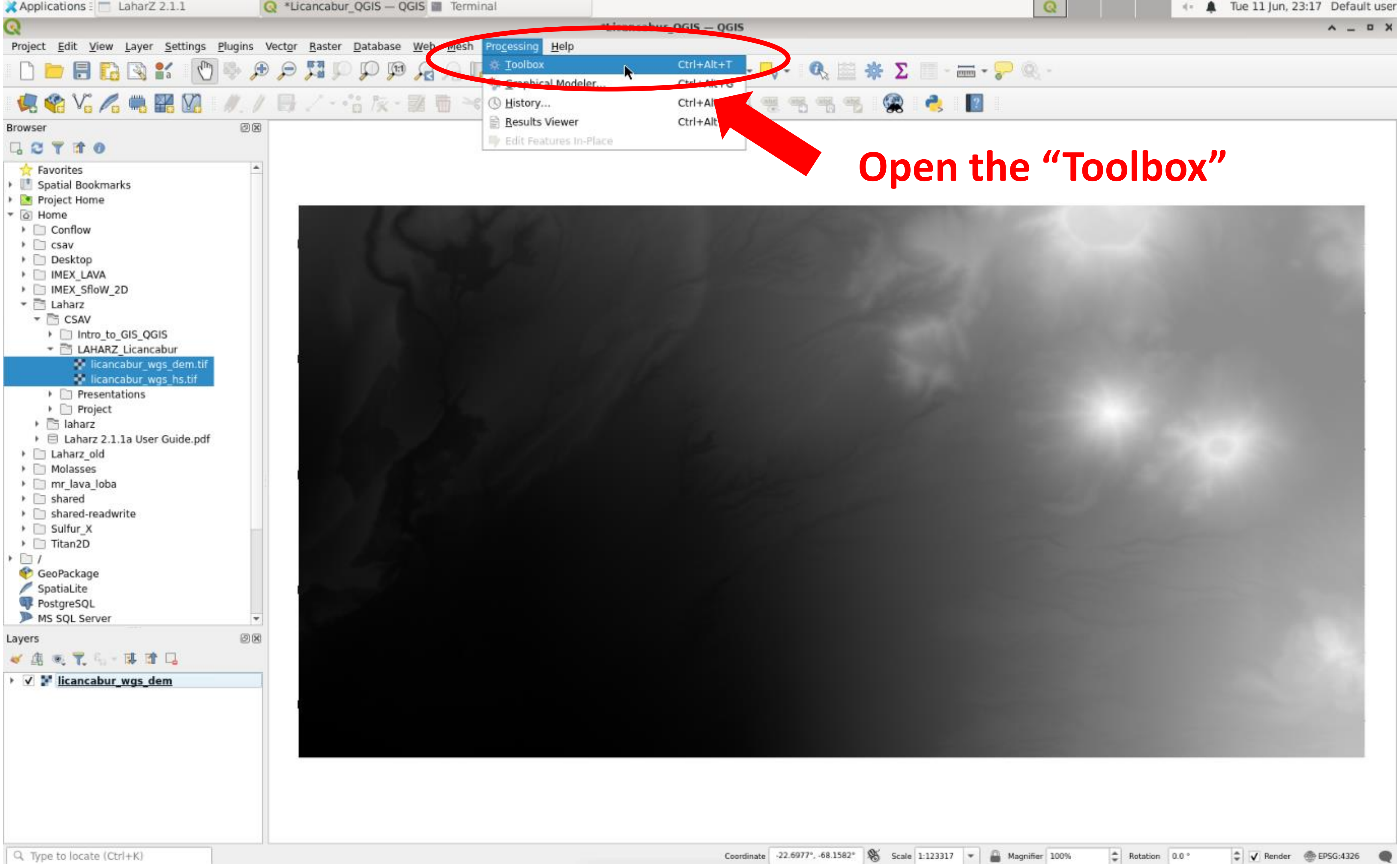
Click on "Save Project"

- **Navigate to /Laharz/laharz/your_volcano_folder**
- **Name it something like "mystery_volcano_QGIS" and click "save"**

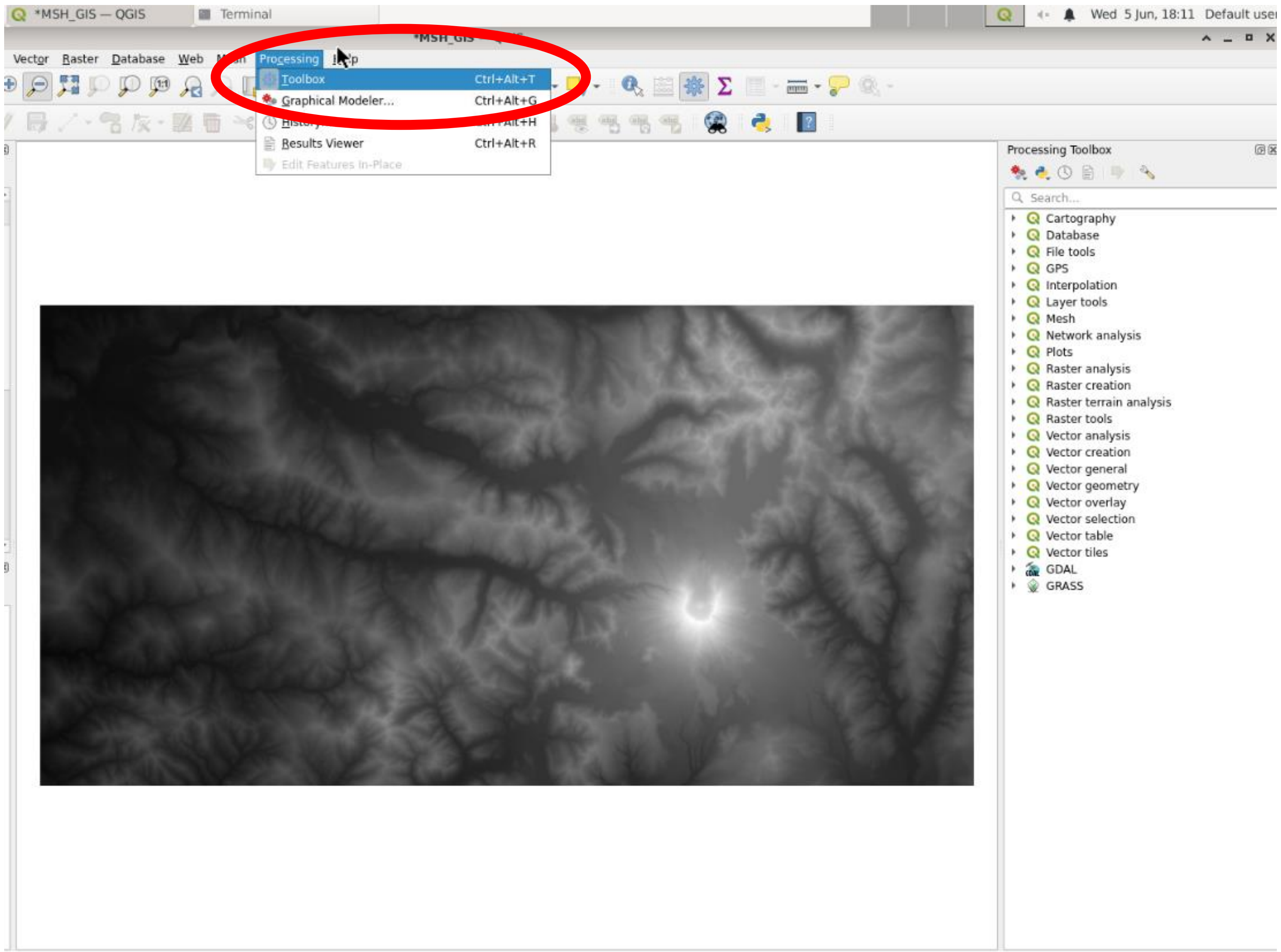


- **Navigate to the Laharz/laharz/mystery_volcano_folder**
- **Drag and drop your DEM file onto your map**

(the image shows the Licancabur example)



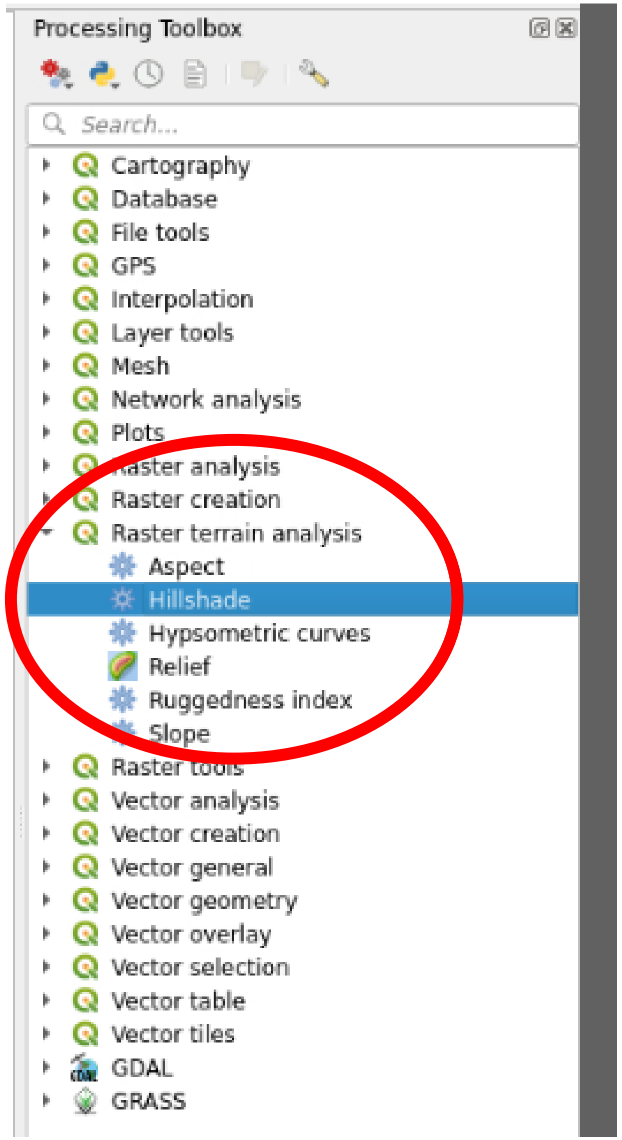
Open the “Toolbox”



- Open the Processing Toolbox

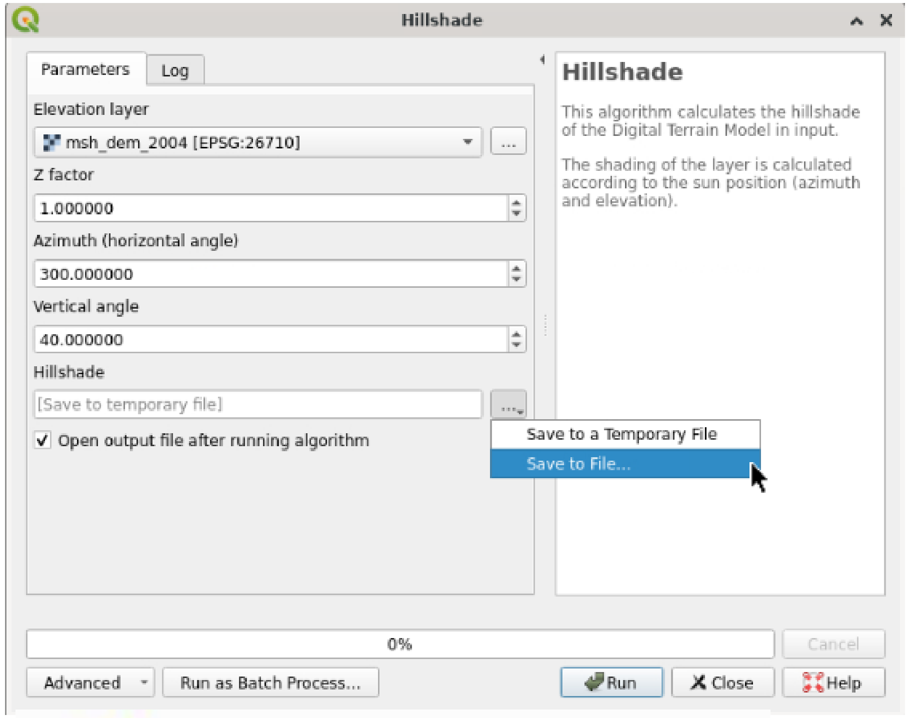
Let's make a hillshade

- A hillshade (or shaded-relief model) is a rendition of shadows across a DEM that helps our brains interpret what we see in DEMs.
- Click the "Raster terrain analysis" menu
- Double click the "Hillshade" tool
- You can also use the "Search" field to search for tools!



Let’s make a hillshade

- This tool let’s you model a “sun” and the “shadows” it would produce on your DEM
- Use the default parameters
    - Z factor is vertical exaggeration (if your z units differ from your x and y units)
    - The azimuth is where the sun is relative to your map. The human brain expects the shadows to be modeled with the sun coming from the upper left (300°)
    - The vertical angle is how high in the sky the modeled sun is. We use the default 40°



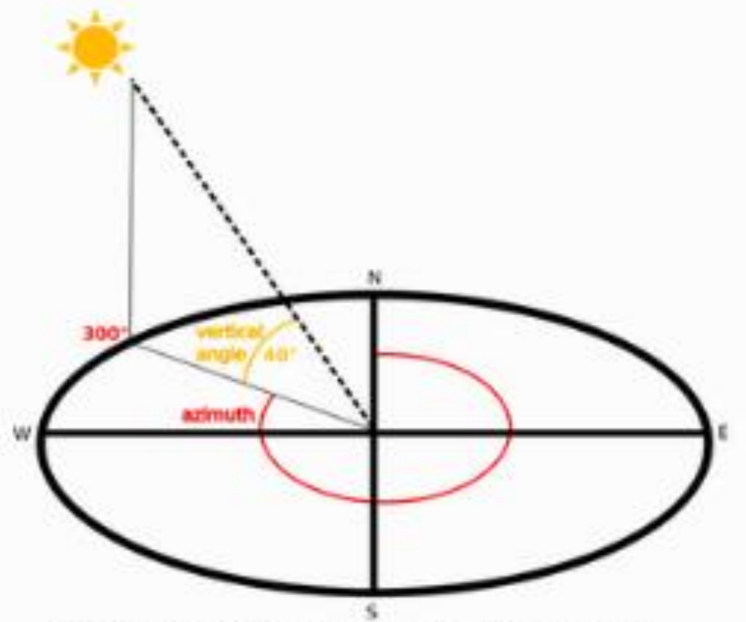
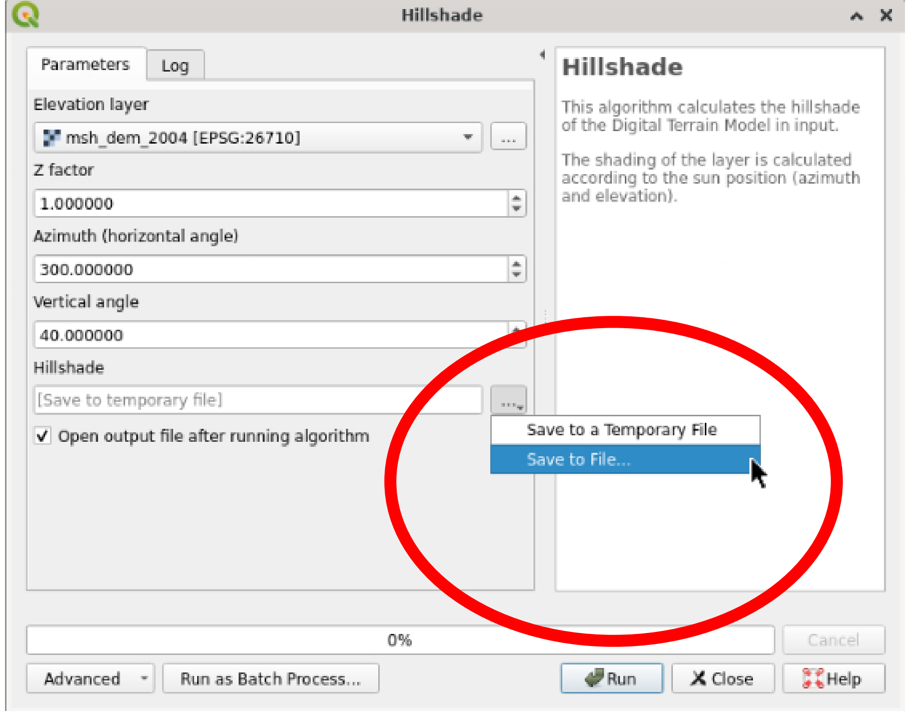


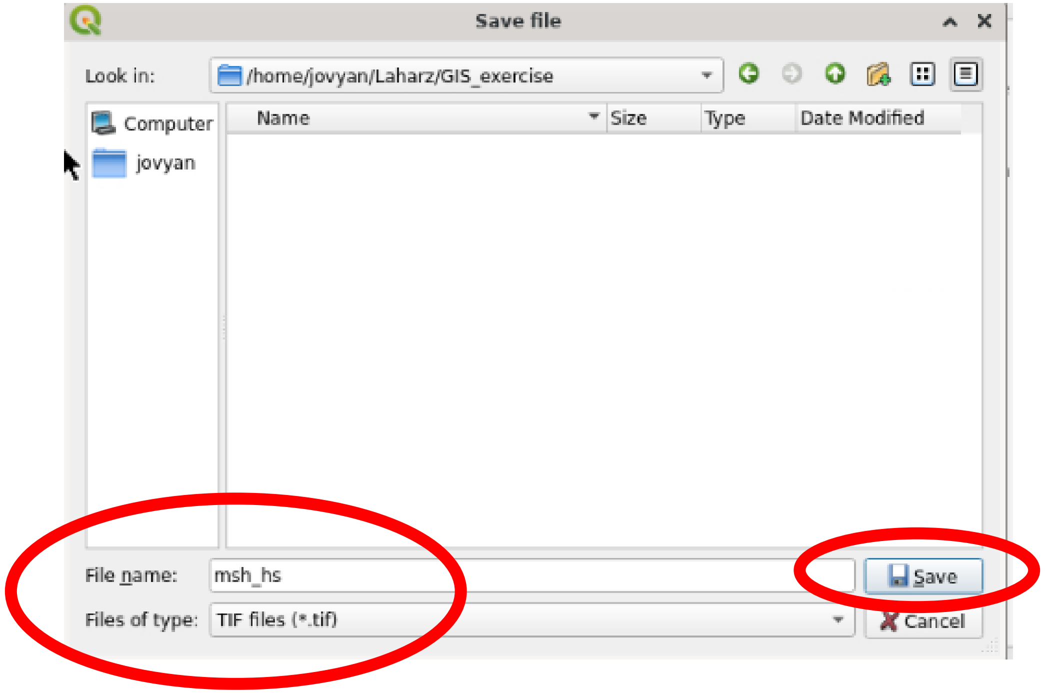
Fig. 28.32 *Azimuth and vertical angle*

Let’s make a hillshade

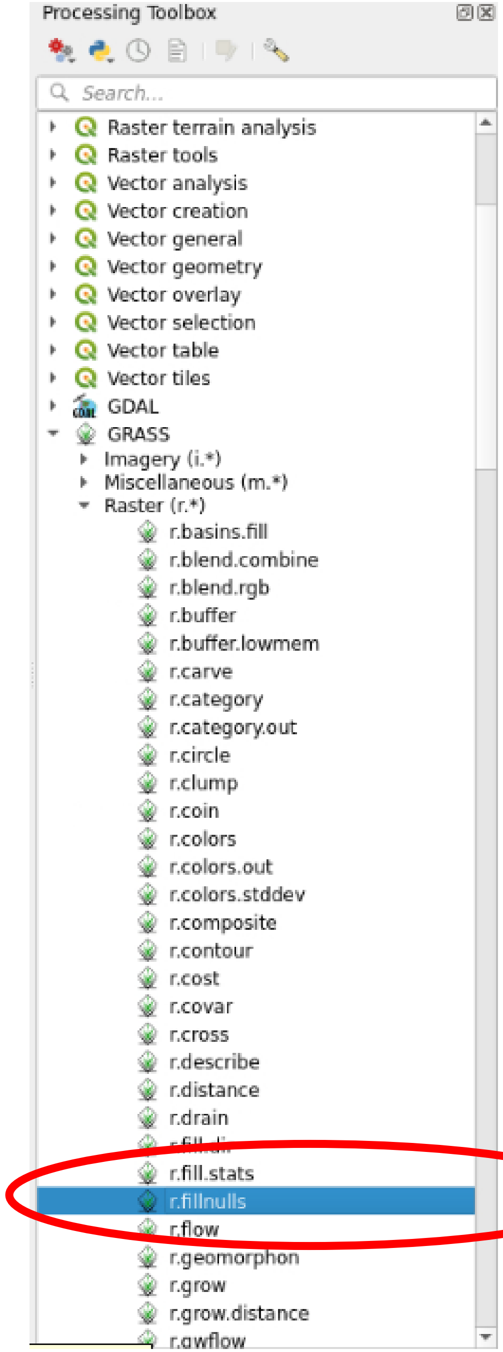
- Click the 3 dot menu next to the Hillshade filename box and select “Save to File”
- Navigate to Laharz/lahar/ mystery_volcano folder
- Enter a filename like “mystery_volcano_hs”
- Click “Save”
- Click “Run”

(this example shows the MSH hillshade)

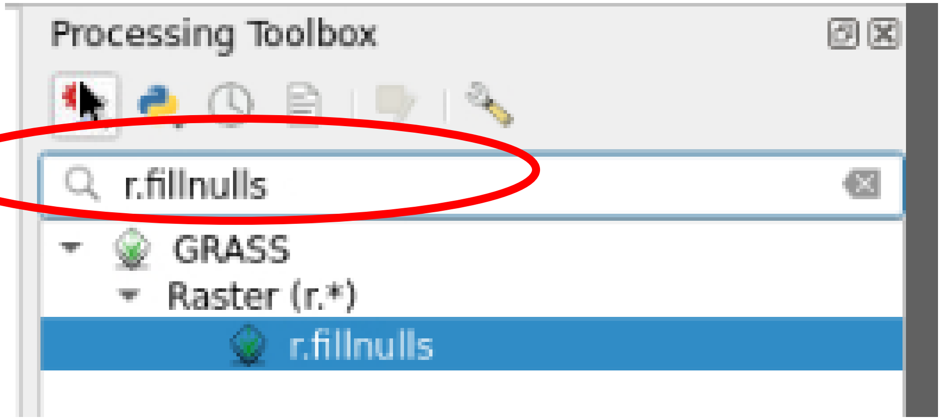




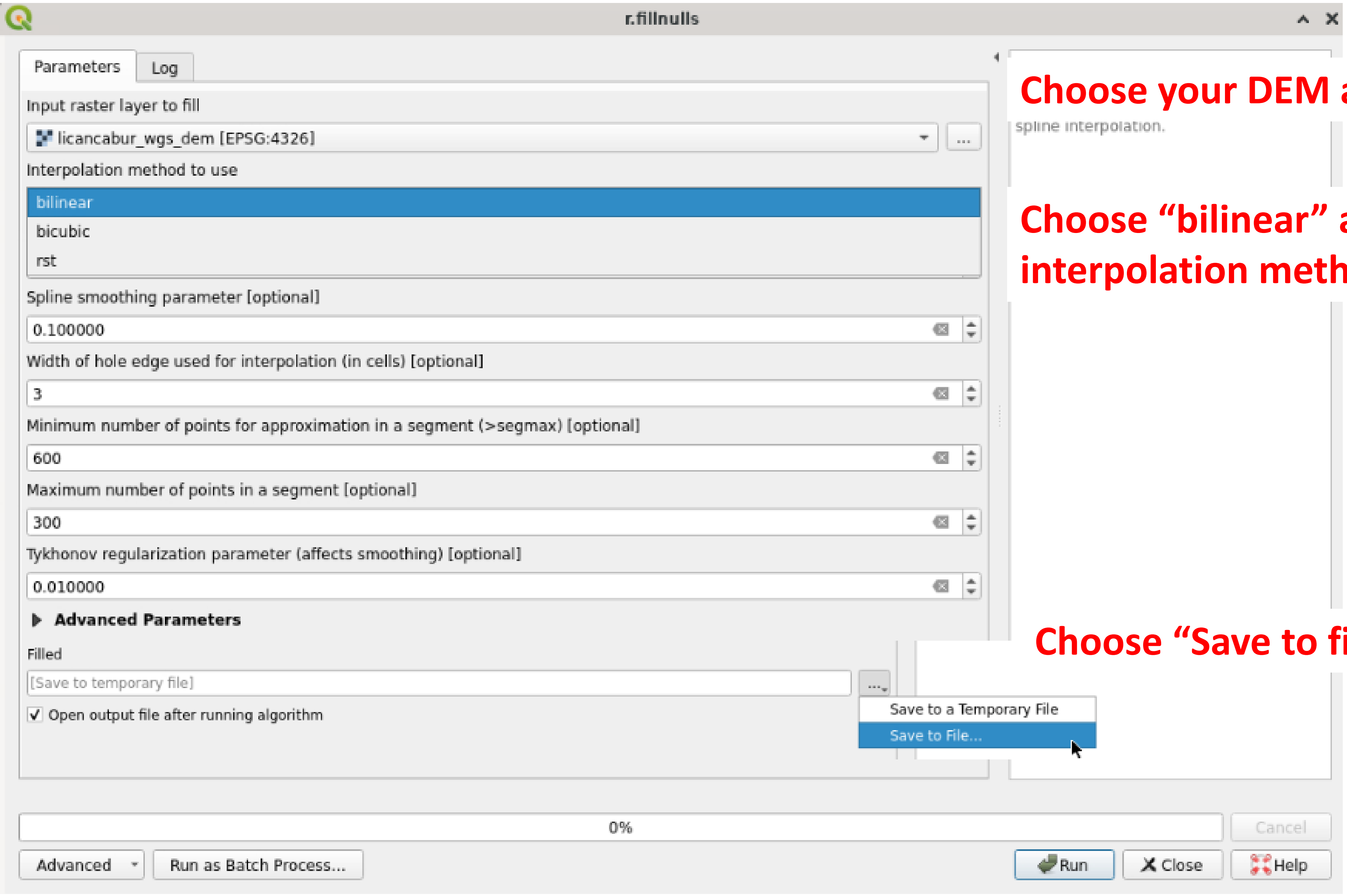
Use QGIS to make a filled DEM



In the Processing Toolbox Navigate to GRASS/Raster/r.fillnulls and double click on it



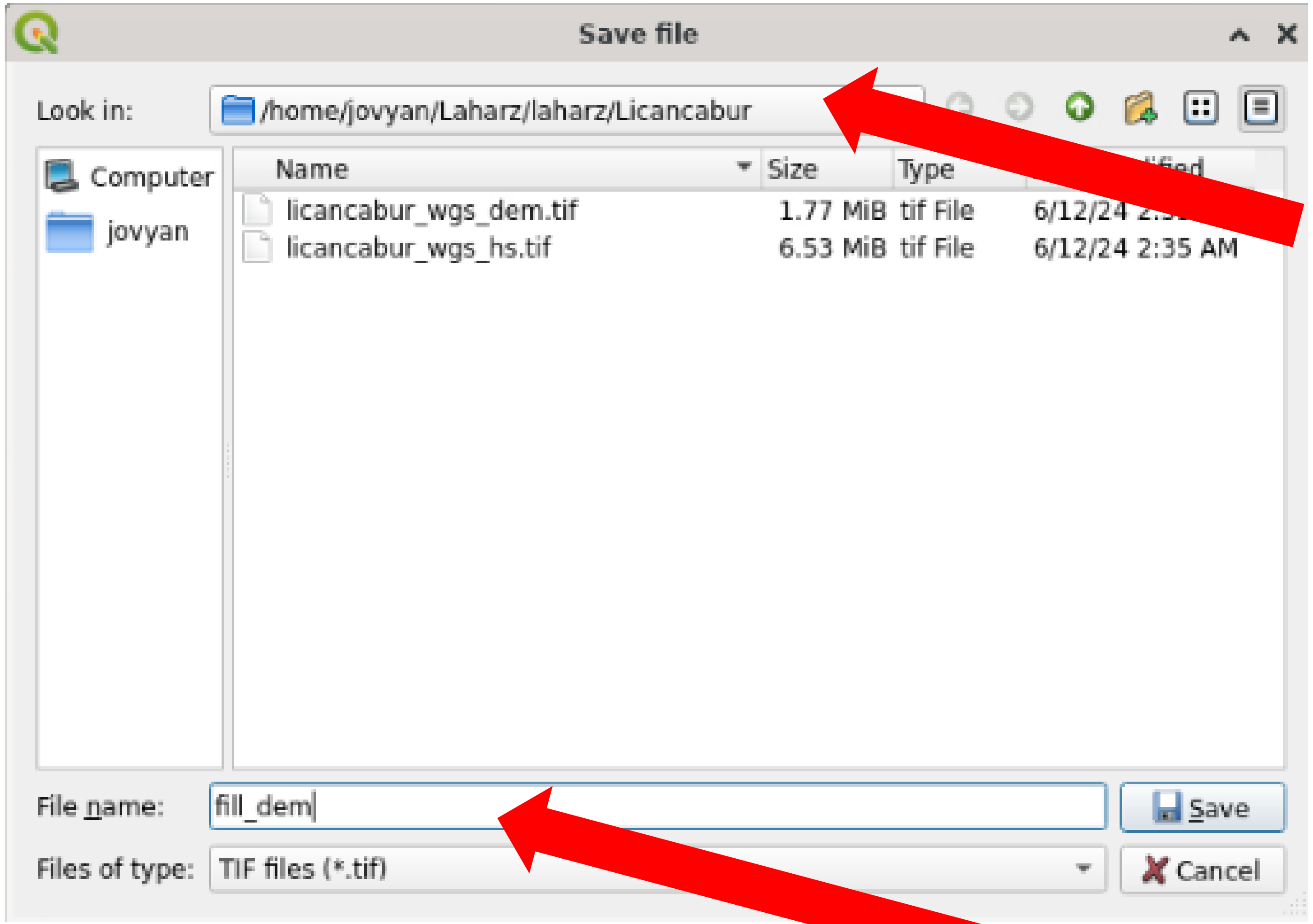
OR type "r.fillnulls" into the search box and double click on it



Choose your DEM as the input raster

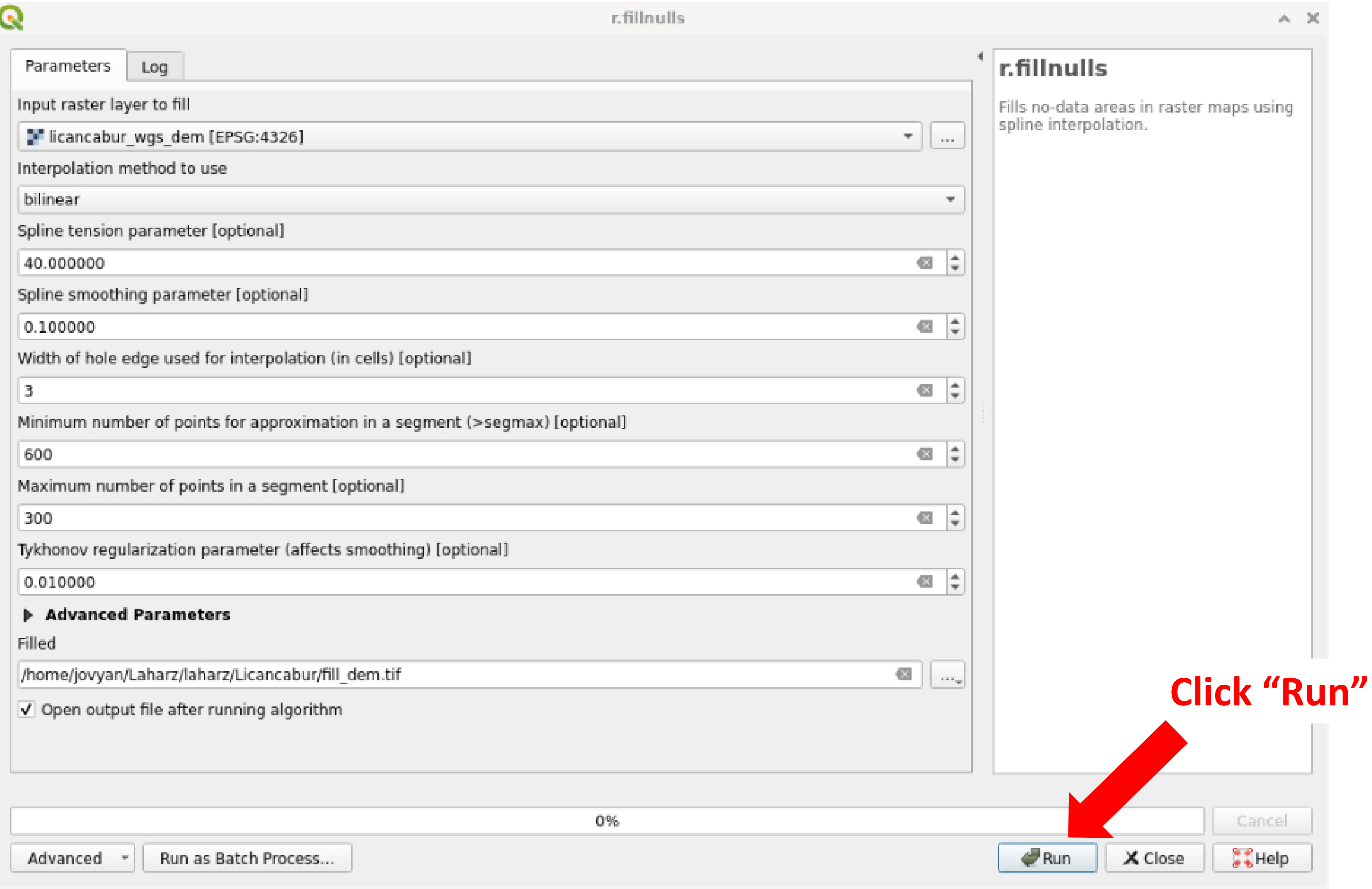
Choose "bilinear" as the interpolation method

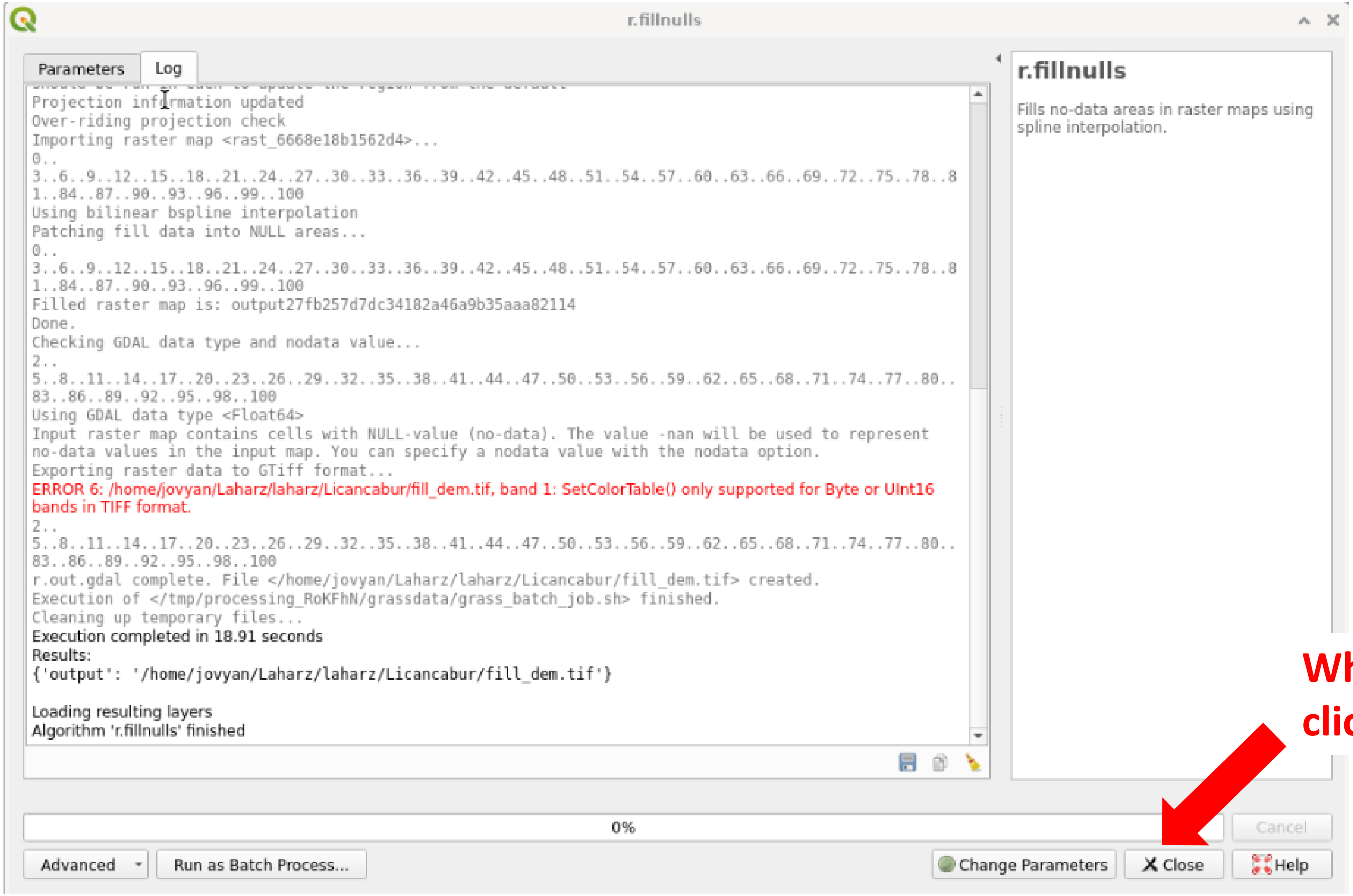
Choose "Save to file"



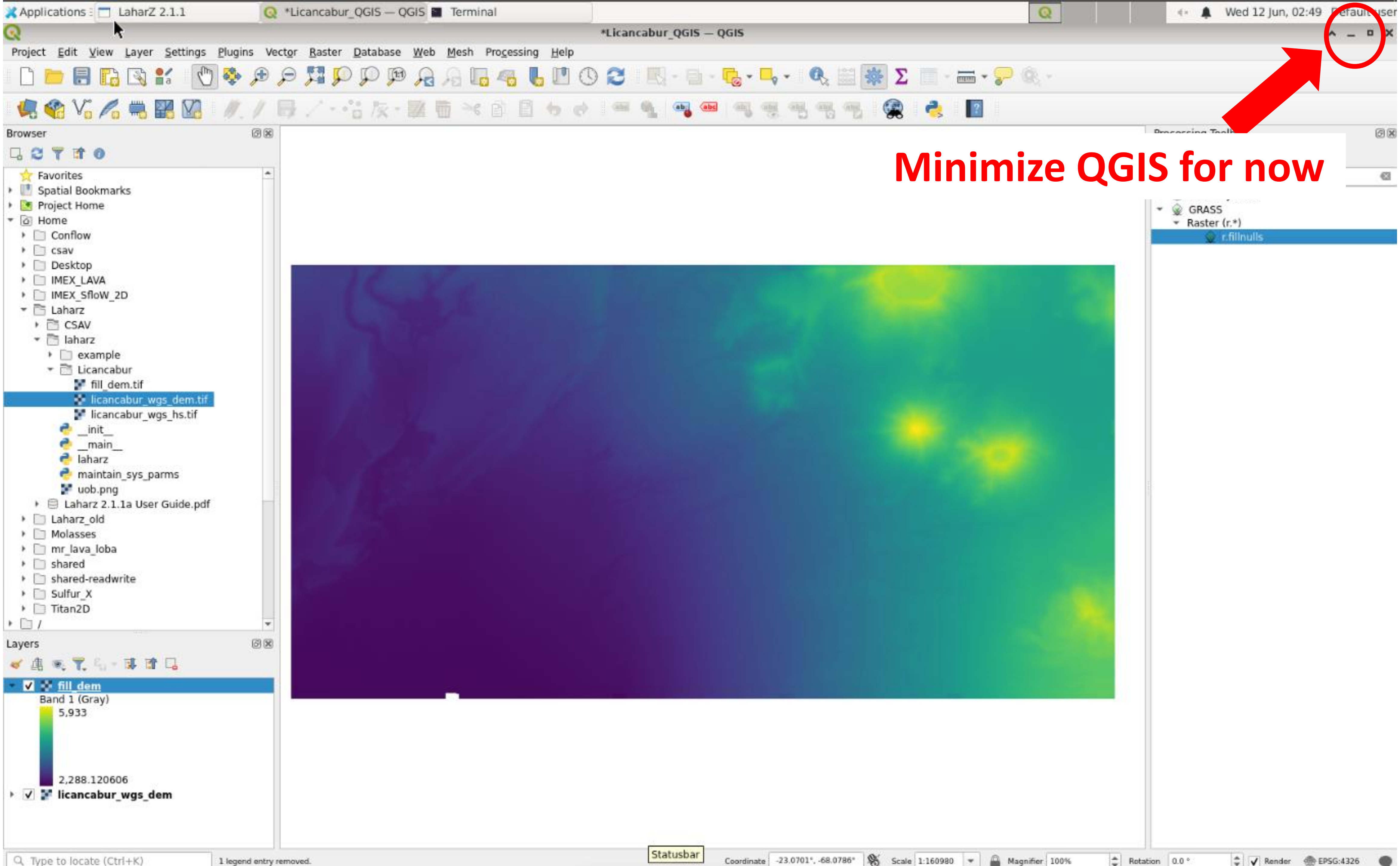
Navigate to the /Laharz/laharz/mystery_volcano folder (image shows Licancabur example)

Name the file "fill_dem"
Click "save"

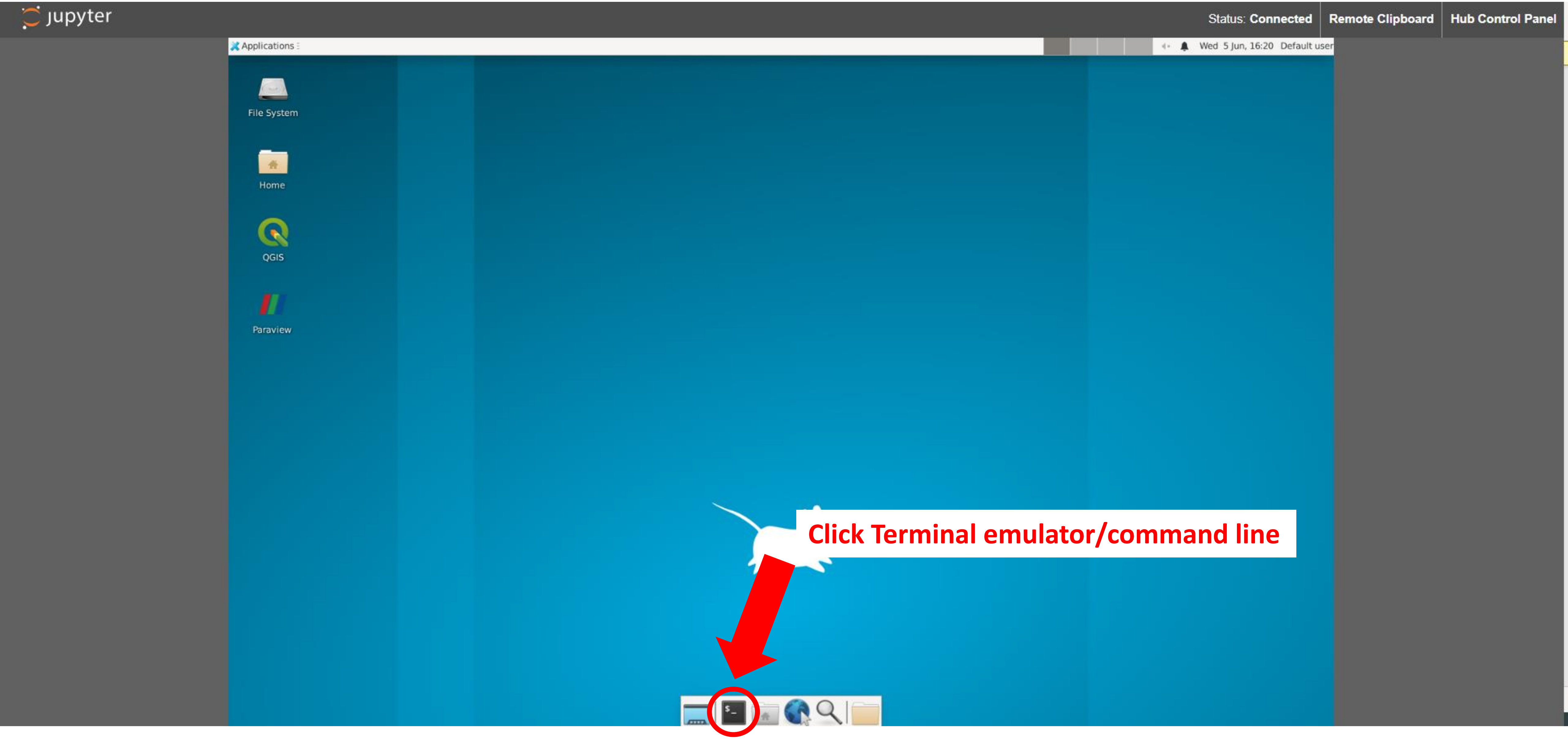

r.fillnulls
Parameters
Log
Input raster layer to fill
licancabur_wgs_dem [EPSG:4326]
Interpolation method to use
bilinear
Spline tension parameter [optional]
40.000000
Spline smoothing parameter [optional]
0.100000
Width of hole edge used for interpolation (in cells) [optional]
3
Minimum number of points for approximation in a segment (>segmax) [optional]
600
Maximum number of points in a segment [optional]
300
Tykhonov regularization parameter (affects smoothing) [optional]
0.010000
Advanced Parameters
Filled
/home/jovyan/Laharz/laharz/Licancabur/fill_dem.tif
Open output file after running algorithm
r.fillnulls
Fills no-data areas in raster maps using spline interpolation.
0%
Cancel
Advanced
Run as Batch Process...
Run
Close
Help
Click "Run"



When it's done, click "Close"

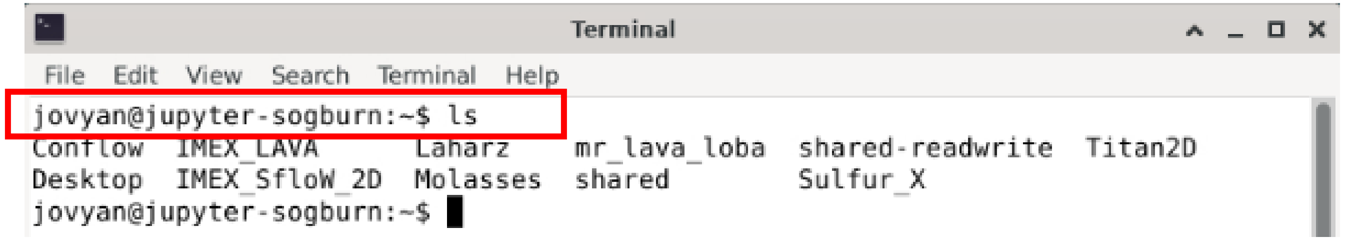

Minimize QGIS for now
Applications
LaharZ 2.1.1
*Licancabur_QGIS — QGIS
Terminal
Wed 12 Jun, 02:49
Project
Edit
View
Layer
Settings
Plugins
Vector
Raster
Database
Web
Mesh
Processing
Help
Browser
Favorites
Spatial Bookmarks
Project Home
Home
Conflow
csav
Desktop
IMEX_LAVA
IMEX_SfloW_2D
Laharz
CSAV
laharz
example
Licancabur
fill_dem.tif
licancabur_wgs_dem.tif
licancabur_wgs_hs.tif
__init__
__main__
laharz
maintain_sys_parms
uob.png
Laharz 2.1.1a User Guide.pdf
Laharz_old
Molasses
mr_lava_loba
shared
shared-readwrite
Sulfur_X
Titan2D
/
Layers
fill_dem
Band 1 (Gray)
5,933
2,288.120606
licancabur_wgs_dem
GRASS
Raster (r.*)
r.fillnulls
Type to locate (Ctrl+K)
1 legend entry removed.
Statusbar
Coordinate
-23.0701°, -68.0786°
Scale
1:160980
Magnifier
100%
Rotation
0.0 °
Render
EPSG:4326

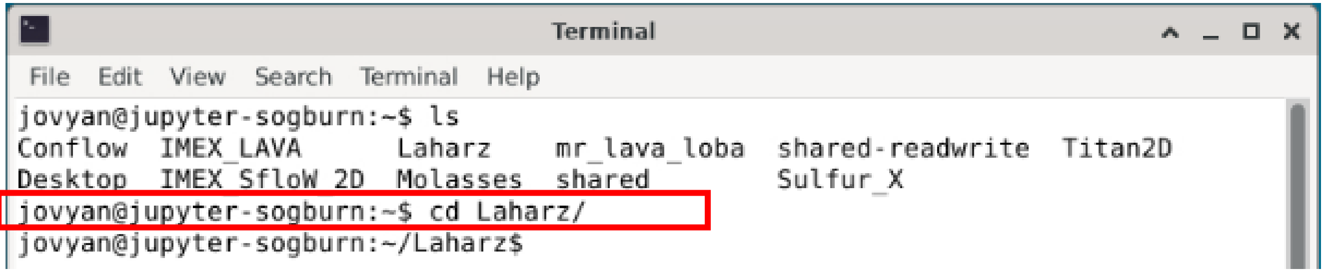
Run LAHARZ

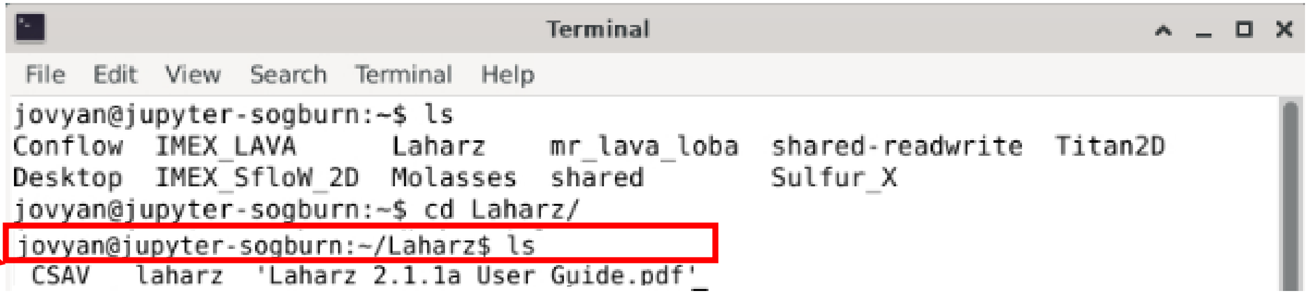

jupyter
Status: Connected
Remote Clipboard
Hub Control Panel
Applications
Wed 5 Jun, 16:20 Default user
File System
Home
QGIS
Paraview
Click Terminal emulator/command line

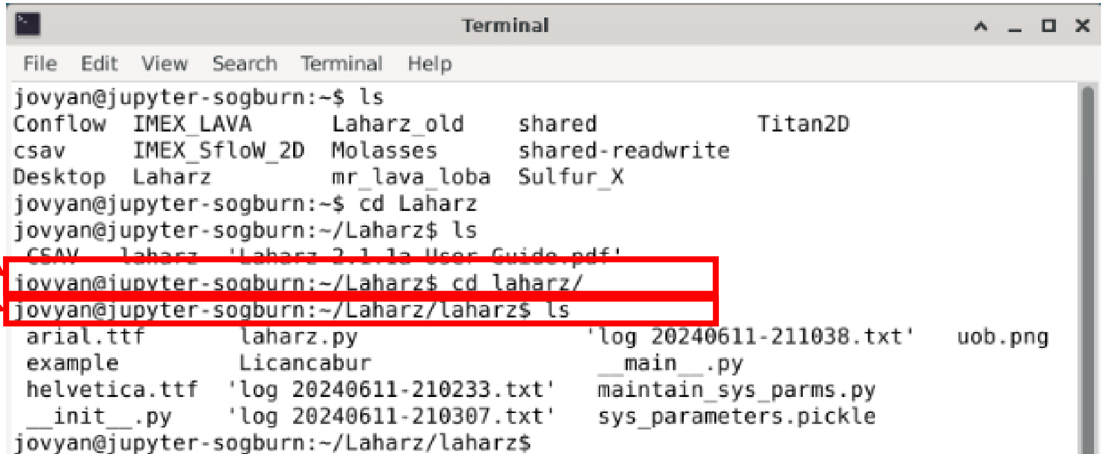
Using the command line

- Type a command and hit “enter”
- To list the files in the folder you’re in: **ls**
- To move to a folder: cd name_of_folder
    - If you type part of a folder name and then click the “tab” key, it will autocomplete
    - Move to Laharz folder: **cd Laharz/**
- List what’s in the “Laharz” folder: **ls**
    - You’ll see a “laharz” folder, the “CSAV” folder you copied there, and the user manual
- Move to “laharz” folder: **cd laharz/**
- List what’s in the “laharz” folder: **ls**
    - You should have the “mystery_volcano_folder” folder that you created earlier and a bunch of other files



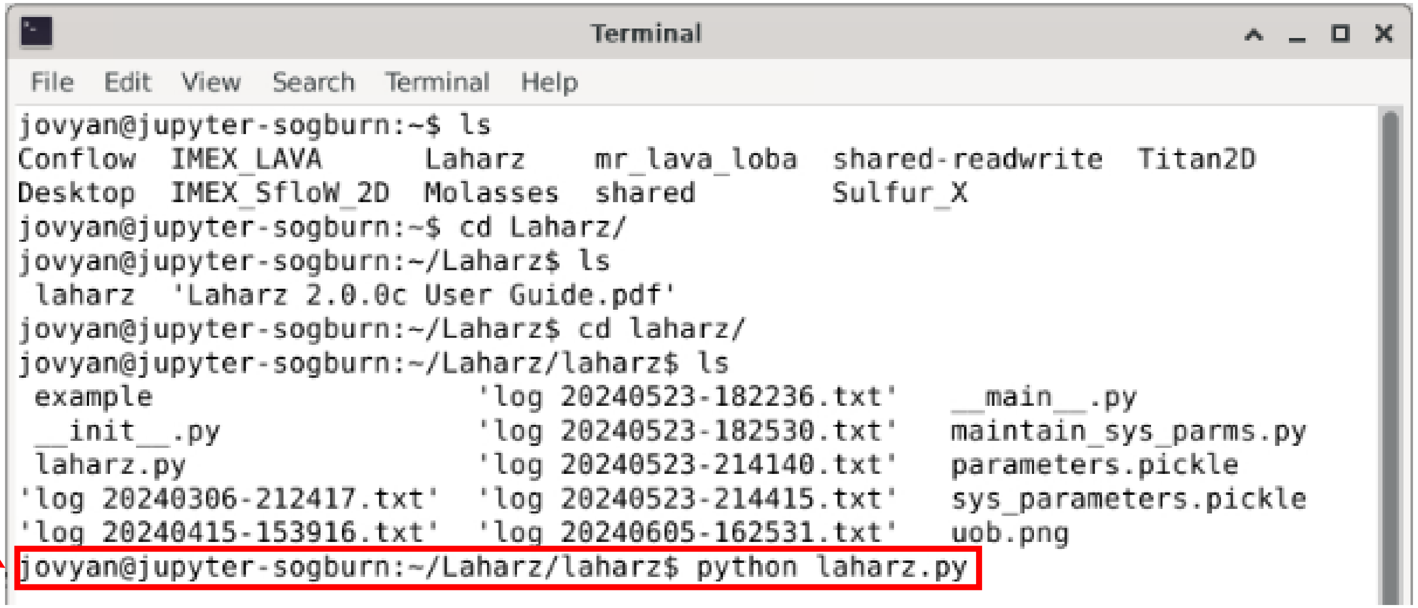




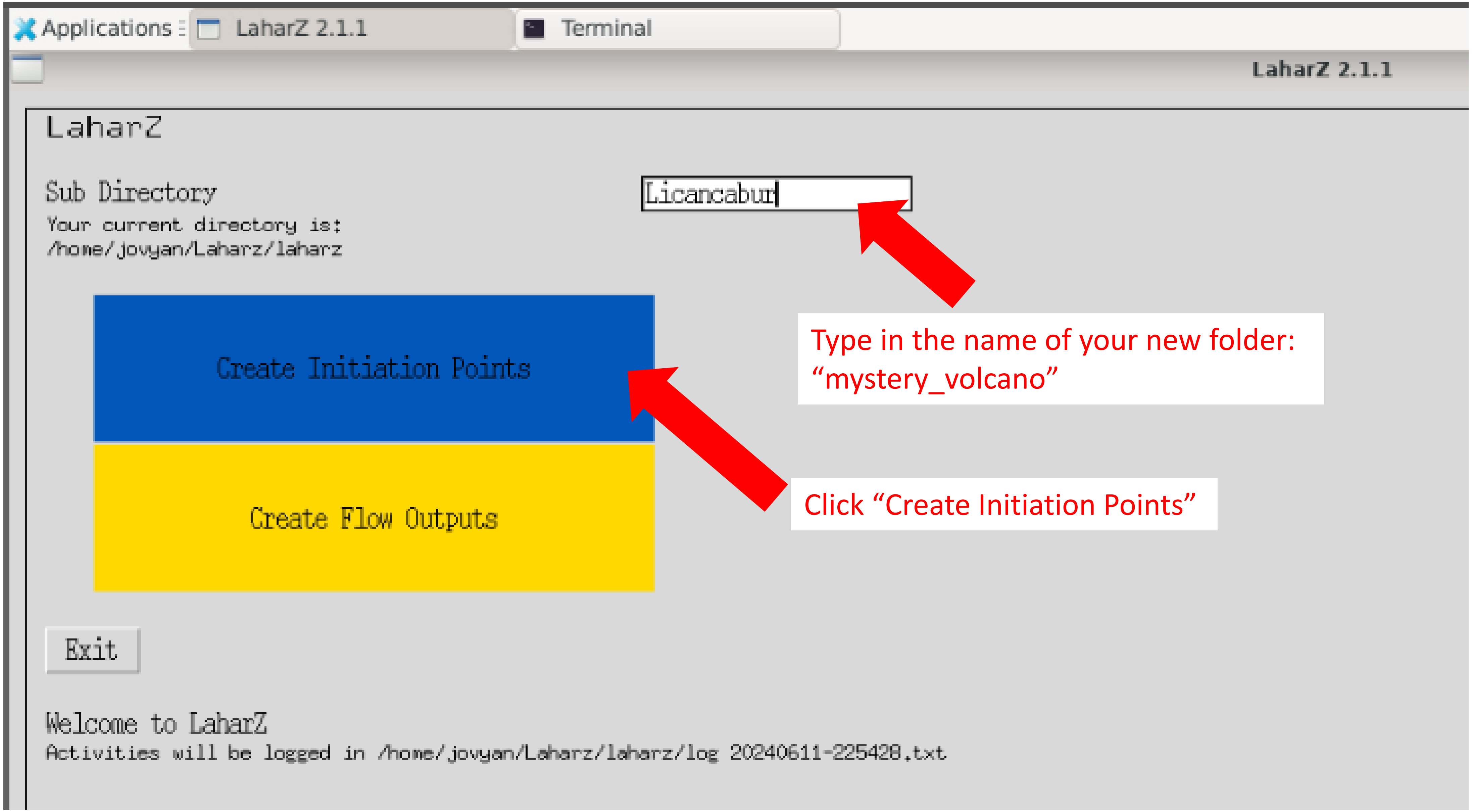


Launch LAHARZ

- Launch LAHARZ: **python laharz.py**

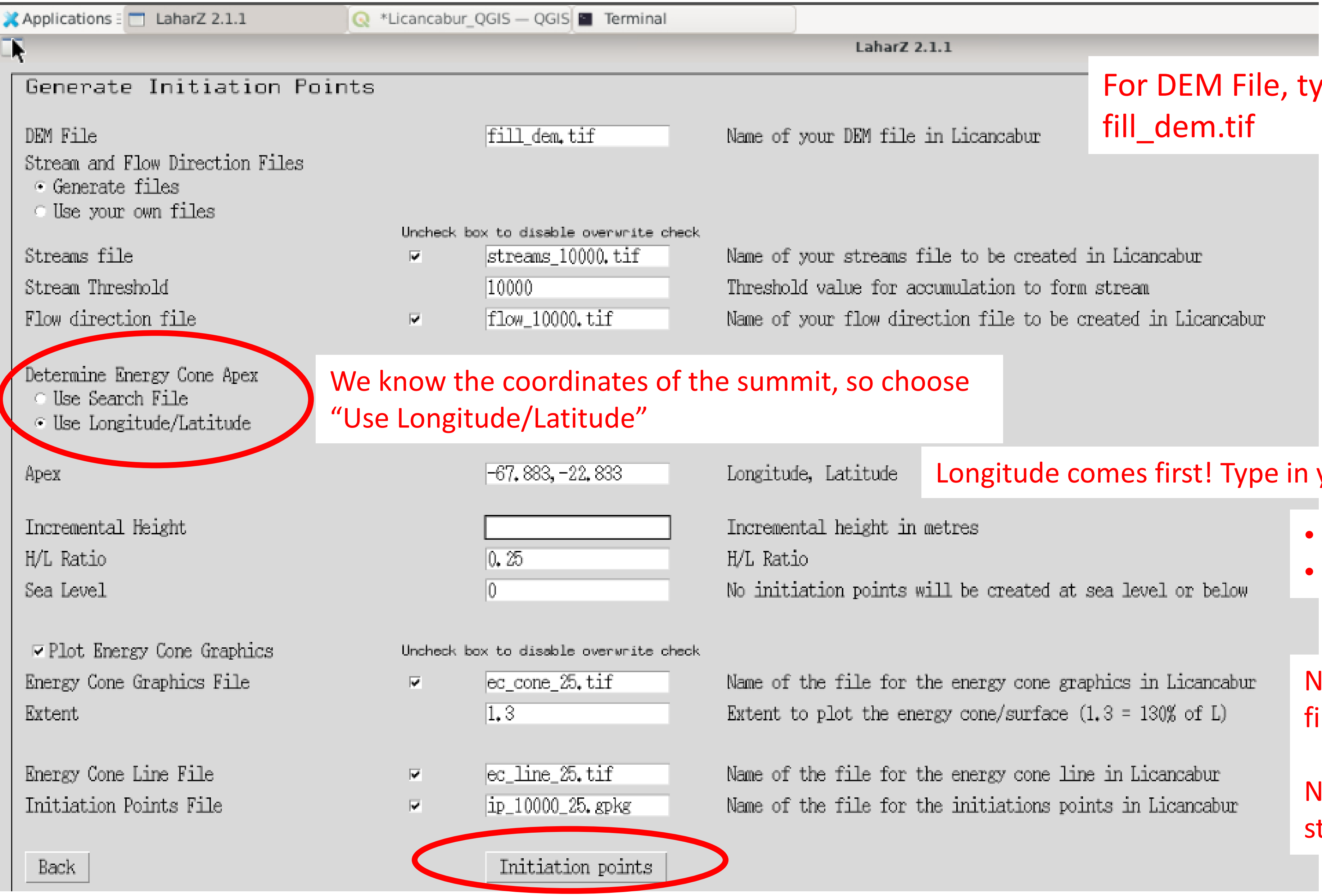


Make a stream network, an H/L cone, and start points



Type in the name of your new folder: “mystery_volcano”

Click “Create Initiation Points”



For DEM File, type the name of your filled DEM: fill_dem.tif

Enter the stream threshold you decided to try and name the "streams file" and "flow direction" file using that number

We know the coordinates of the summit, so choose "Use Longitude/Latitude"

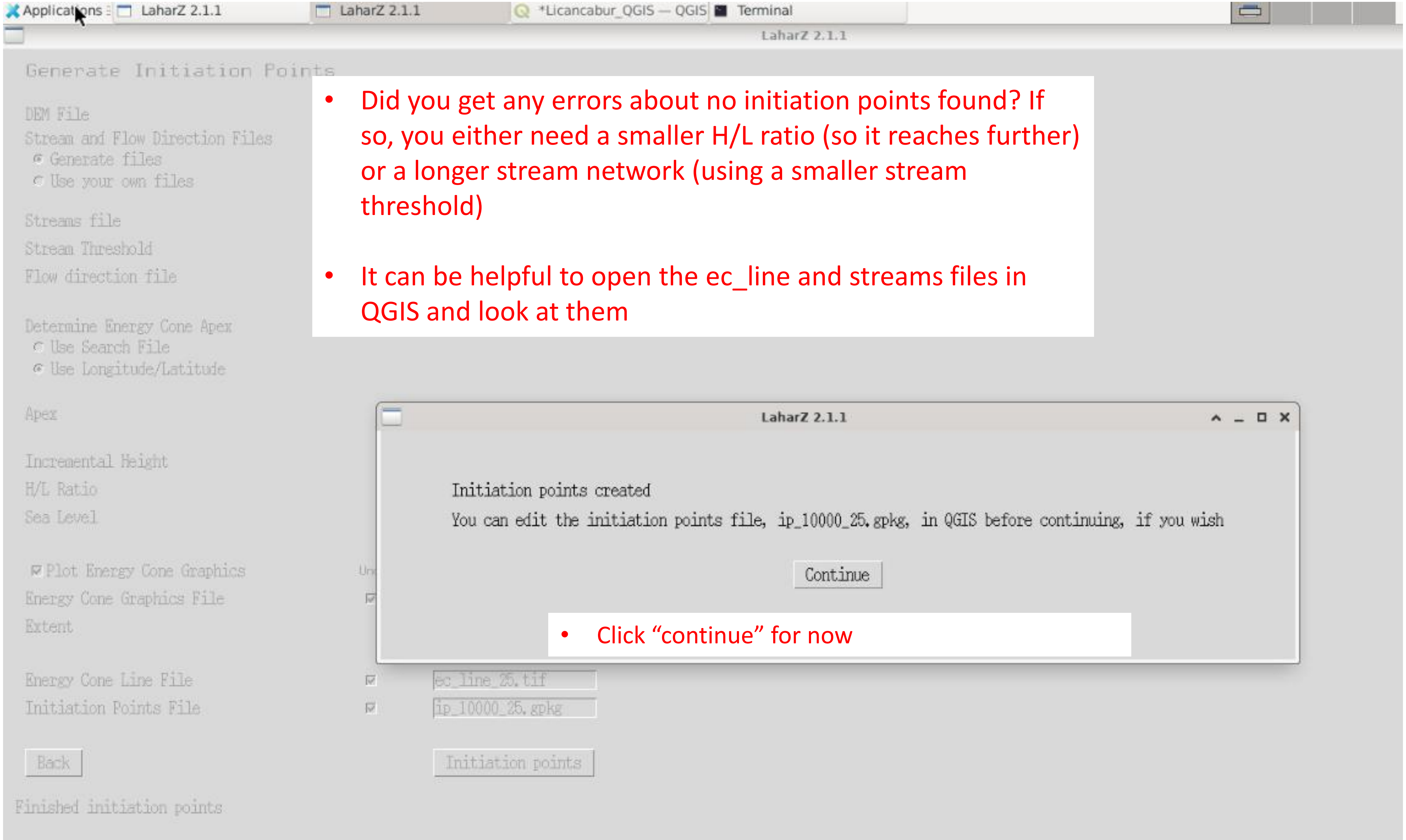
Longitude comes first! Type in your Longitude, Latitude

- Add height if you need to
- Enter the H/L ratio you decided to try

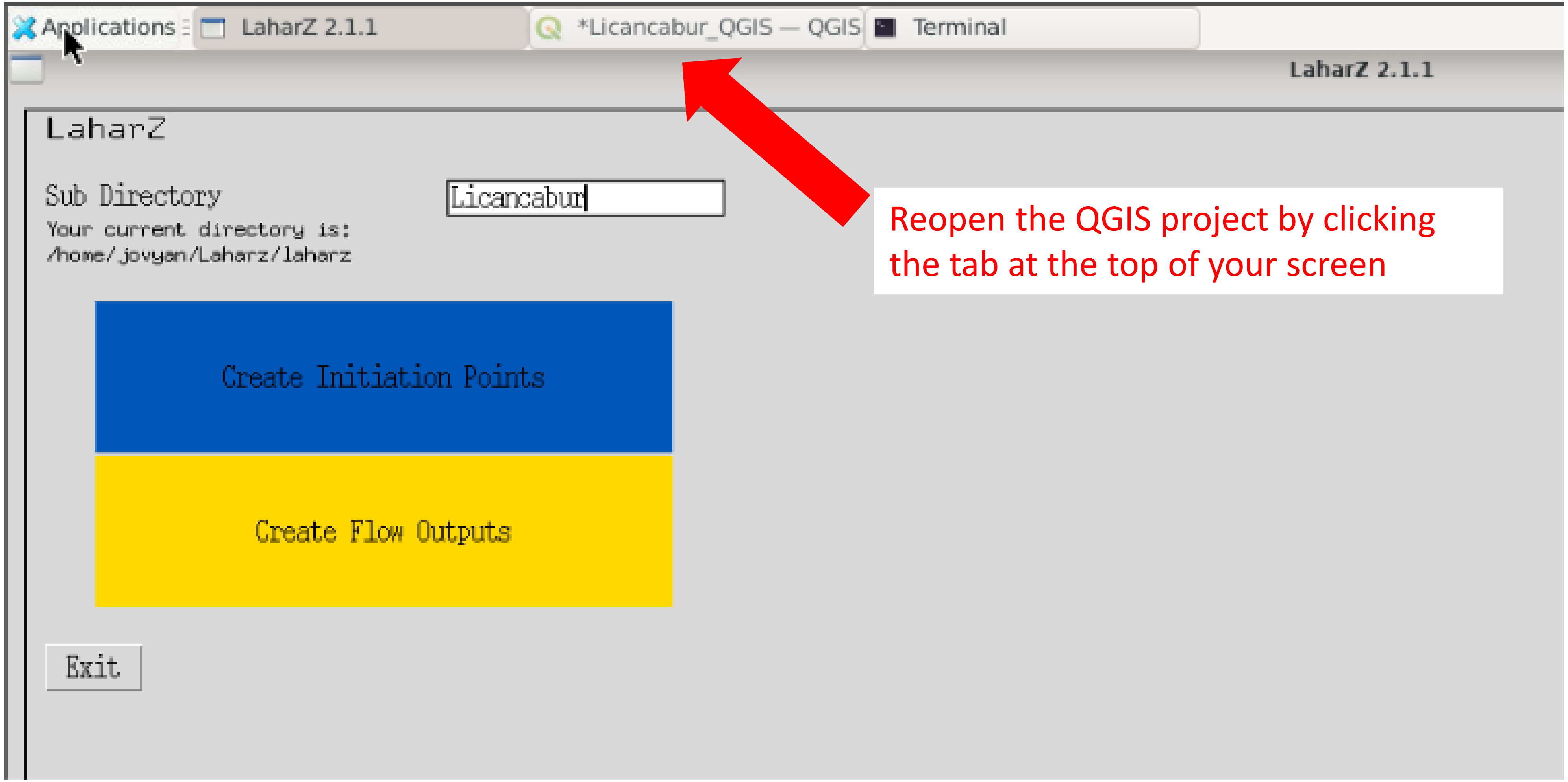
Name the energy cone and energy line files with the H/L ratio used

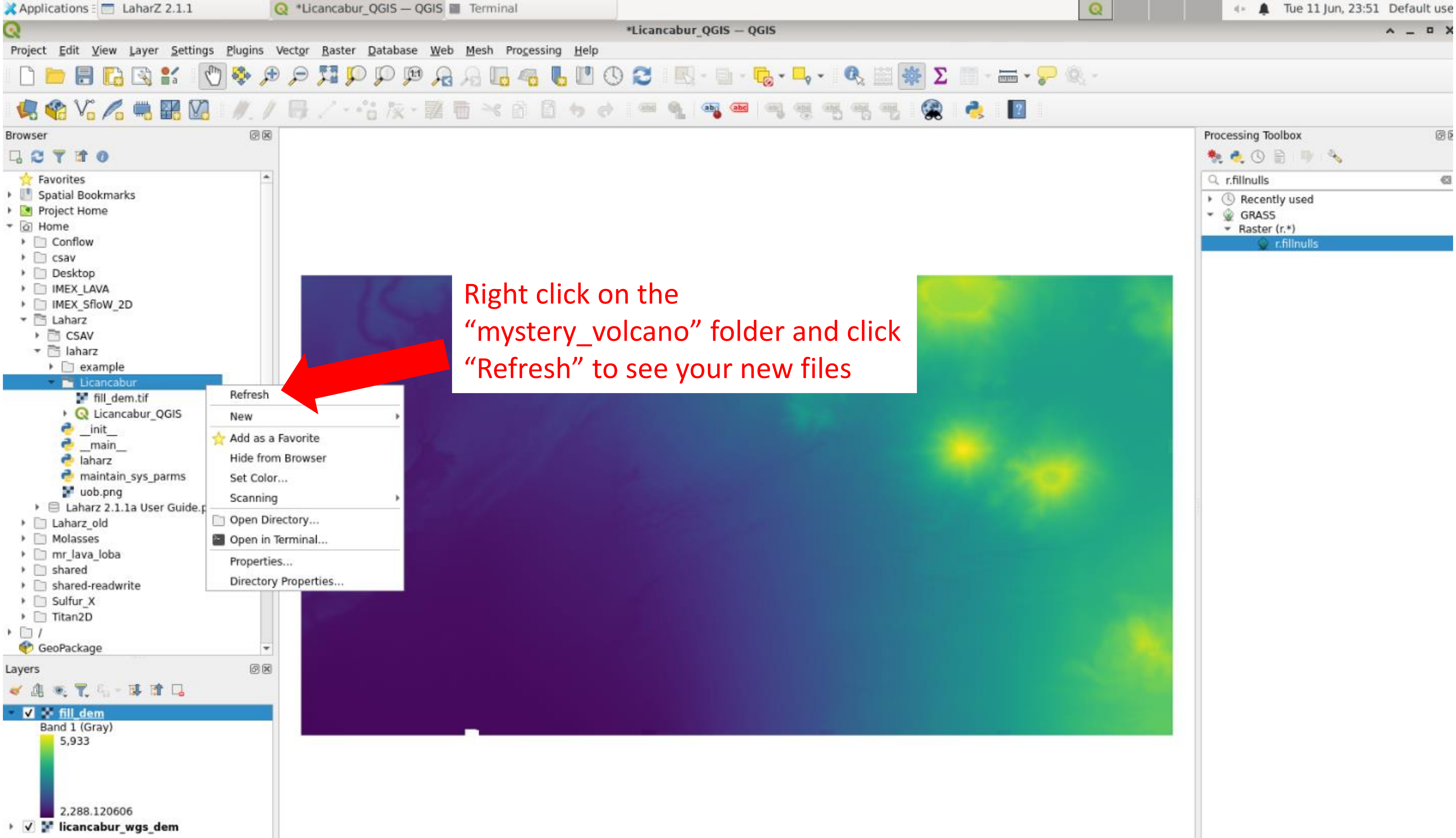
Name the initiation points with the stream threshold and H/L ratio you used

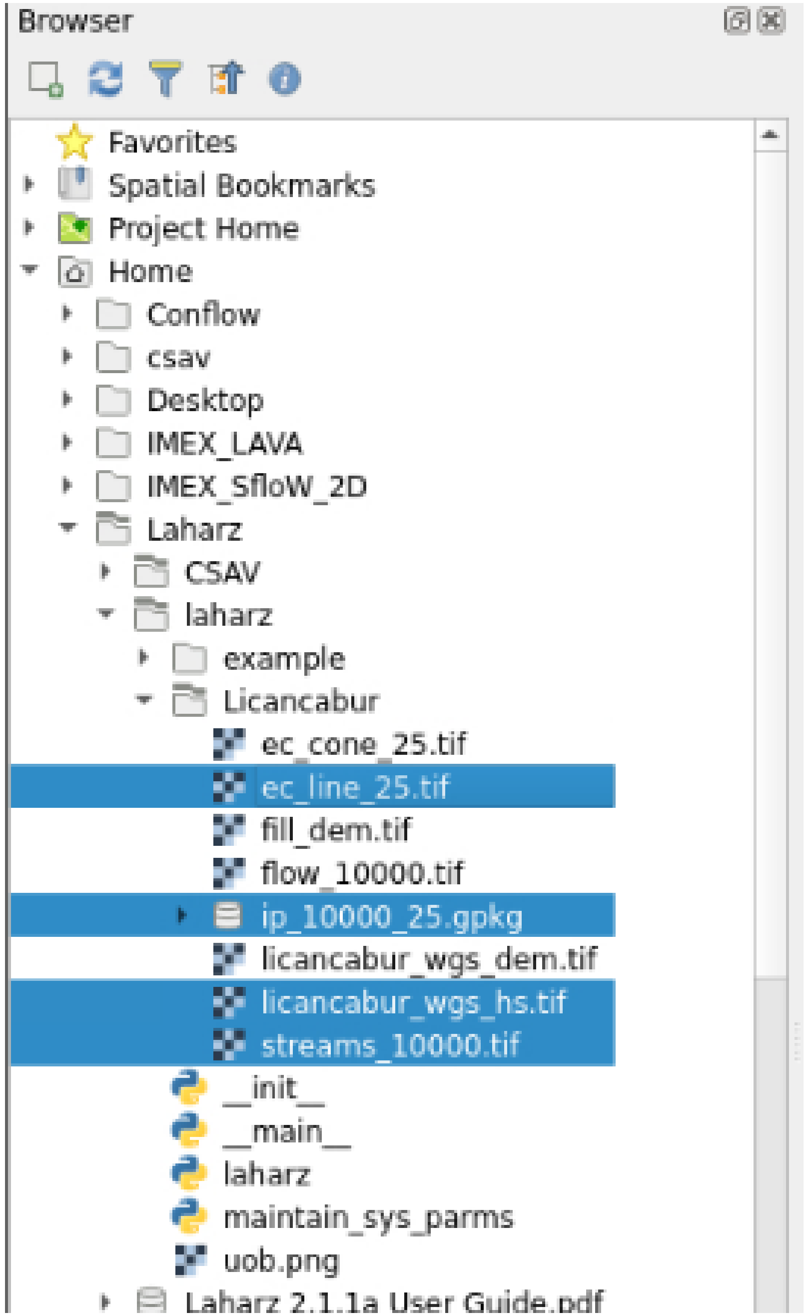
When you're done adding these choices, click "Initiation points" button



- Did you get any errors about no initiation points found? If so, you either need a smaller H/L ratio (so it reaches further) or a longer stream network (using a smaller stream threshold)
- It can be helpful to open the ec_line and streams files in QGIS and look at them
- Click "continue" for now

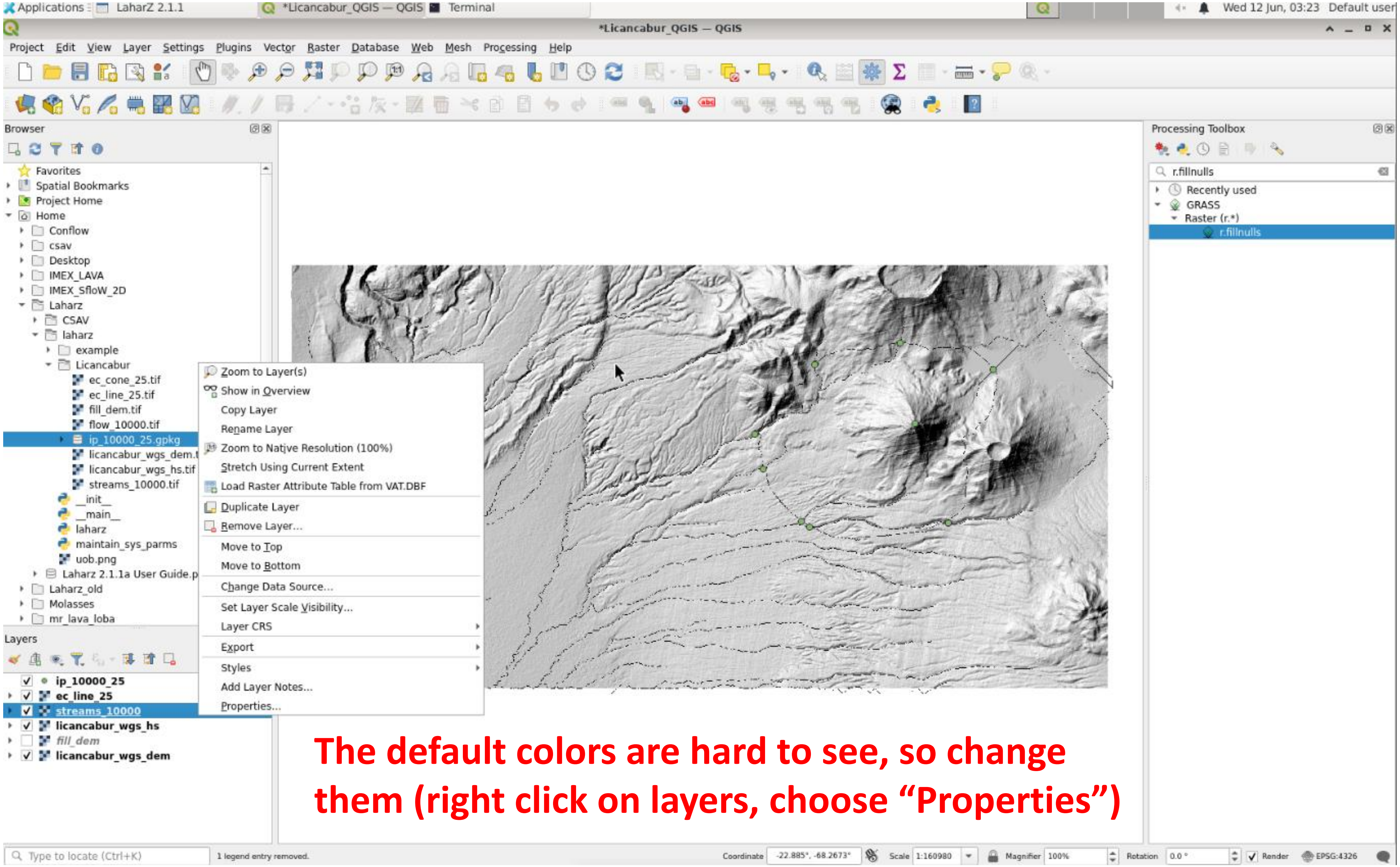

Applications
LaharZ 2.1.1
*Licancabur_QGIS — QGIS
Terminal
LaharZ 2.1.1
LaharZ
Sub Directory
Licancabur
Your current directory is:
/home/jovyan/Laharz/laharz
Create Initiation Points
Create Flow Outputs
Exit
Reopen the QGIS project by clicking the tab at the top of your screen


Right click on the "mystery_volcano" folder and click "Refresh" to see your new files
*Licancabur_QGIS — QGIS
Browser
Favorites
Spatial Bookmarks
Project Home
Home
Conflow
csav
Desktop
IMEX_LAVA
IMEX_SfloW_2D
Laharz
CSAV
laharz
example
Licancabur
fill_dem.tif
Licancabur_QGIS
__init__
__main__
laharz
maintain_sys_parms
uob.png
Laharz 2.1.1a User Guide.p
Laharz_old
Molasses
mr_lava_loba
shared
shared-readwrite
Sulfur_X
Titan2D
GeoPackage
Refresh
New
Add as a Favorite
Hide from Browser
Set Color...
Scanning
Open Directory...
Open in Terminal...
Properties...
Directory Properties...
Layers
fill_dem
Band 1 (Gray)
5,933
2,288.120606
licancabur_wgs_dem
Processing Toolbox
r.fillnulls
Recently used
GRASS
Raster (r.*)
r.fillnulls



Let's look at the stream network, H/L cone (ec_line), and the initiation points we created!

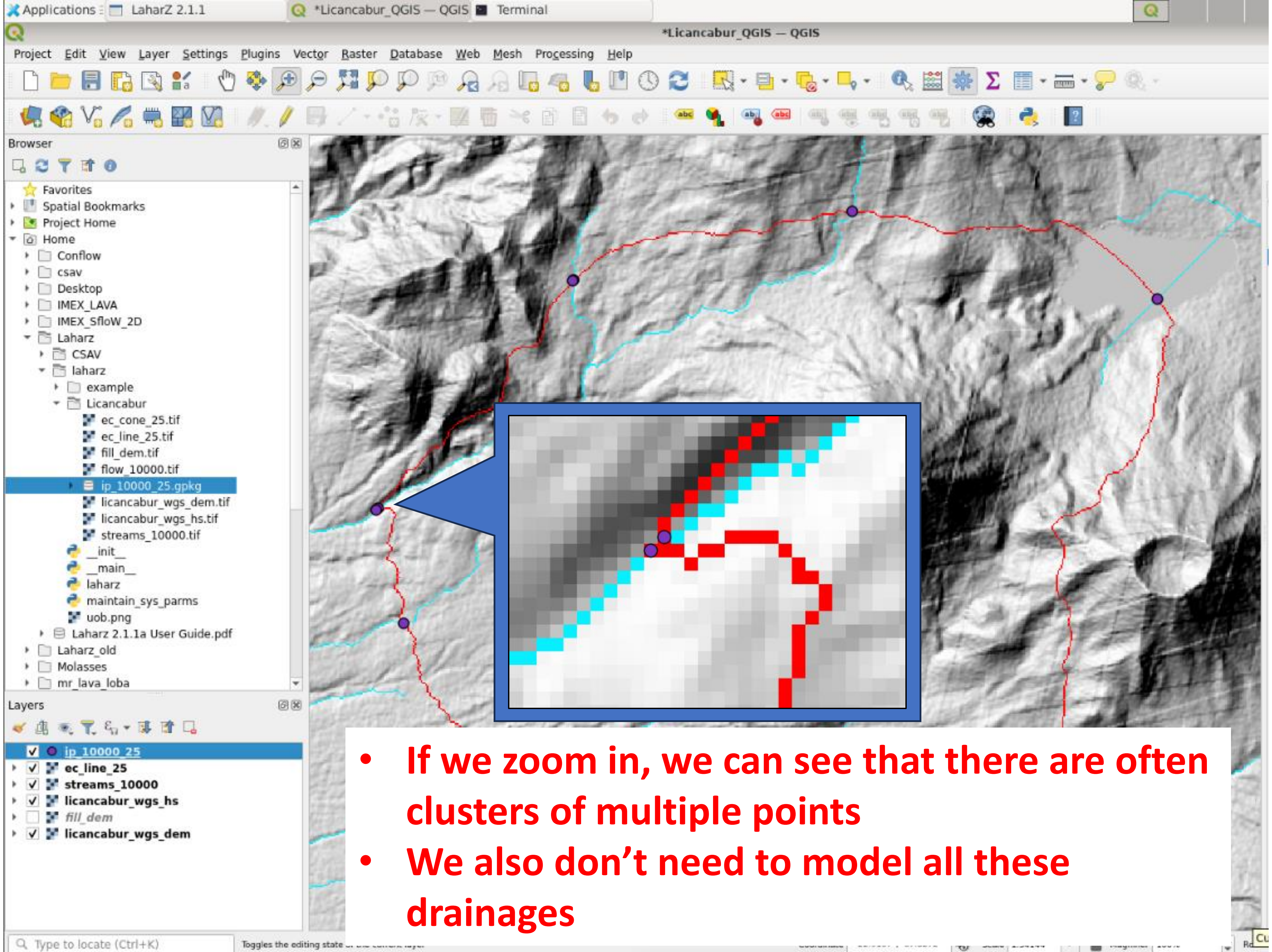
Add these files to your map:
Your_volcano_hs.tif
streams_x.tif
ec_line_x.tif
ip_x_x.gpkg



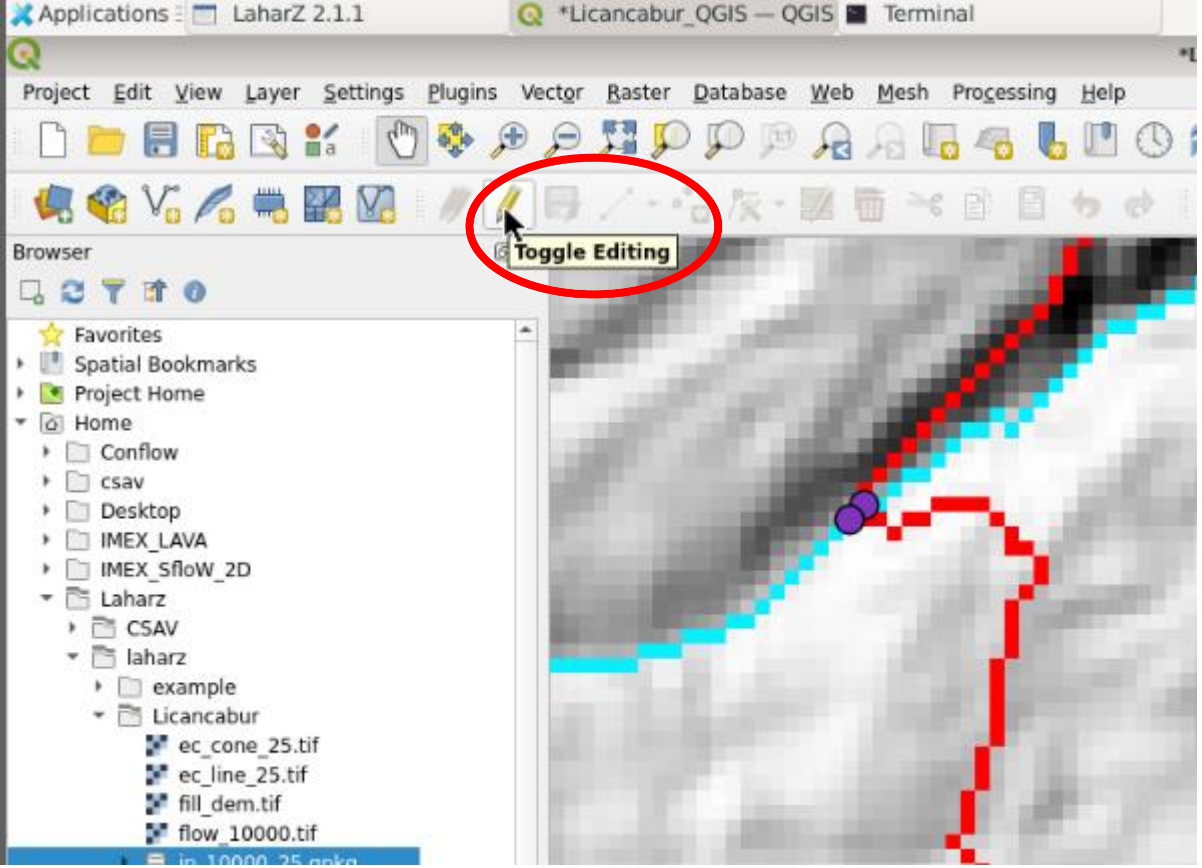
The default colors are hard to see, so change them (right click on layers, choose “Properties”)

Adjust if needed!

- Examine the H/L cone (ec_line), the stream network (streams), and the initiation points (ip)
- Did you get any errors about no initiation points found? If so, you either need a smaller H/L ratio (so it reaches further) or a longer stream network (using a smaller stream threshold)
- Compare the stream network to any topographical or geological maps you have
- Does the stream network look good?
  - Is it too short? If so, try a smaller stream threshold number
  - Is it too long with lots of crazy branches? Try a larger stream threshold number
- Are there too many initiation points? Are they clustered? Delete some!
- Ask me questions!

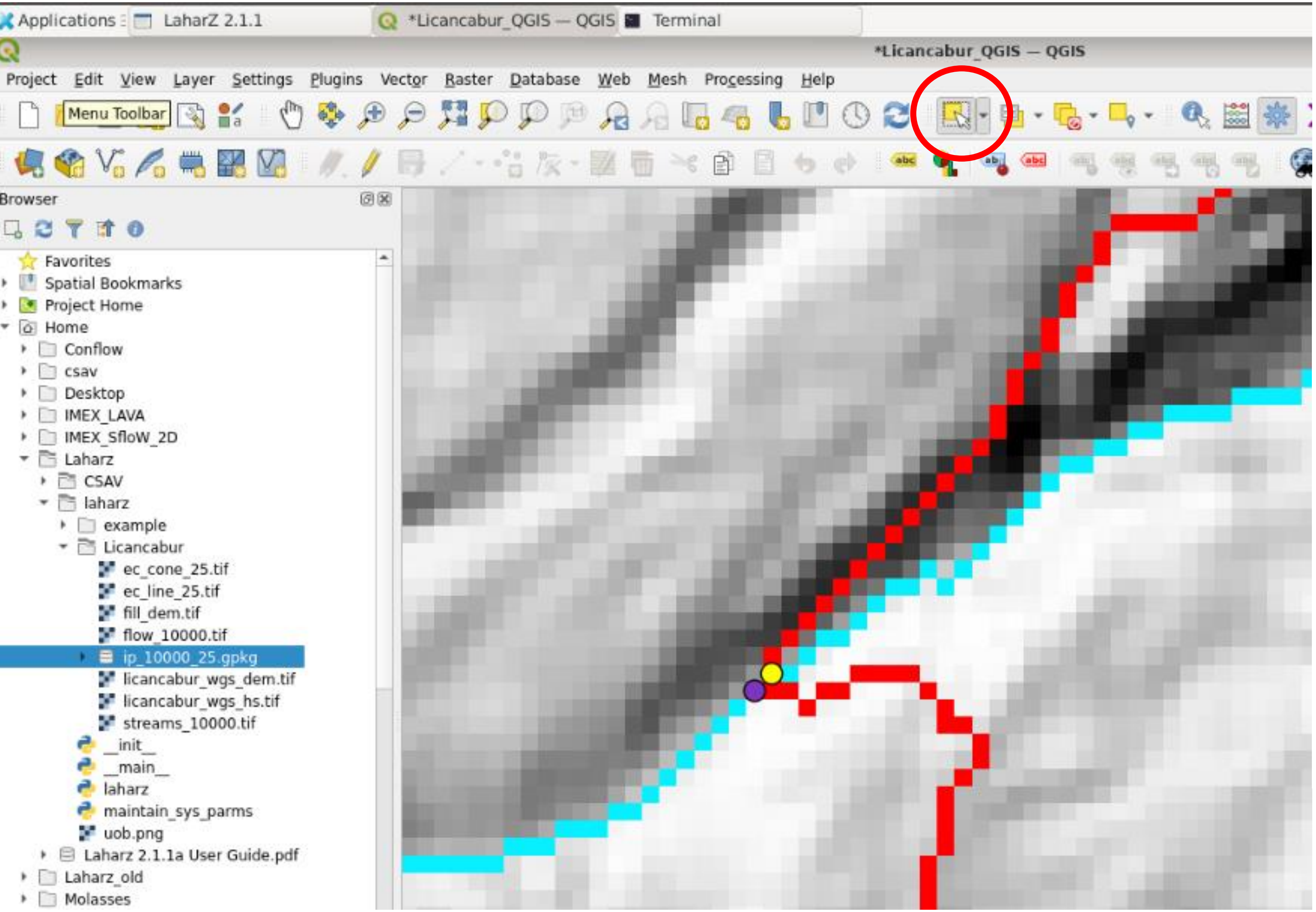


- If we zoom in, we can see that there are often clusters of multiple points
- We also don't need to model all these drainages



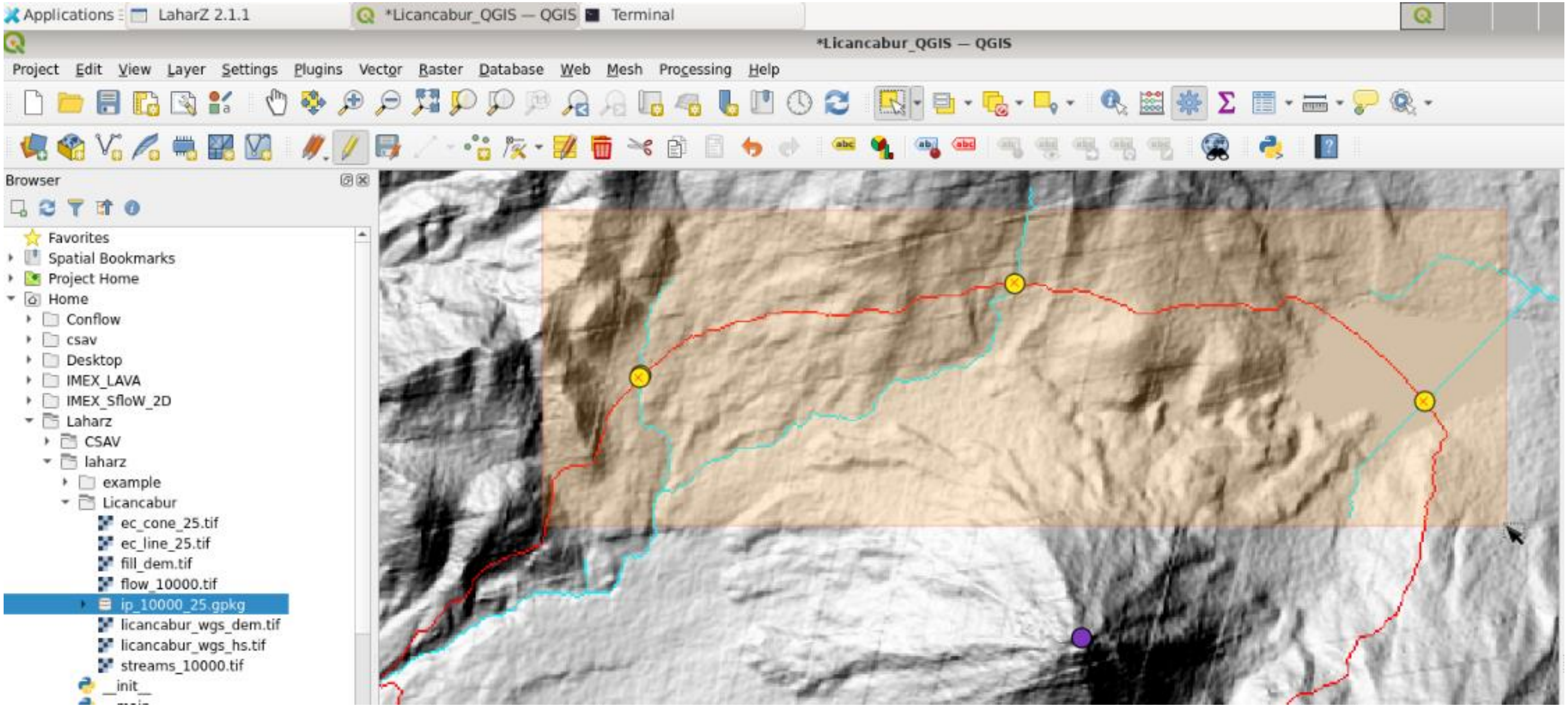
Let's delete some points

- Turn editing on by clicking the pencil "Toggle editing"



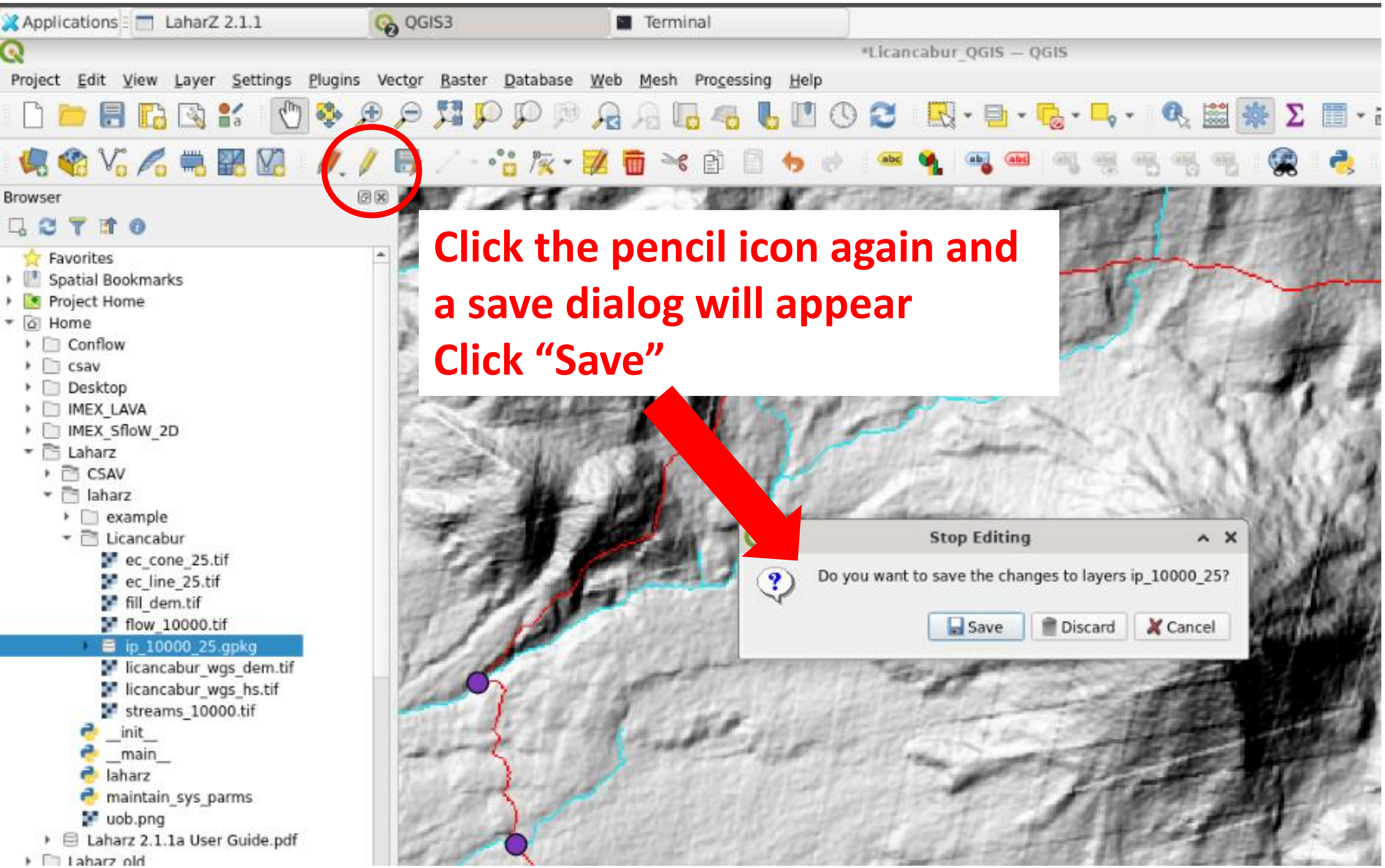
Let's delete some points

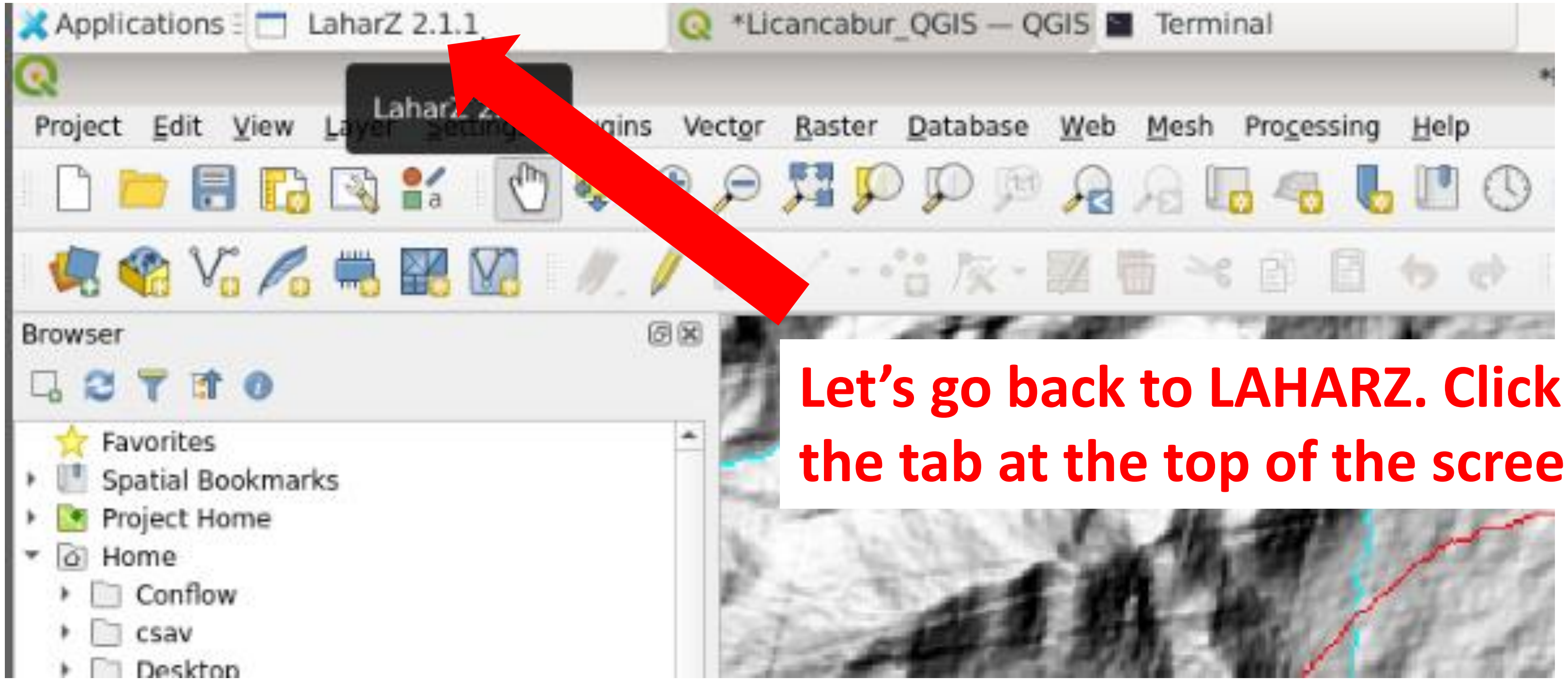
- Turn on the "Select features by area" tool and zoom in to each point cluster
- Click on the point you want to delete
- It will turn yellow
- The press the "delete" key on your keyboard



Let’s delete some points

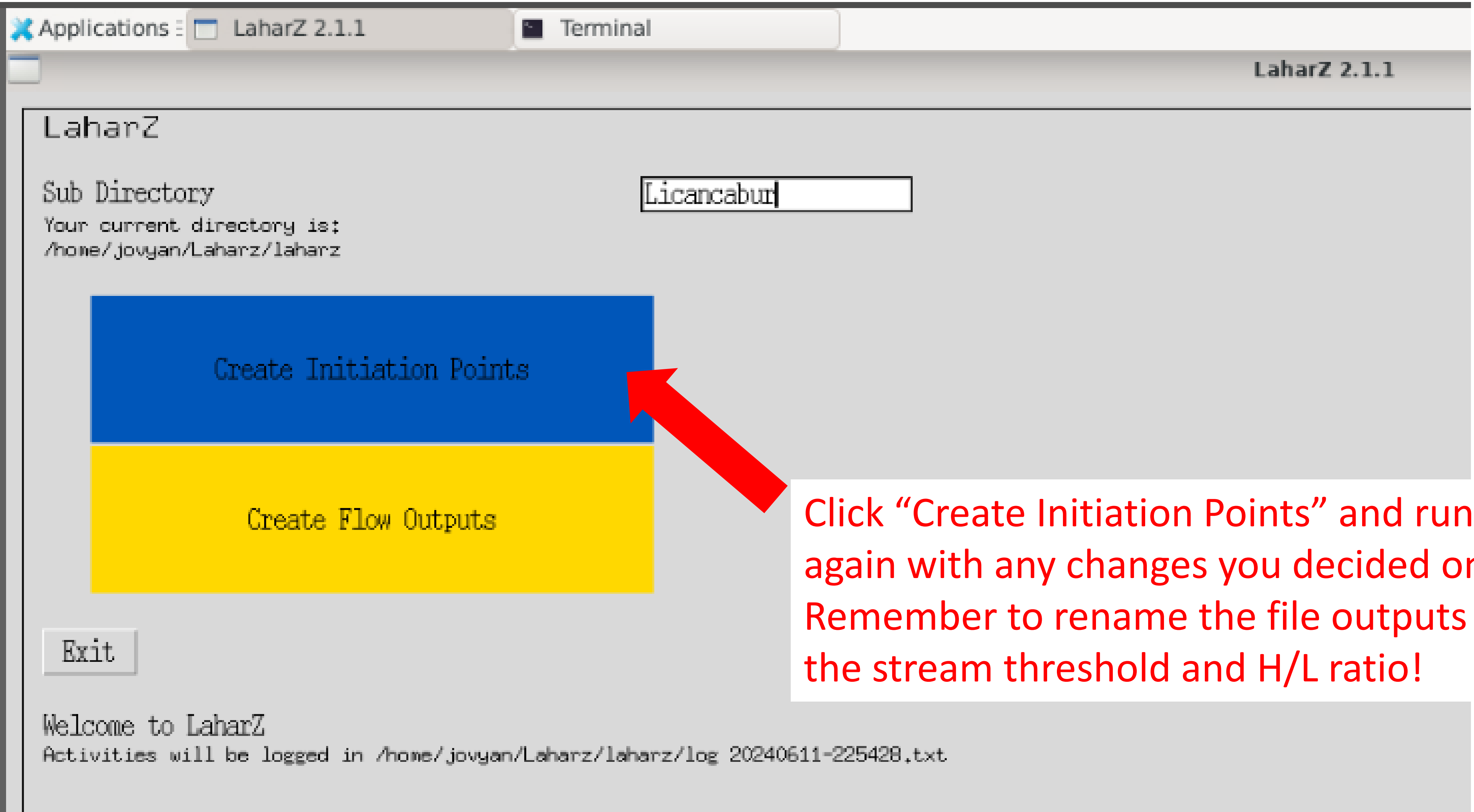
- You can also use the “Select features by area” tool to drag a selection box around multiple points
- They will turn yellow
- The press the “delete” key on your keyboard


Click the pencil icon again and
a save dialog will appear
Click “Save”
Stop Editing
Do you want to save the changes to layers ip_10000_25?
Save
Discard
Cancel

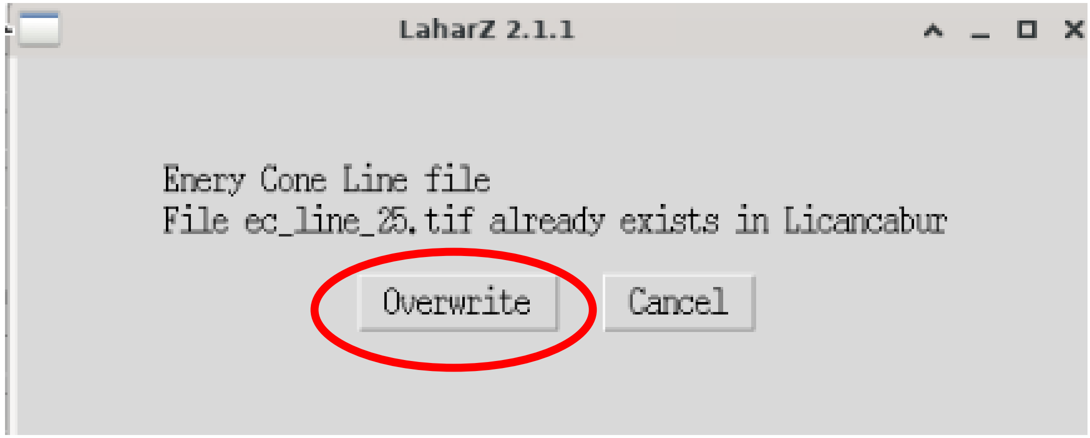


Let's go back to LAHARZ. Click the tab at the top of the screen

Let’s explore what a different stream threshold does.

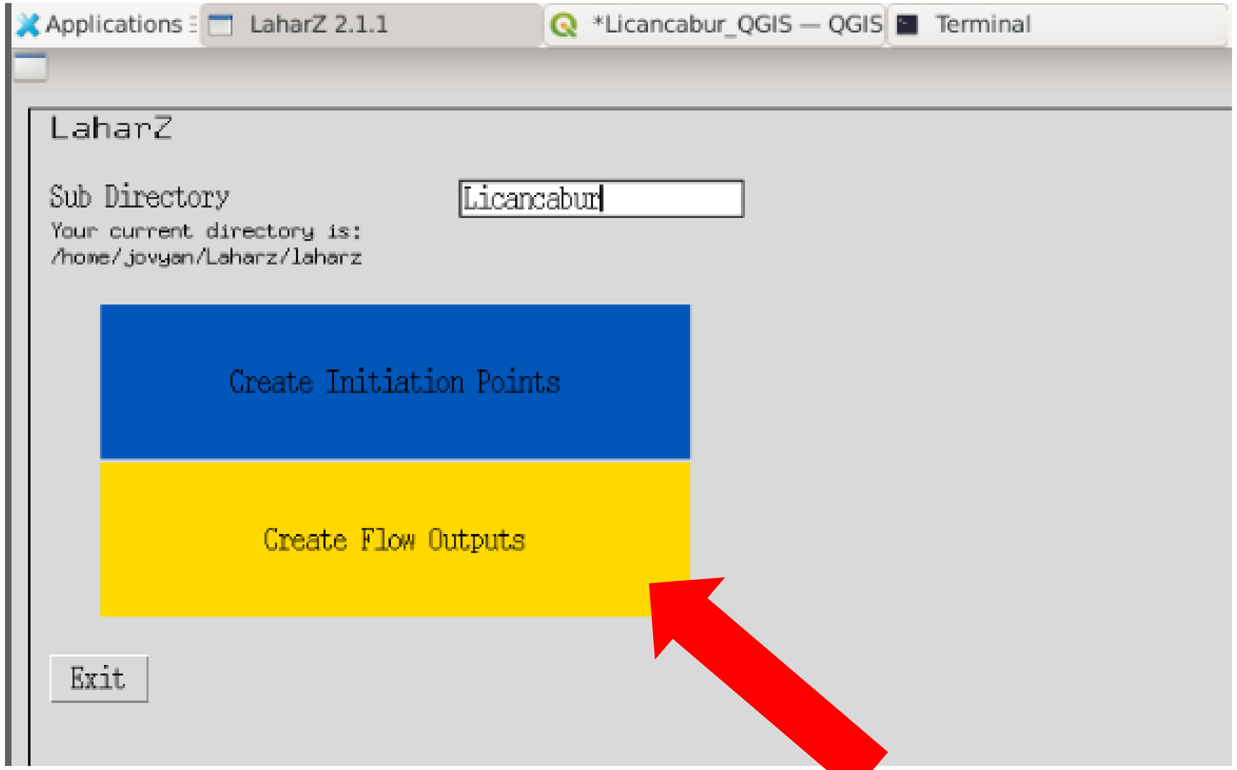


Click “Create Initiation Points” and run it again with any changes you decided on. Remember to rename the file outputs with the stream threshold and H/L ratio!

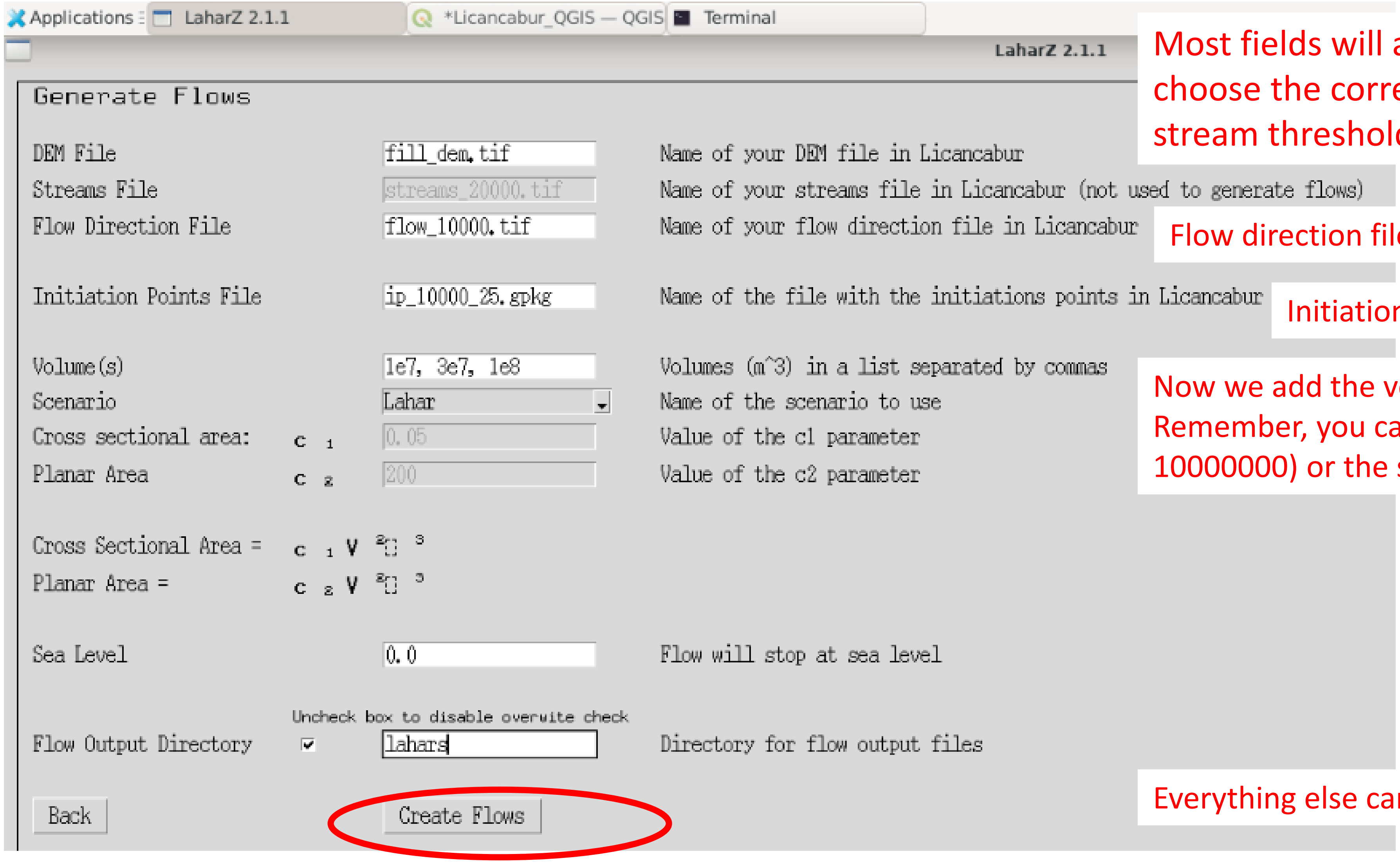


Choose “Overwrite” in the dialog boxes that pop up

Let’s run LAHARZ!



Click “Create Flow Outputs”



Most fields will already be filled in, but you need to choose the correct files that correspond to the stream threshold and H/L ratio you decided on (x)

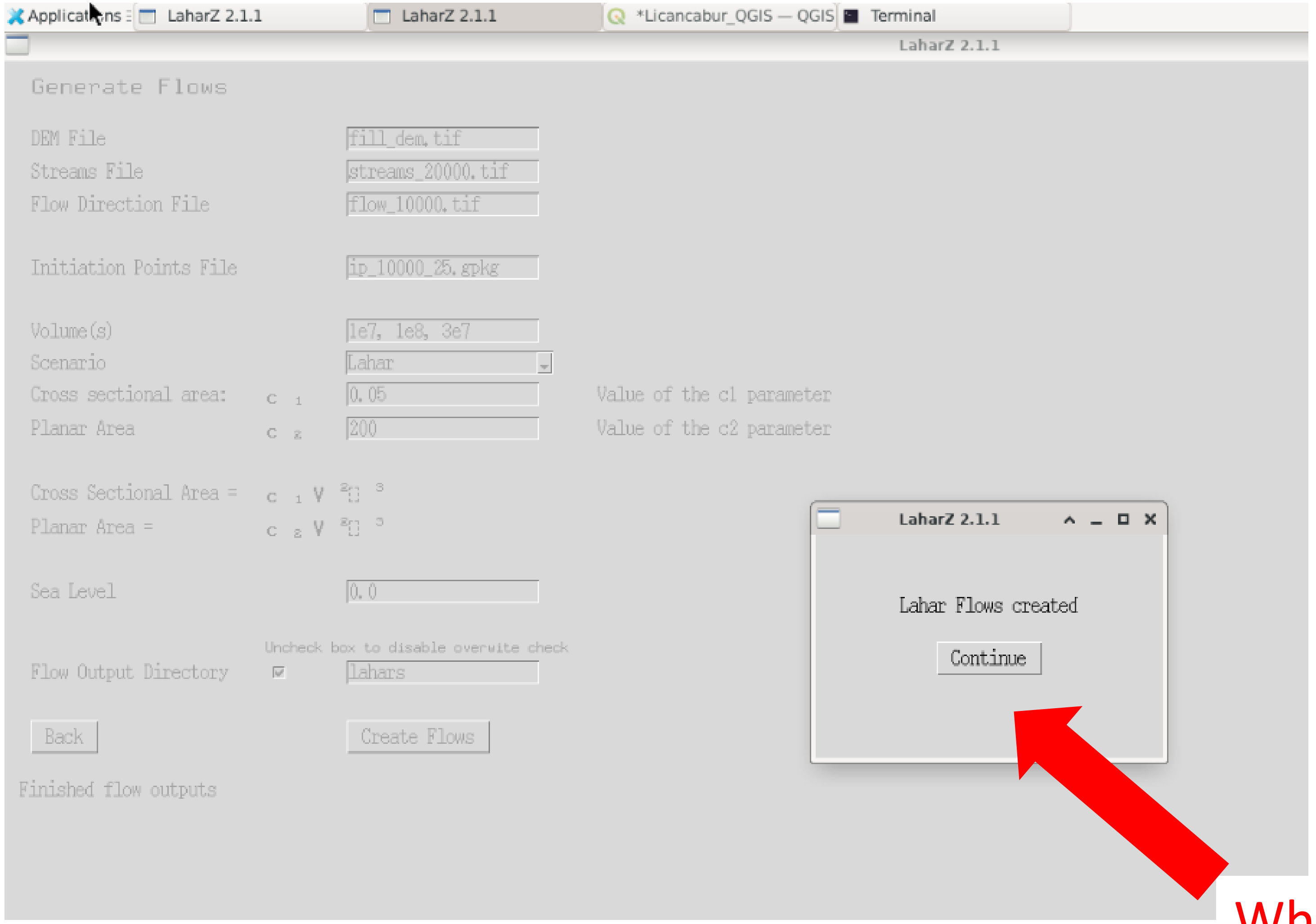
Flow direction file: flow_x.tif

Initiation points file: ip_x_x.gpkg

Now we add the volumes you decided on
Remember, you can type out the whole number (e.g., 10000000) or the scientific notation version (e.g., 1e7)

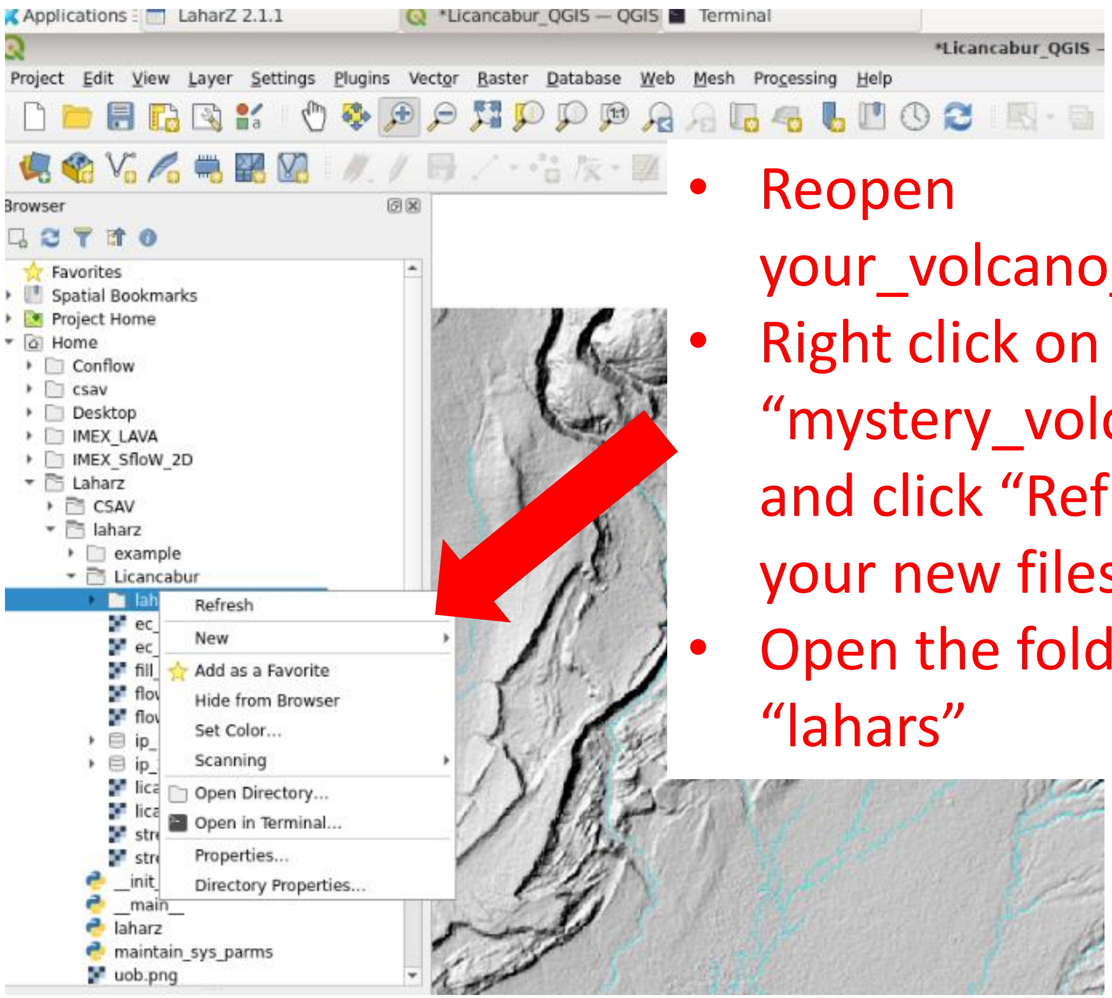
Everything else can stay as the default

When you're done adding these choices, click "Create flows" button

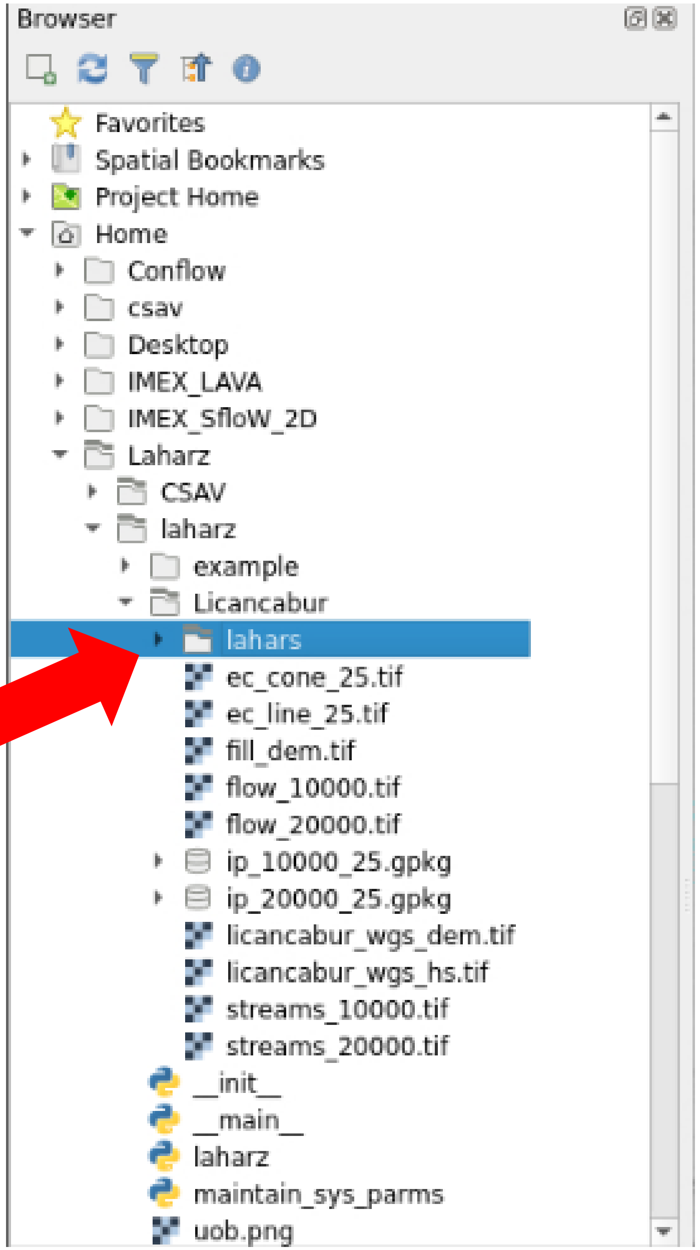


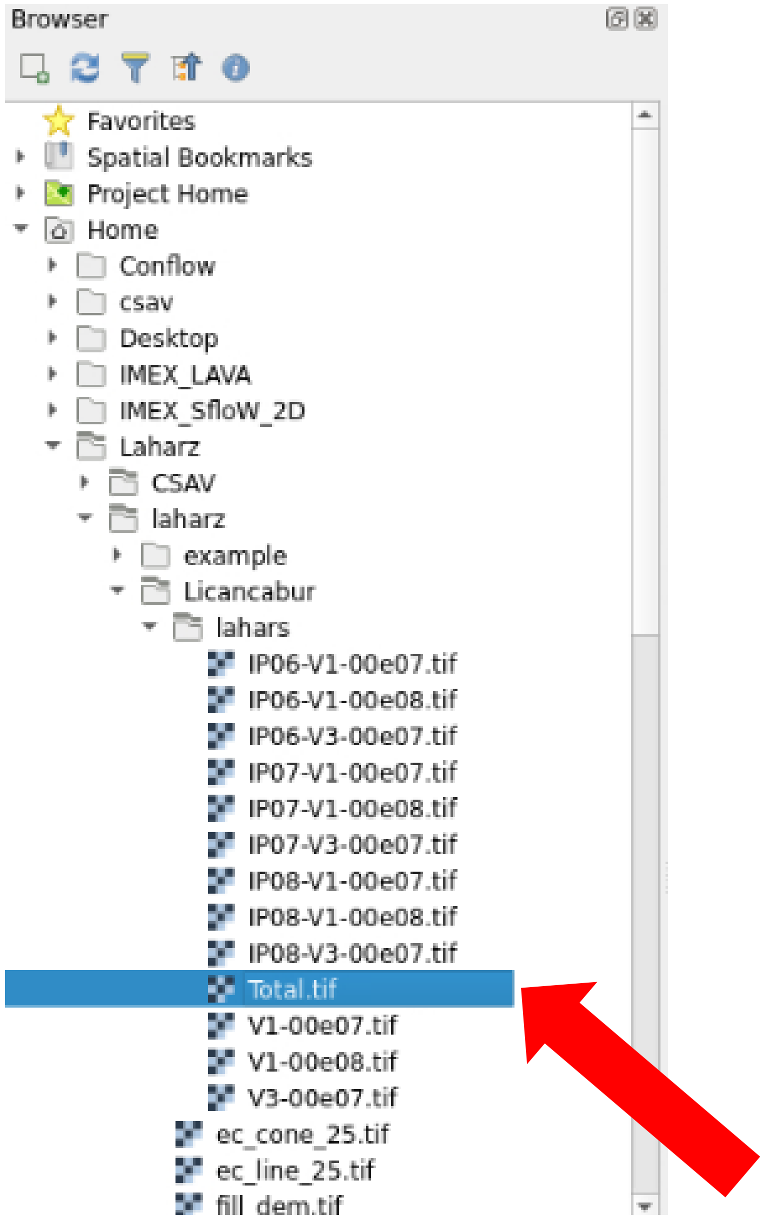
When it's done, click "Continue"

Let's look at the new files.



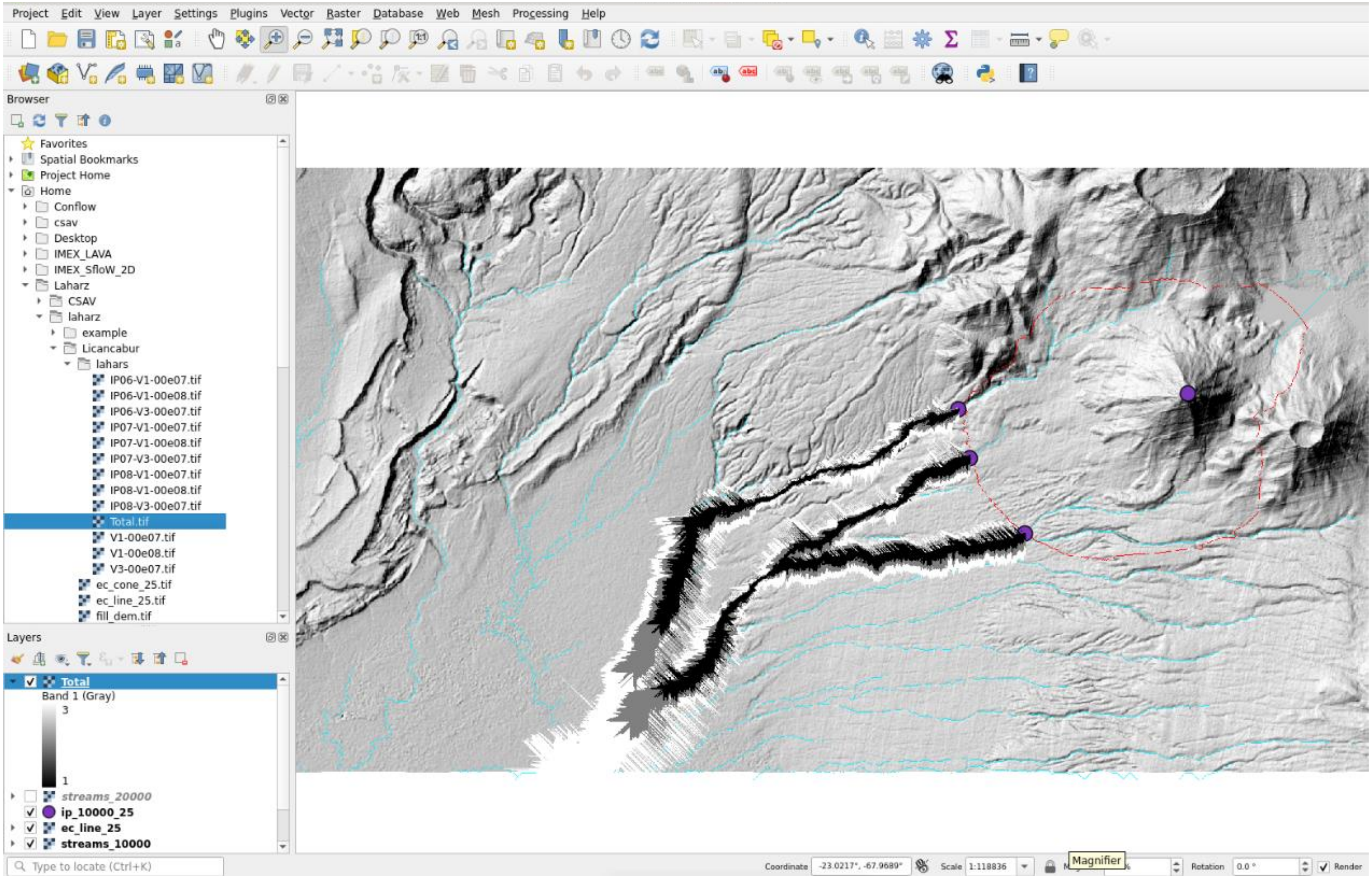
- Reopen your_volcano_QGIS
- Right click on the "mystery_volcano" folder and click "Refresh" to see your new files
- Open the folder called "lahars"

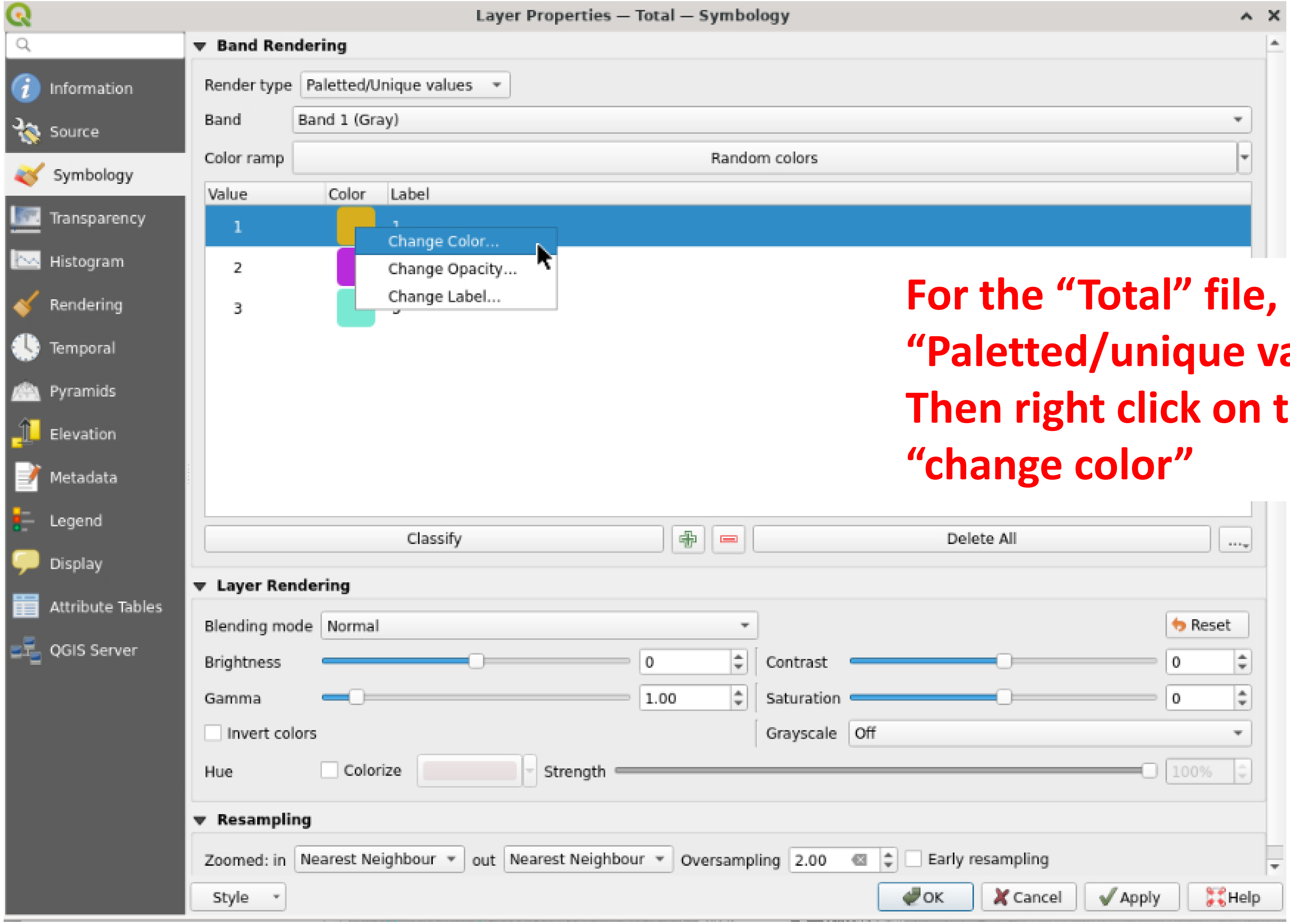




- LAHARZ made a bunch of different files
- There is a separate file for each volume/initiation point combination (these start with "IP"
- LAHARZ also made a file for each volume, merging all the flows from each initiation point (these start with "V")
- And finally, there is a "Total" file that has all the volumes from all the initiation points in one file
- Add the "total.tif" file to your map

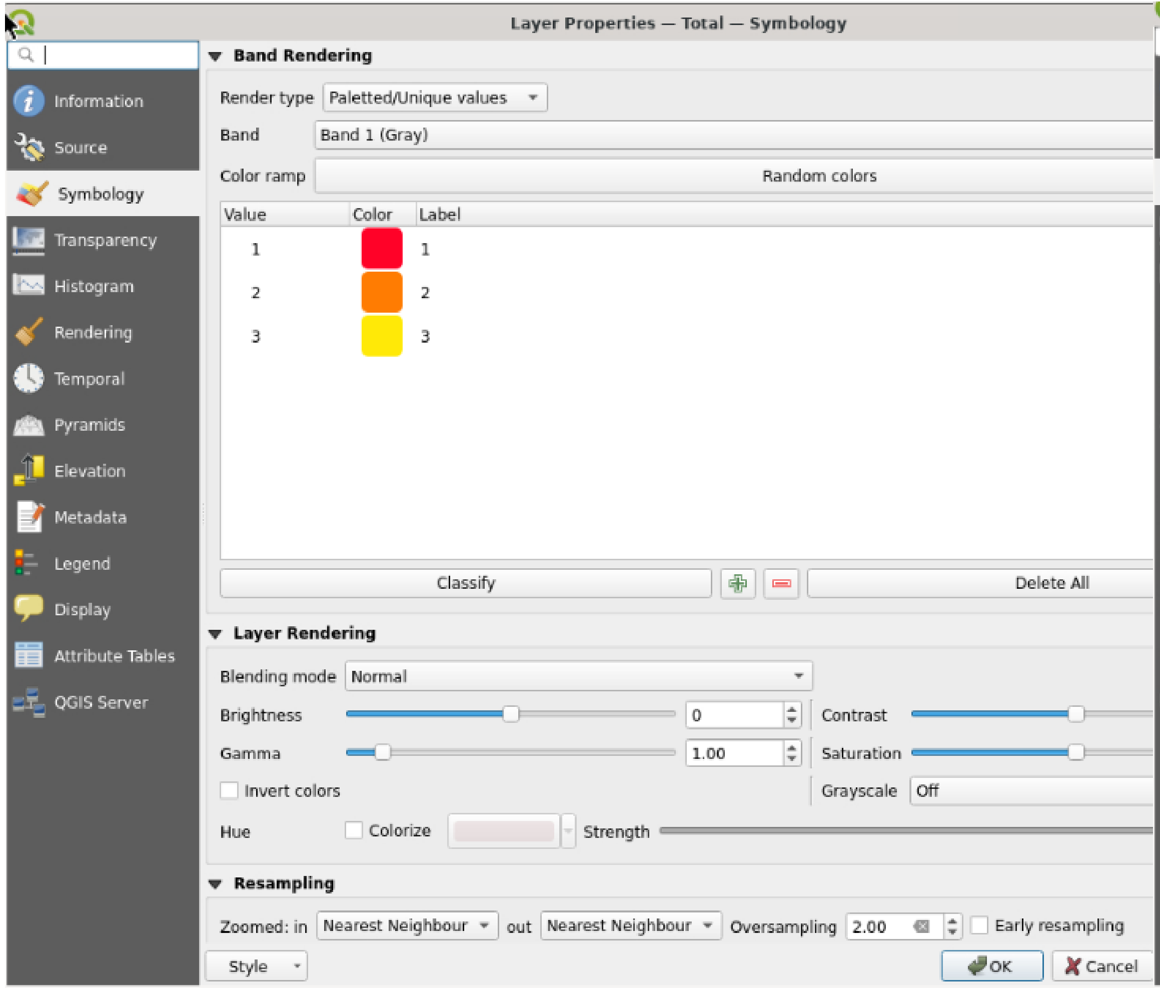
Change the colors and make the map prettier.

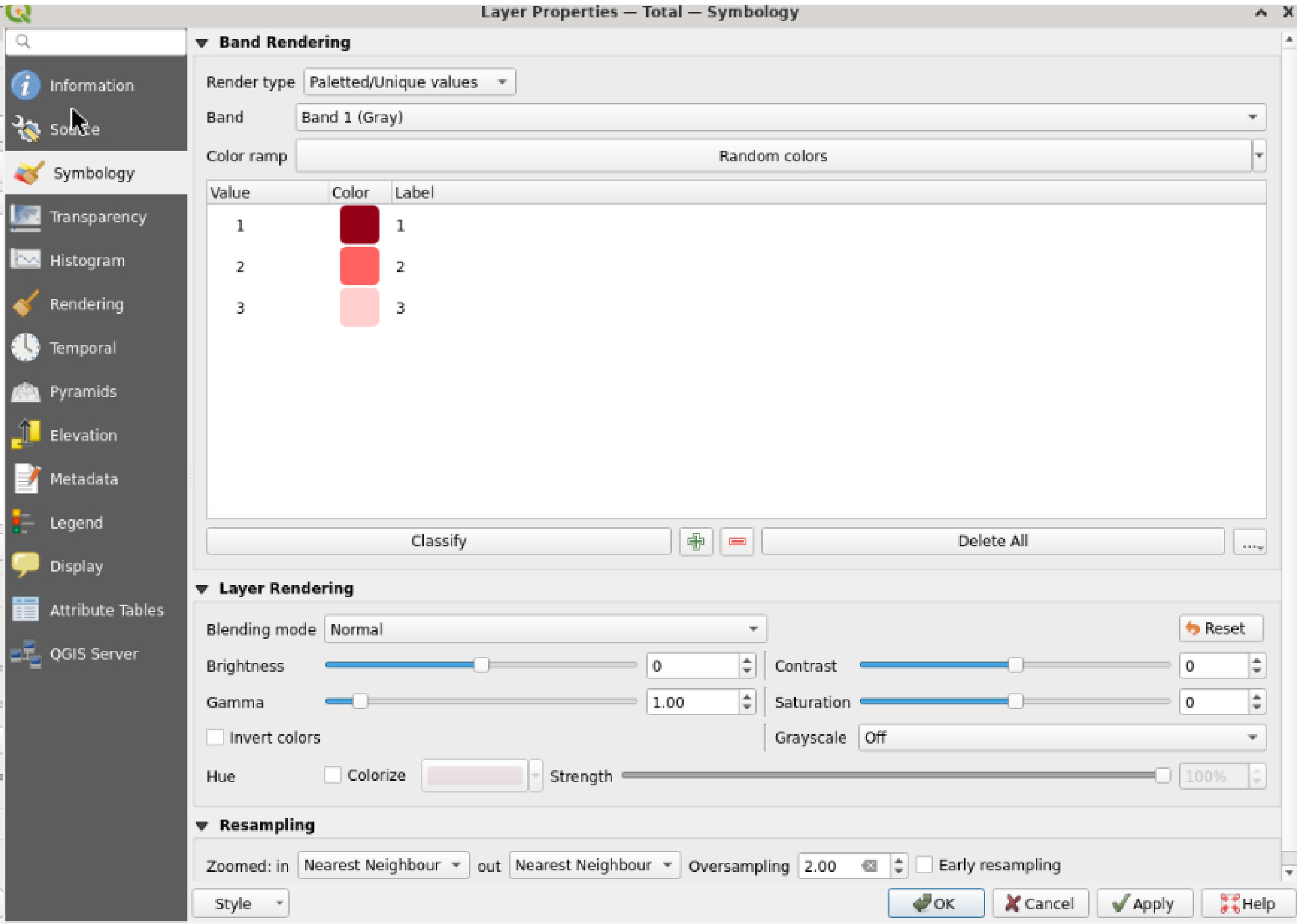


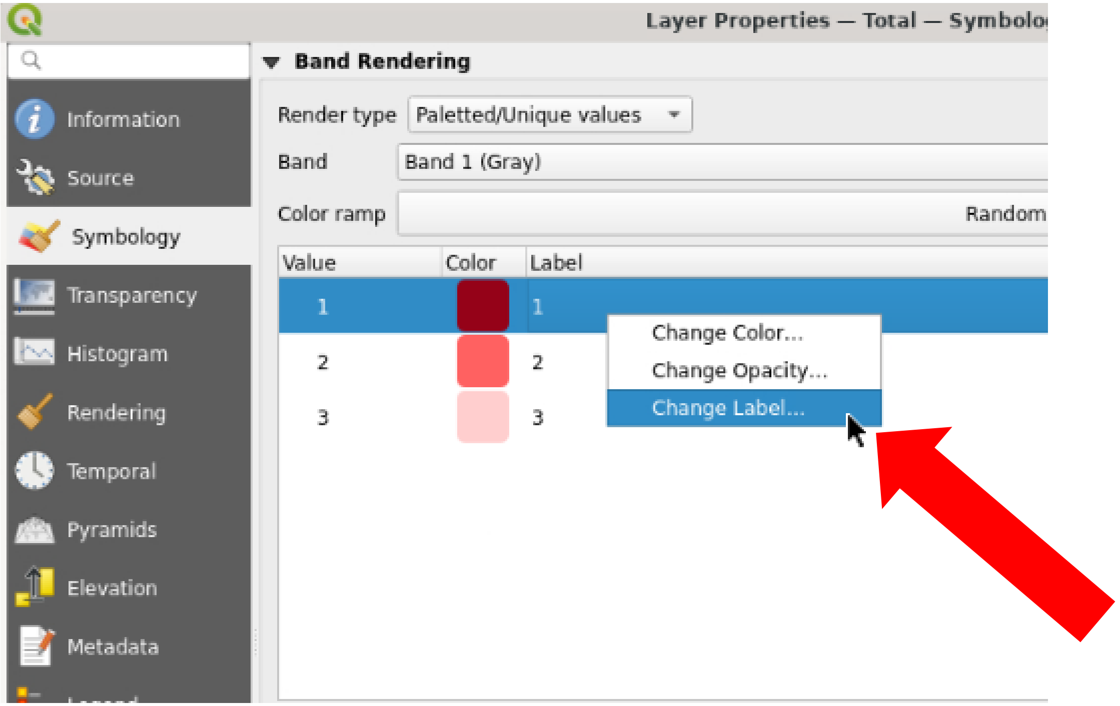


For the "Total" file, change the "render type" to "Paletted/unique values" and click "classify". Then right click on the listed value and choose "change color"

- **The lowest value (1) is the lowest volume**
- **This is the highest hazard zone because smaller lahars occur more often.**
- **Choose a color scheme that makes sense, like red-orange-yellow or dark red to light red or anything else we discussed.**

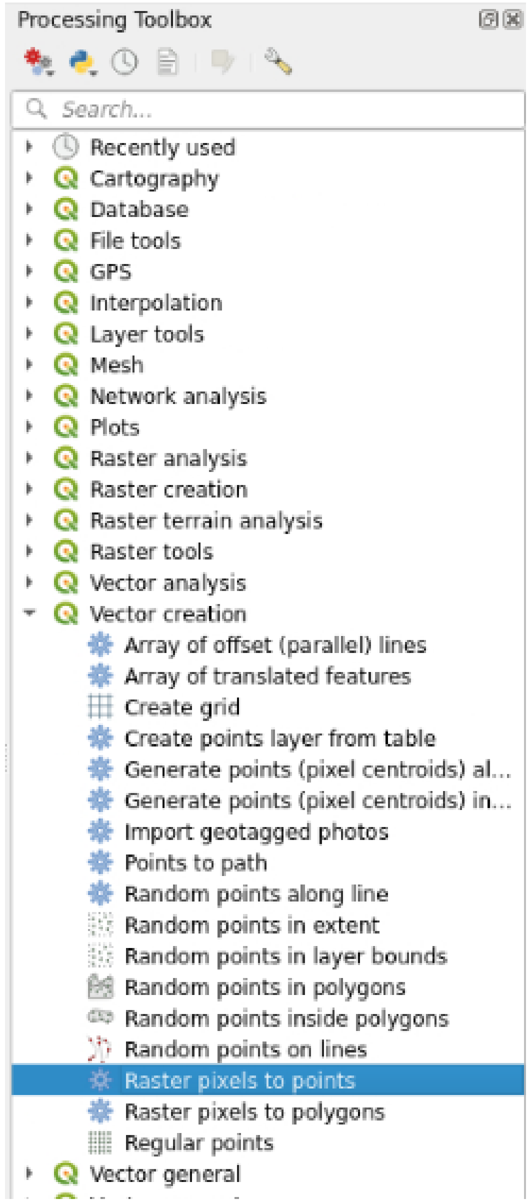






You can also change the label for each layer to the volumes you used

Let’s make a proximal hazard zone layer



Open the “Raster pixels to points” tool

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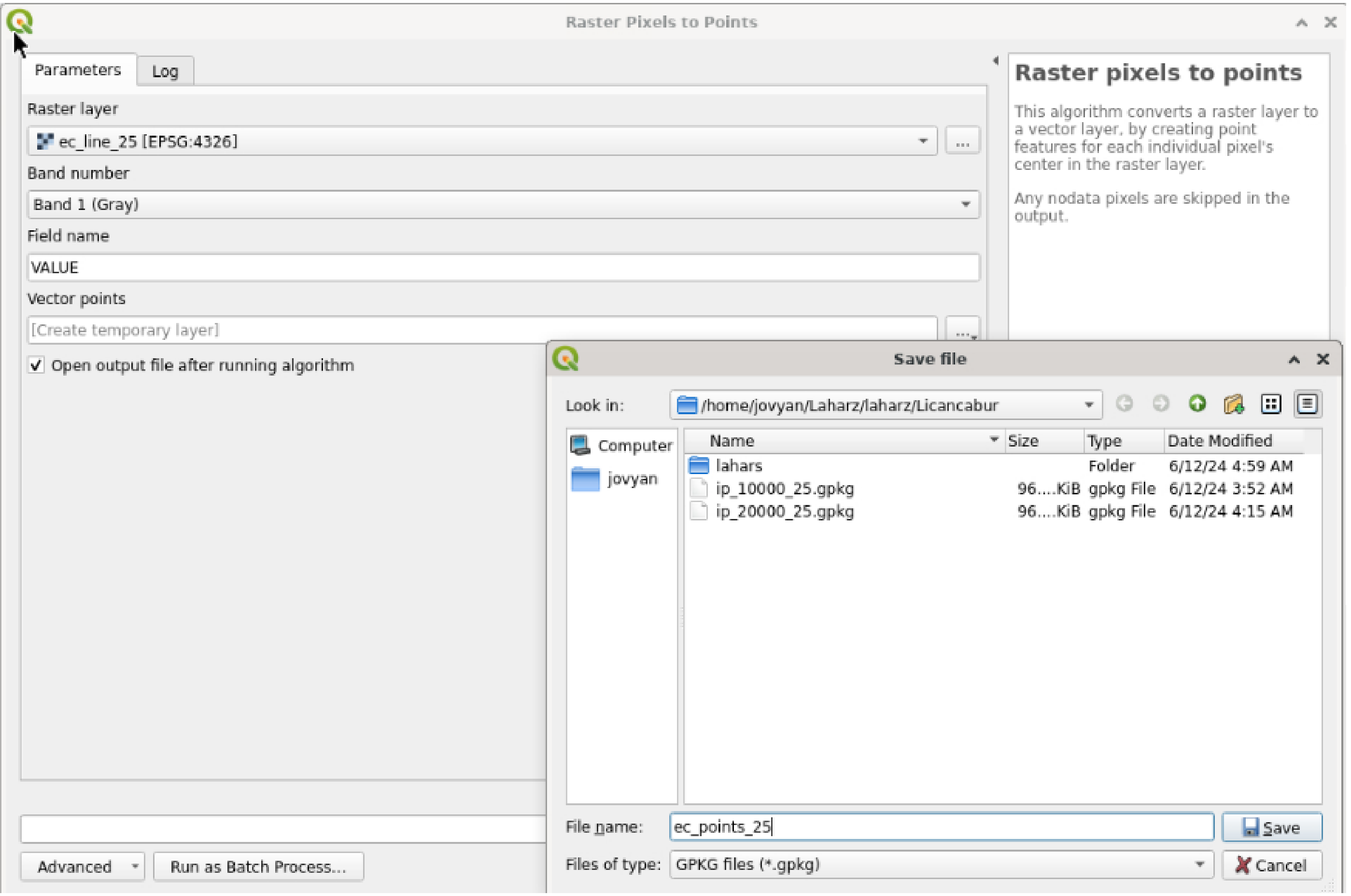
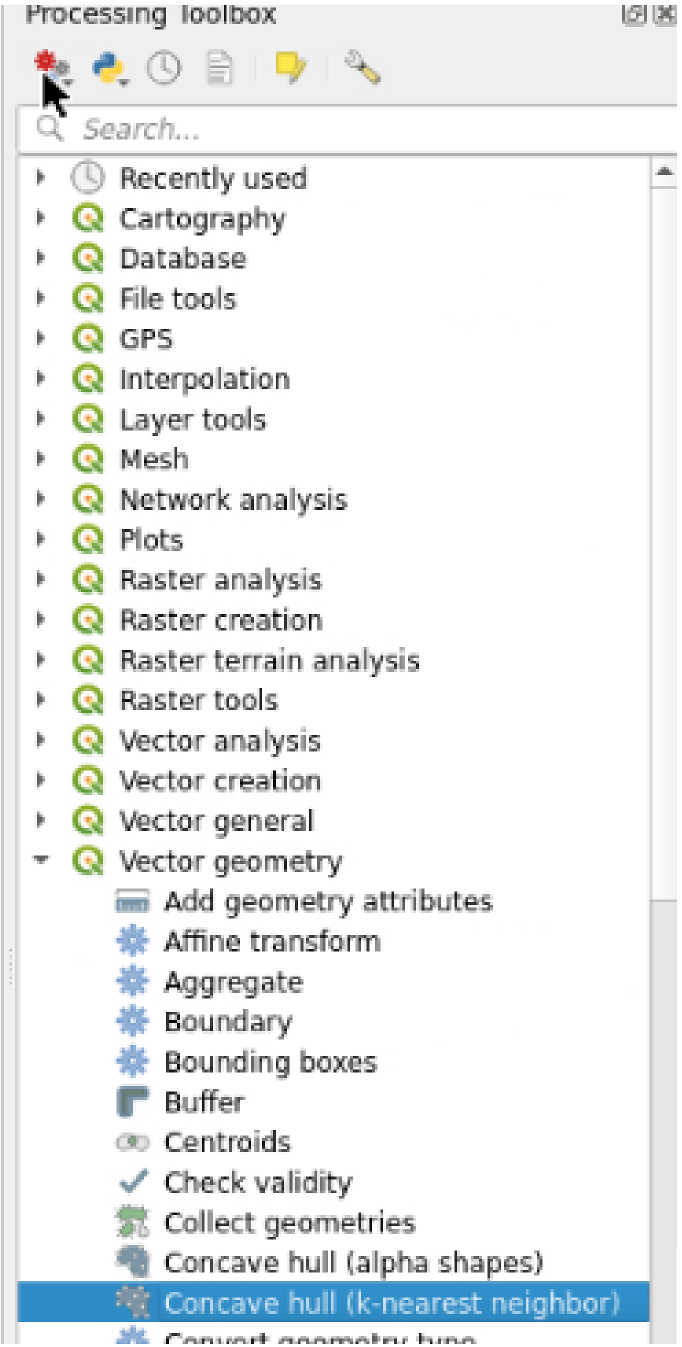


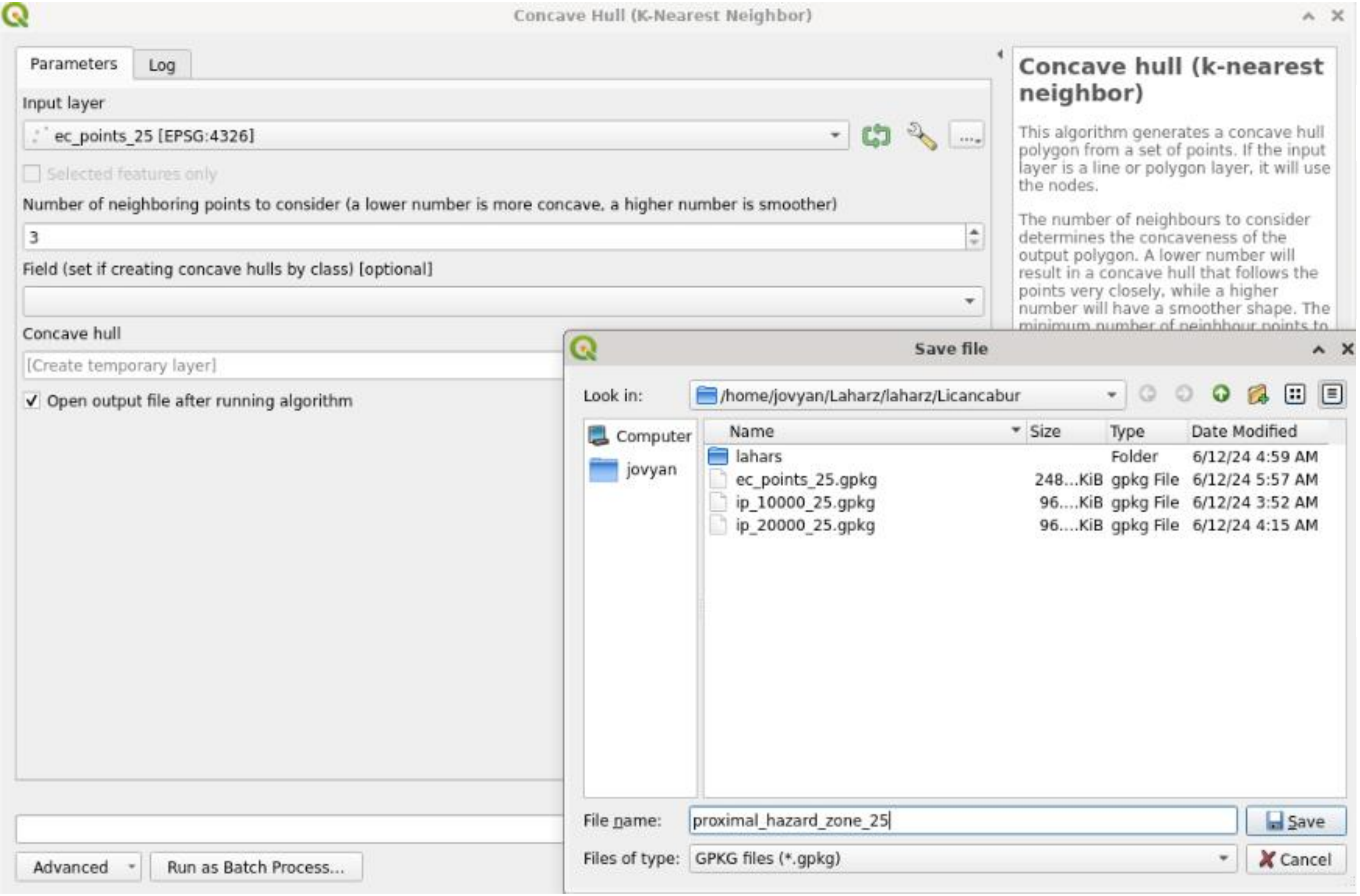
Raster layer: ec_line_x

- **Vector points: Save to File**
- **File name: ec_points_x**
- **Click "save"**
- **Click "run"**



Open the "Concave hull (k-nearest neighbor)" tool

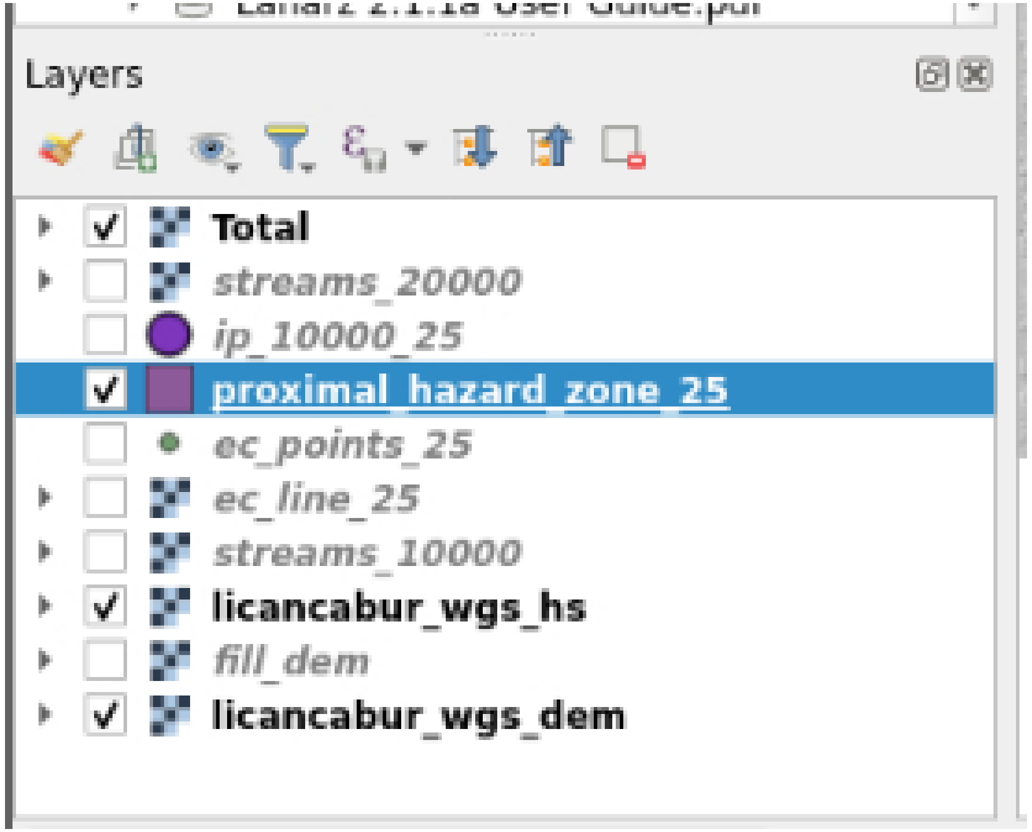
Open the “Concave hull (k-nearest neighbor)” tool



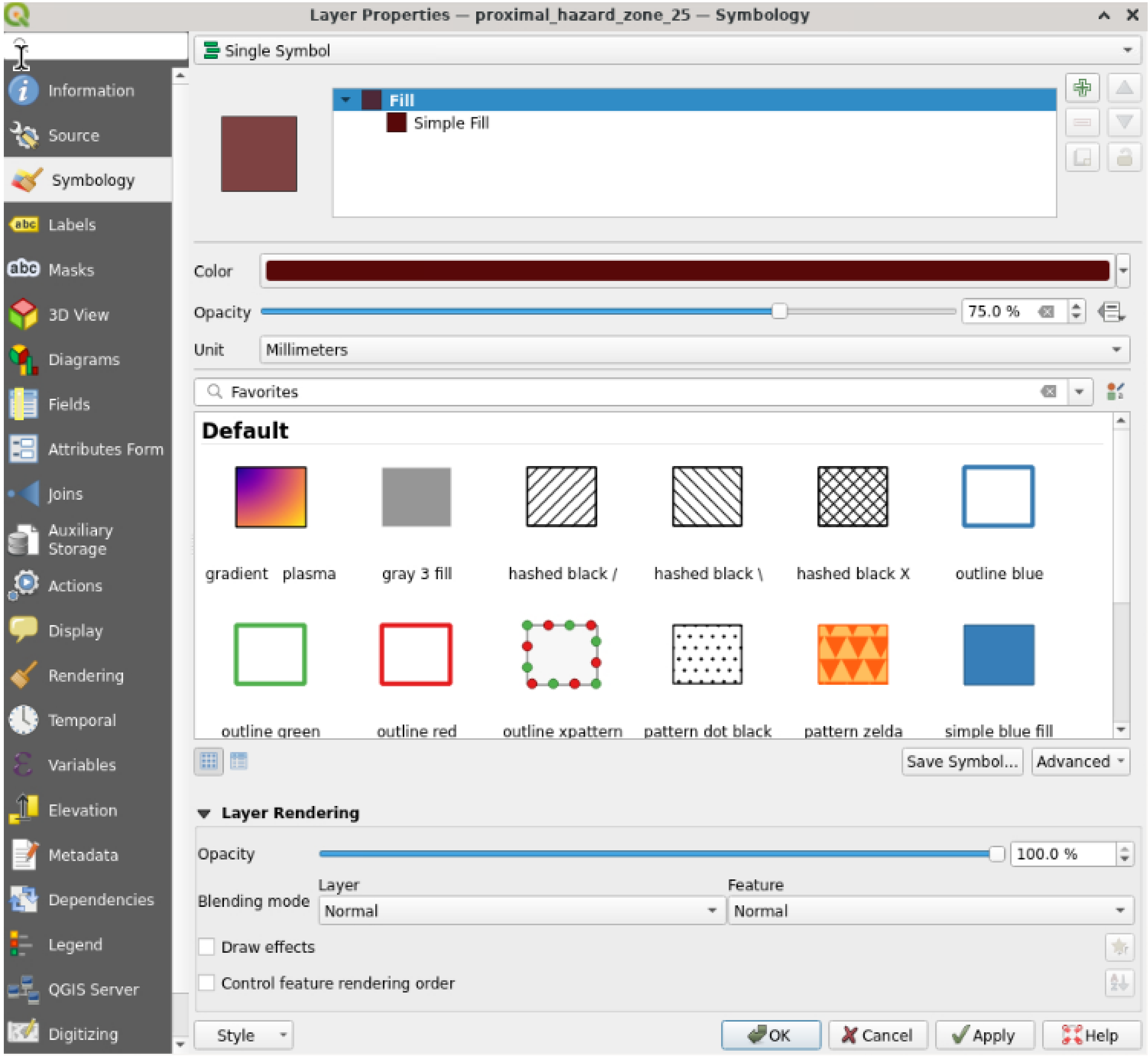
Input layer: ec_points_x

- **Concave hull: Save to File**
- **File name: proximal_hazard_zone_x**
- **Click “save”**
- **Click “run”**

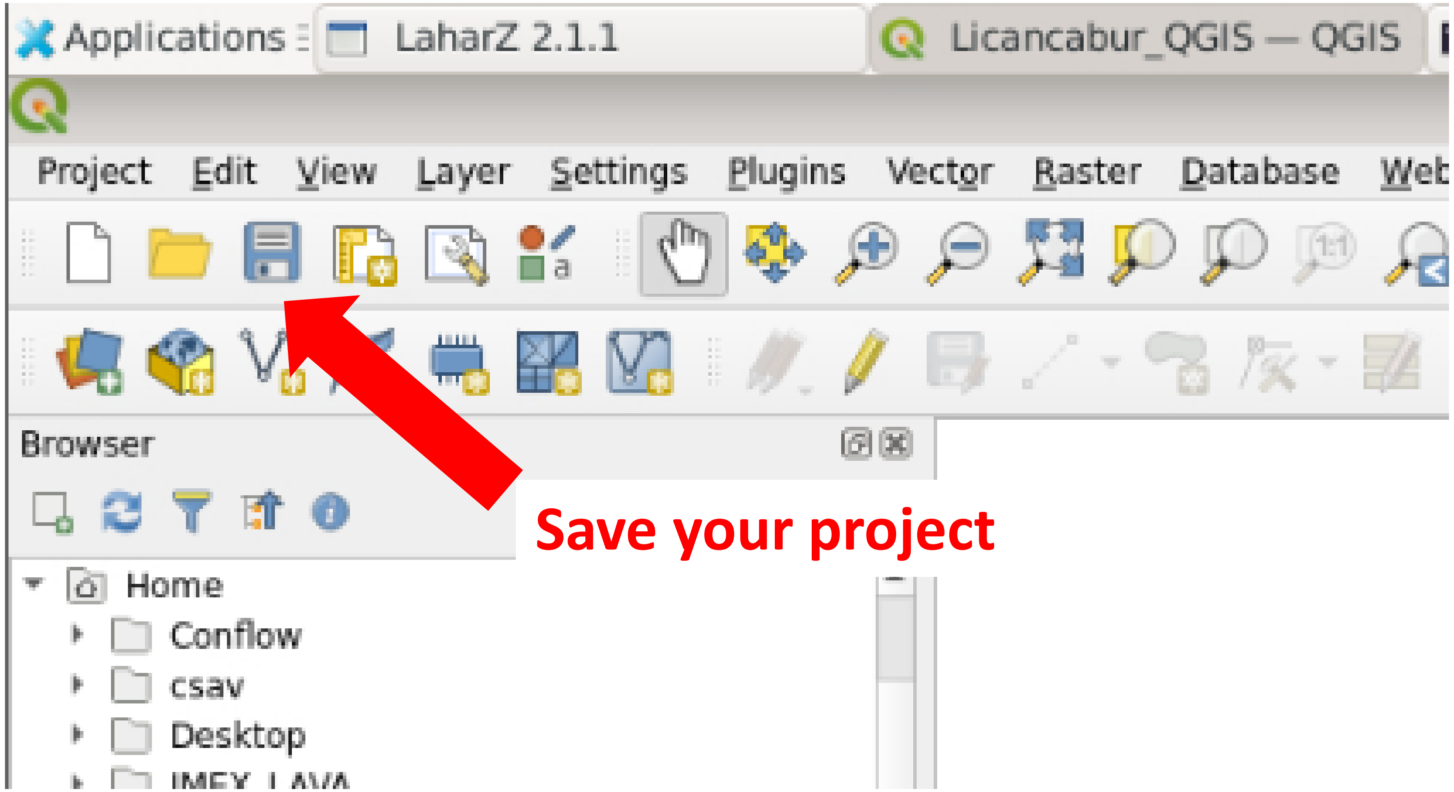
Let's make the map neater



- Turn off layers that aren't needed



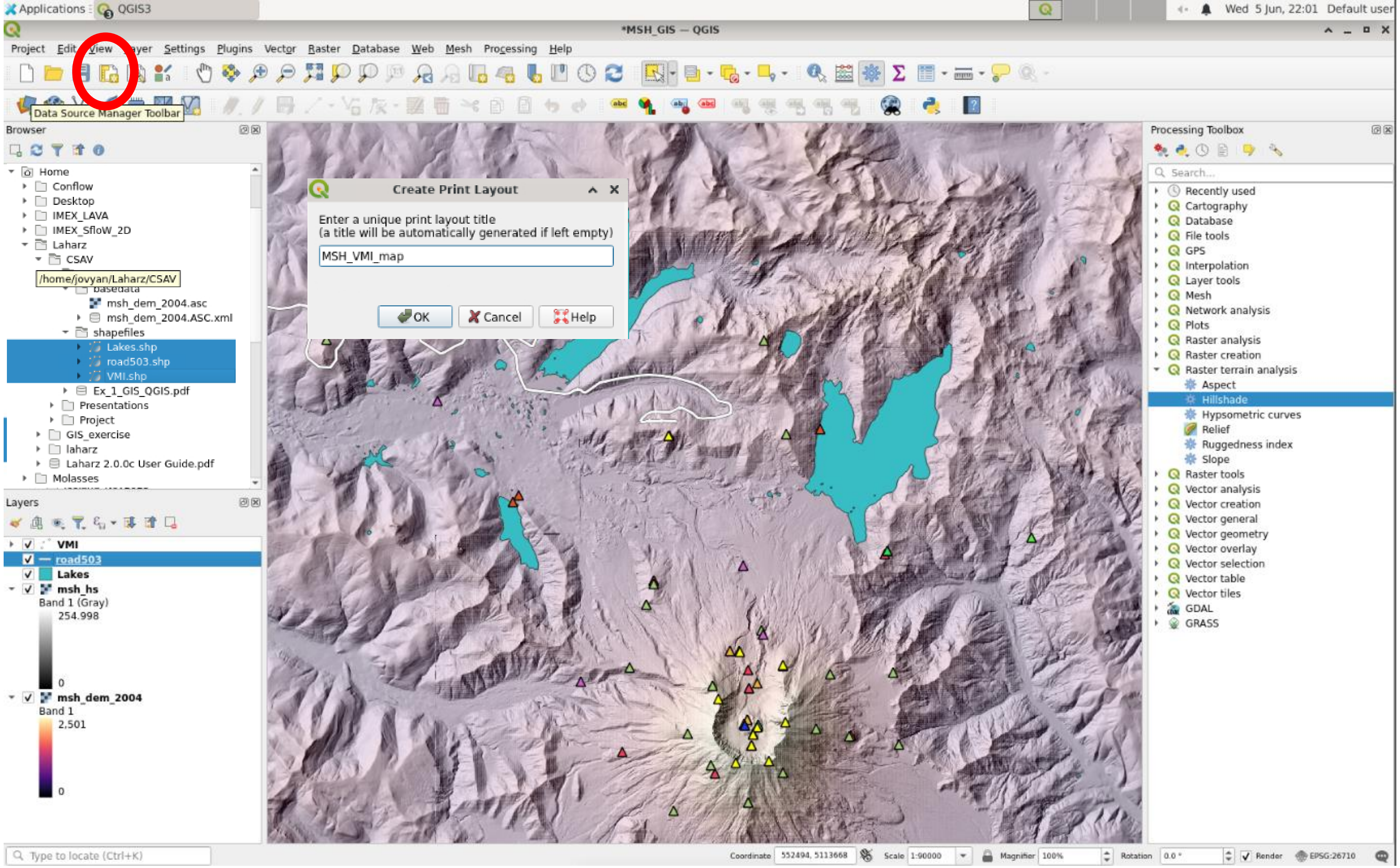
Change the color and opacity of the proximal hazard zone layer



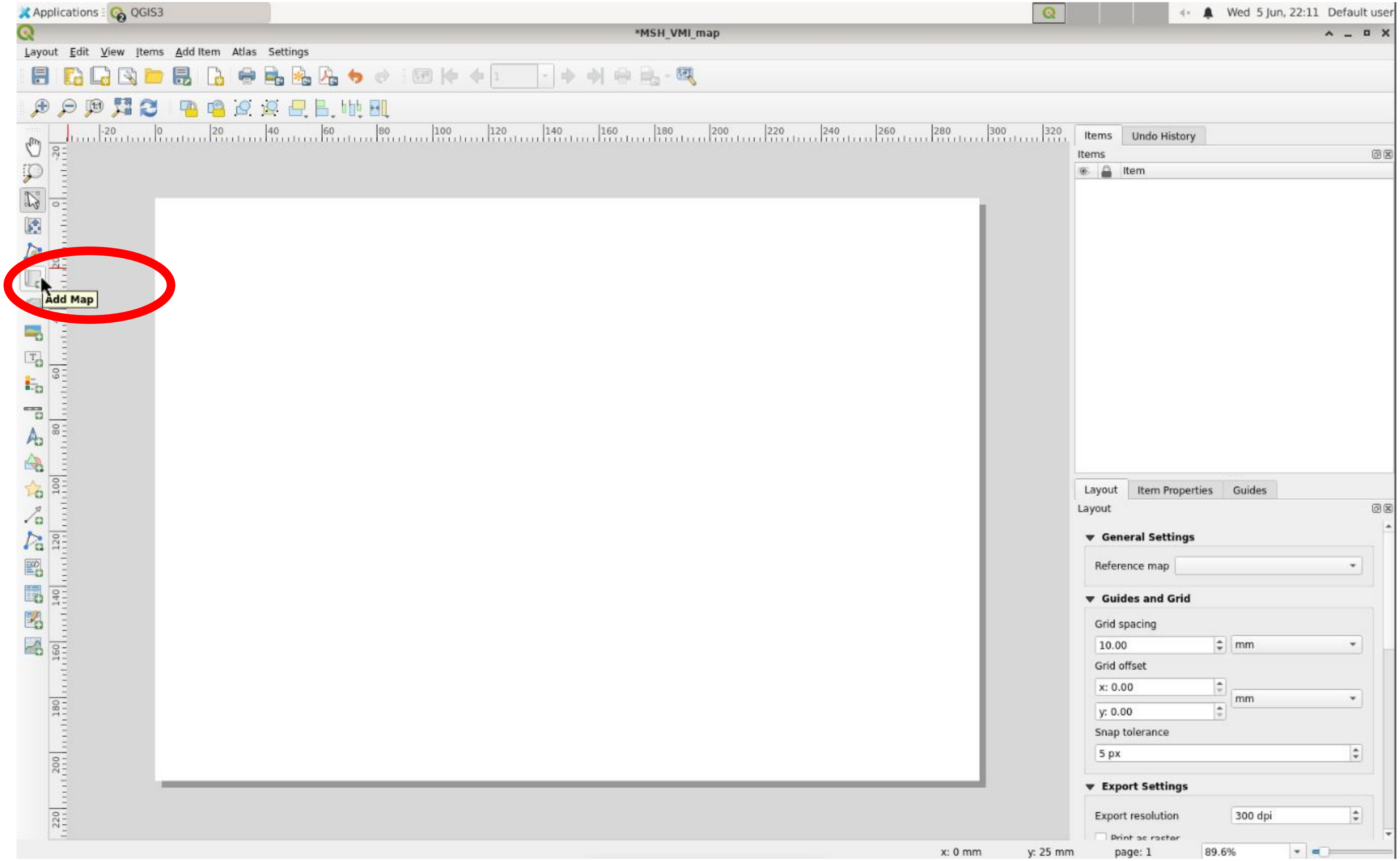
Save your project

Make a useful map!

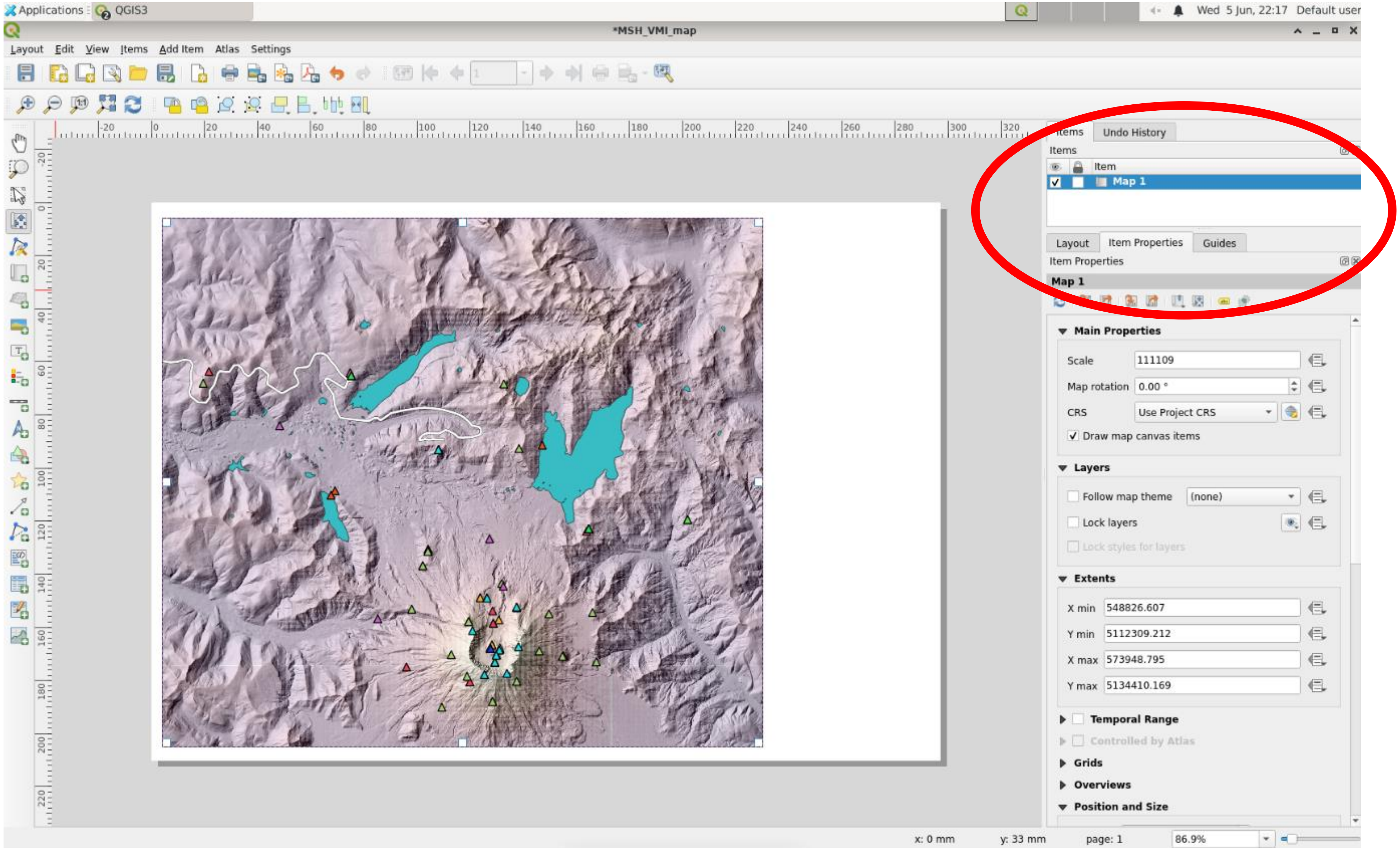
- Think about designing for your audience
- Refer to the “Ex_1_GIS_QGIS” exercise for instructions on:
    - Making the hillshade transparent and overlaying on the DEM
    - Making a map (“New Print Layout”)
    - Adding a title, legend, scale bar, north arrow, or any other elements you want to include!
    - I’ve included a few slides after this one from the GIS exercise



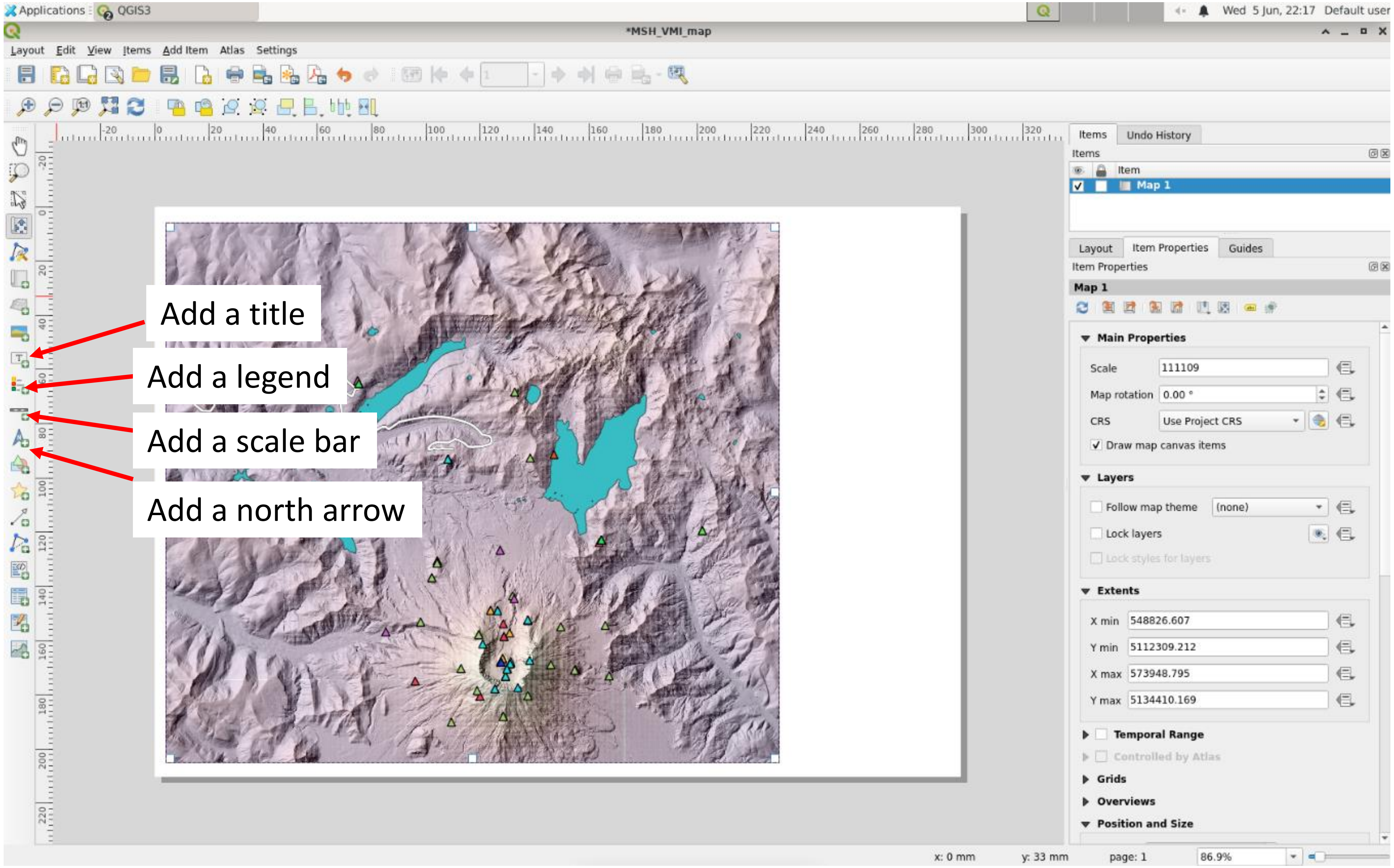
- Let's make a map!
- Click the "New Print Layout" button
- Enter a name for the new map. "mystery_volcano_map"
- Click "OK"



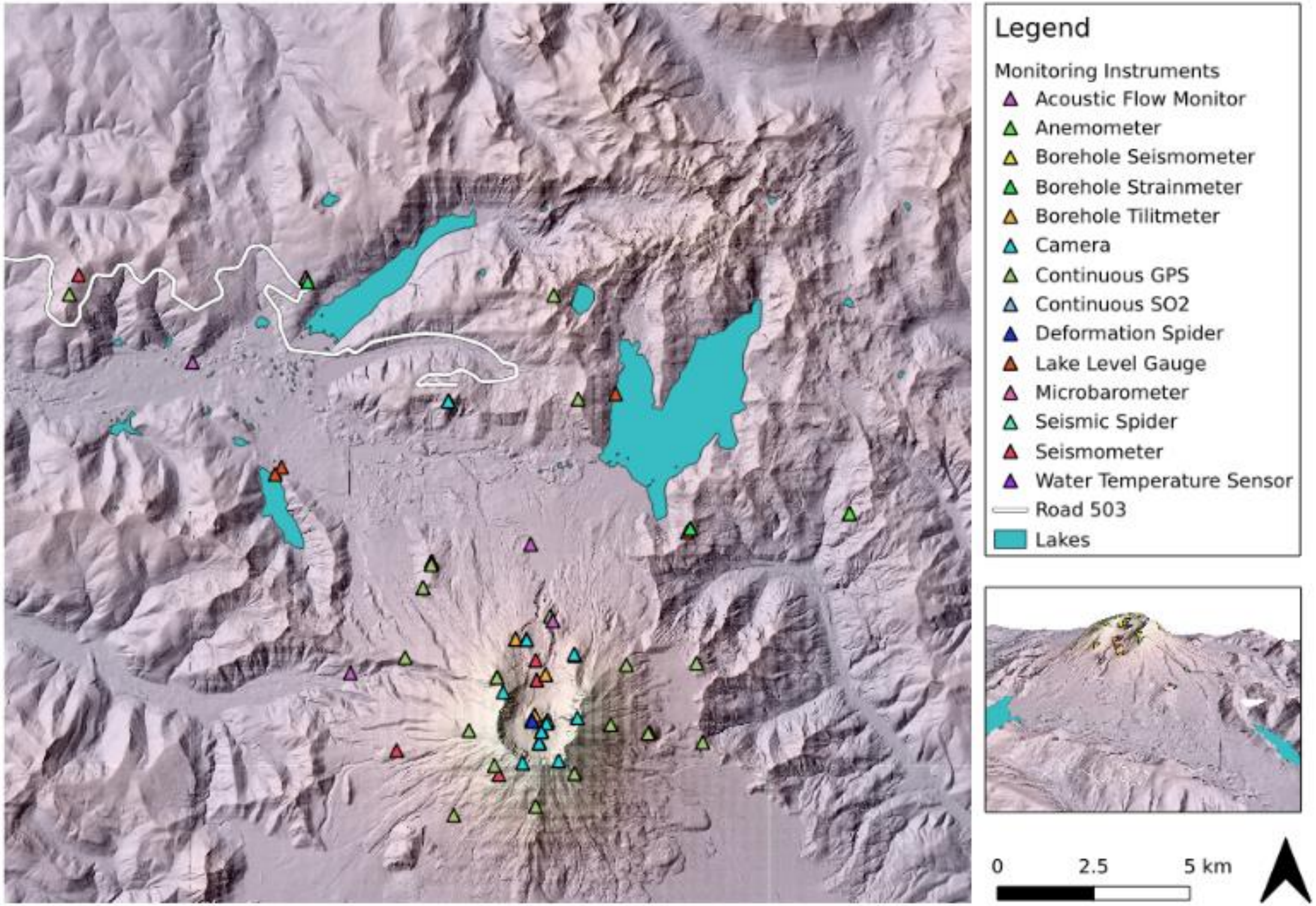
- First you need to add your map to the layout
- Click the “Add map” button
- Then draw a rectangle on the blank page to add the map



- You can adjust the scale and extent and other properties by selecting the map layer and "Item properties"



- You can add a legend for the stations and other layers
- Add a scale bar and/pr coordinates
- Add a north arrow
- Play around with the options to make your map



- Save your final work!
- You can also export the map as an image